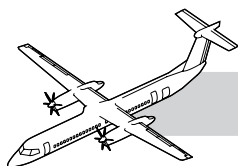


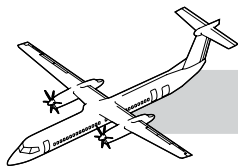
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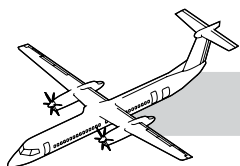
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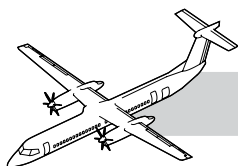
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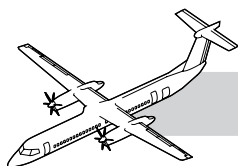


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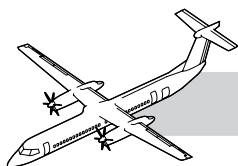


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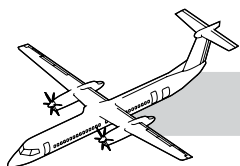
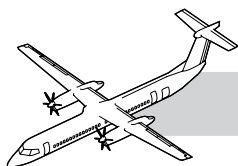
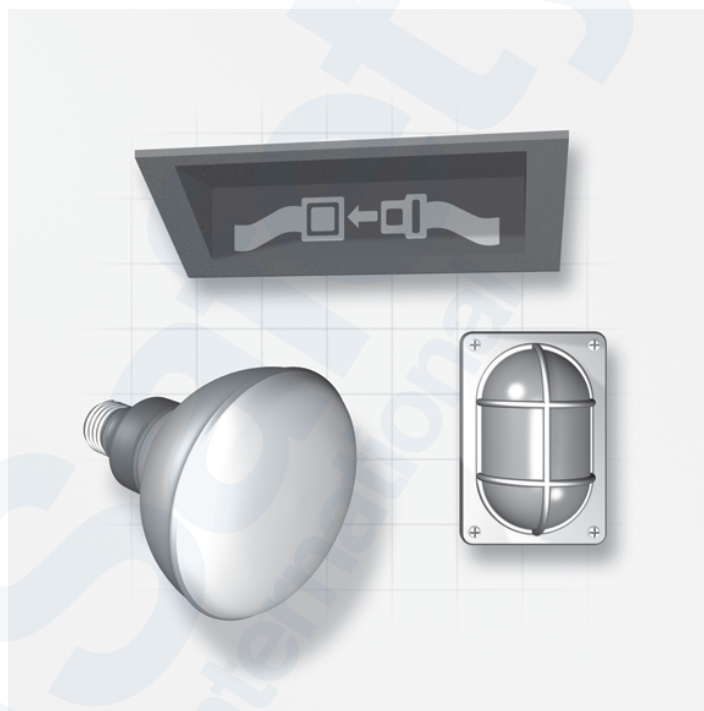


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CHAPTER 33 LIGHTING



33-00-00 INTRODUCTION

The lighting systems supply sufficient illumination for the convenience of the passengers and crew, as well as for the safe operation of the aircraft under all lighting conditions.

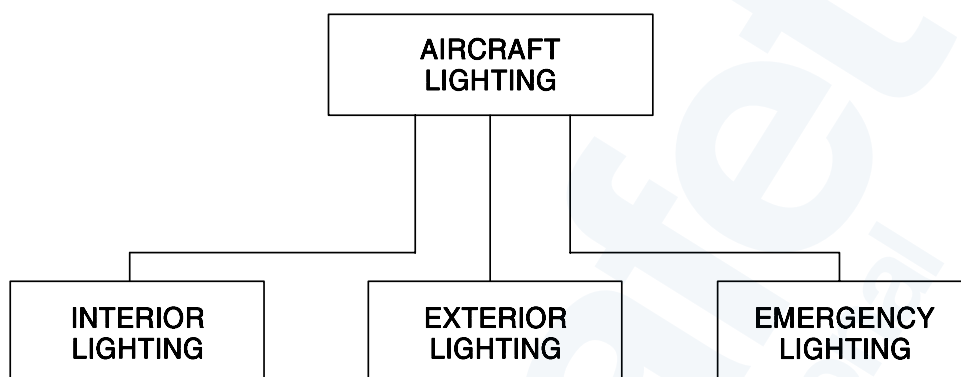
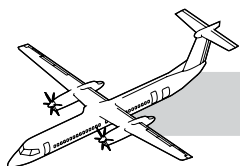


Figure 33-1. Lighting Block Diagram

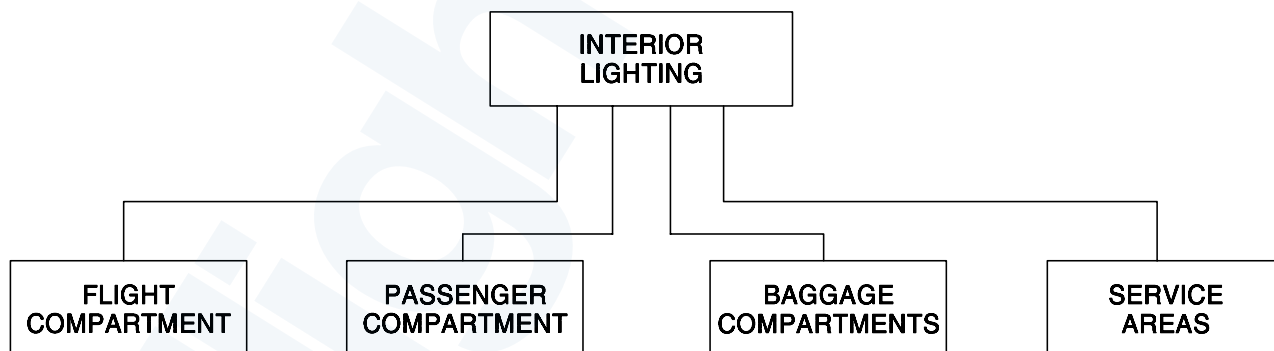
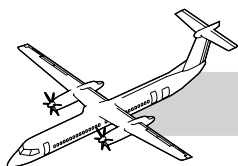


Figure 33-2. Interior Lighting Block Diagram



GENERAL

NOTES

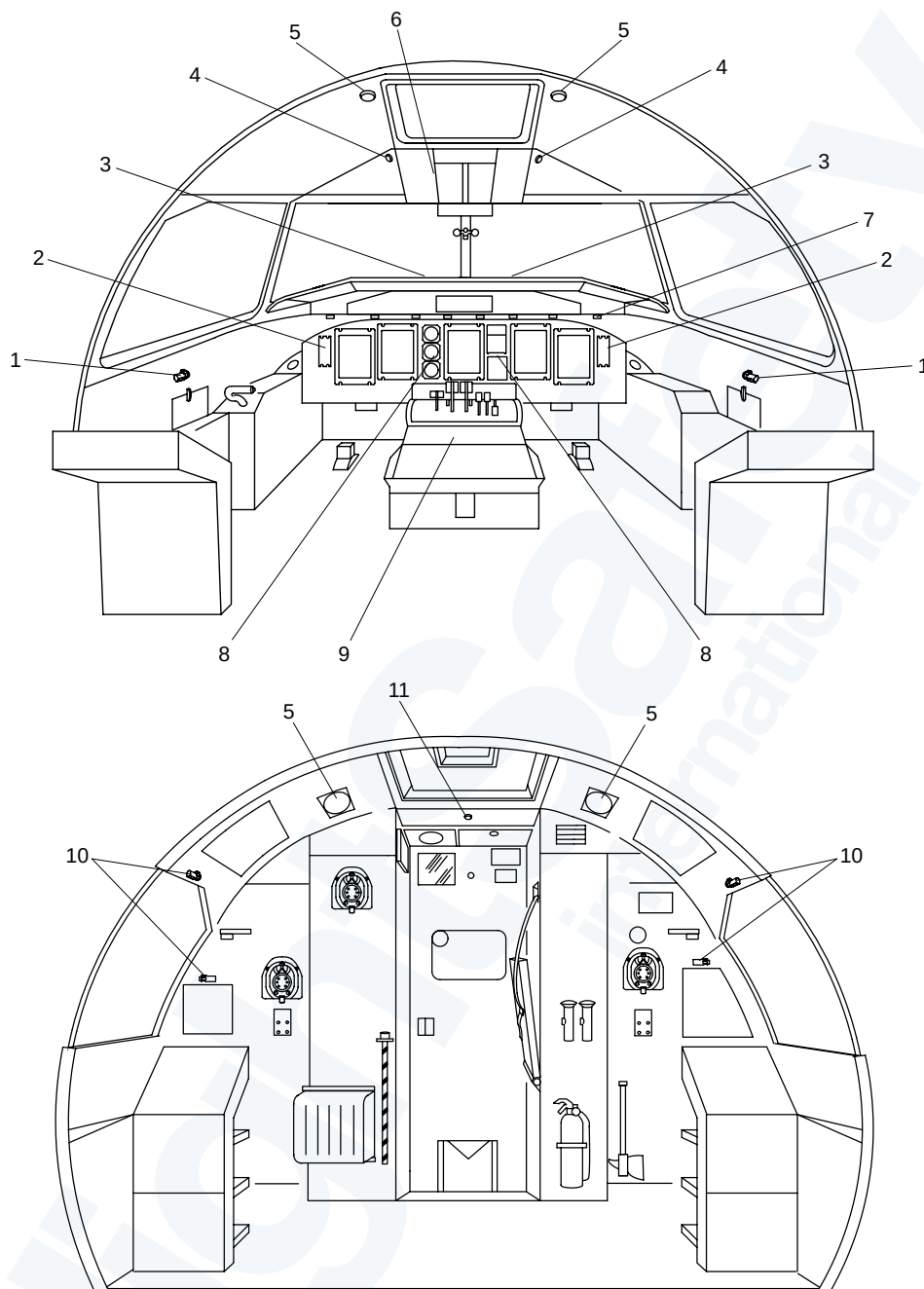
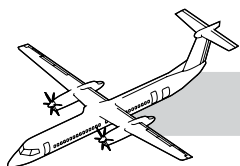
The aircraft lighting system is comprised of the following sub-systems:

- Flight compartment
- Passenger compartment
- Baggage and service compartments
- Exterior
- Emergency.

GENERAL DESCRIPTION

Interior lighting includes (Refer to Figure 33-2):

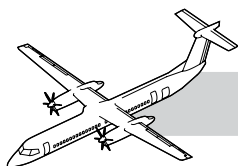
- Flight compartment
- Passenger compartment
- Baggage compartments
- Service areas.



LEGEND

- | | |
|------------------------|---------------------------------|
| 1. Map Light | 7. Thunderstrom Lights |
| 2. Index Control Panel | 8. Instrument Panel |
| 3. Ice Detector Light | 9. Center Console Panel |
| 4. Reading Light | 10. Circuit Breaker Panel Light |
| 5. Dome Light | 11. Observer Reading Light |
| 6. Overhead Panel | |

Figure 33-3. Panel and Instrument Lights



33-10-00 FLIGHT COMPARTMENT LIGHTS

NOTES

INTRODUCTION

The flight compartment lights supply illumination of the flight compartment for the safe operation of the aircraft.

GENERAL

Refer to Figure 33-3. Panel and Instrument Lights.

The flight compartment has the lights that follow:

- Panel and instrument
- Circuit breaker panel
- Dome
- Thunderstorm
- Map
- Reading
- Windshield wiper ice detection.

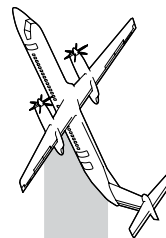
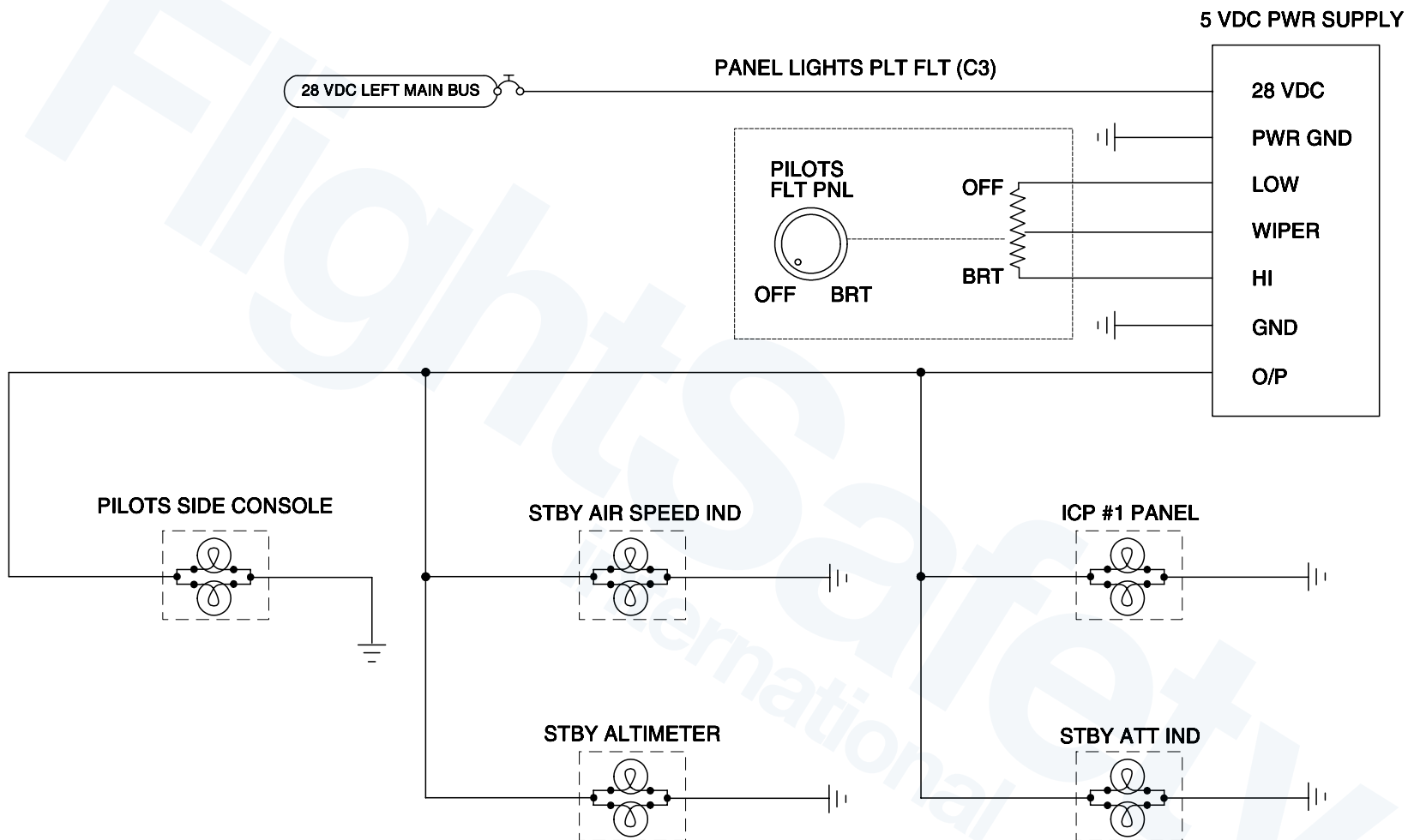
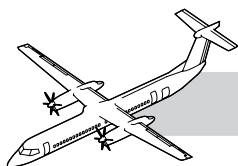


Figure 33-4. Pilot's Flight Panel Lights Wiring Schematic



33-11-00 PANEL AND INSTRUMENT LIGHTS

NOTES

INTRODUCTION

Lamp assemblies are installed internally in the plexiglass facing the panel.

GENERAL

Refer to Figure 33-4. Pilot's Flight Panel Lights Wiring Schematic.

A variable resistance is supplied from a potentiometer in the lighting control panel to the power supply to change the output voltage.

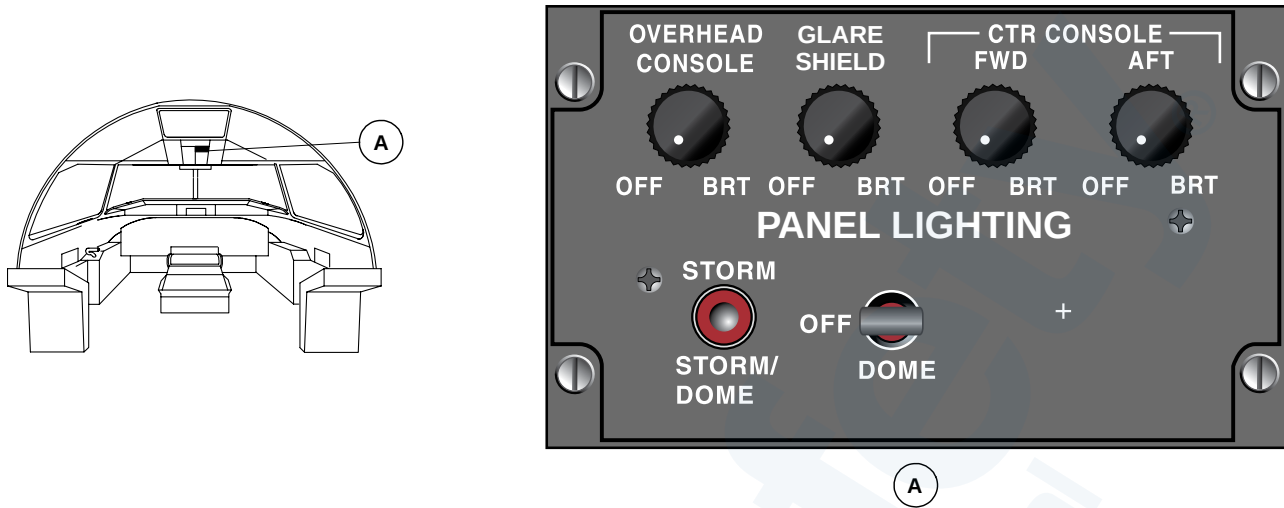
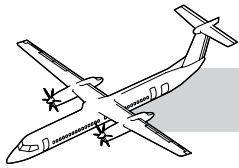


Figure 33-5. Overhead Console Lighting Panel

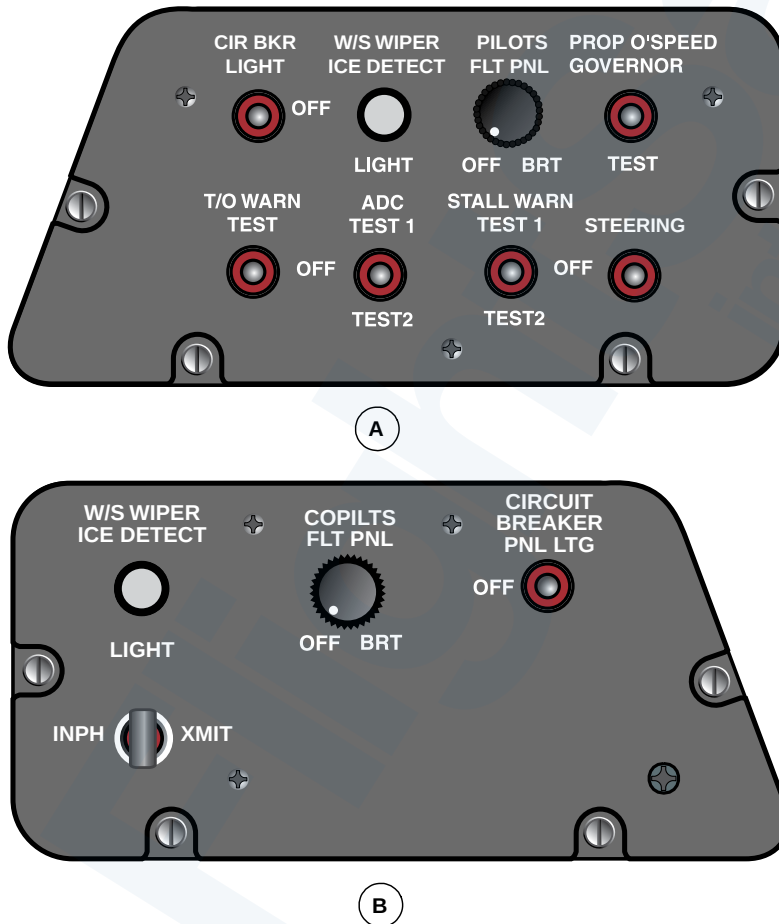
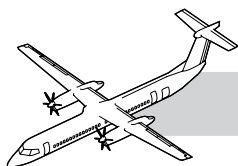


Figure 33-6. Side Console Lighting Panel



COMPONENT DESCRIPTION

Overhead Console Panel

Refer to Figure 33-5. Overhead Console Lighting Panel.

The overhead console lighting control panel is installed in the overhead console.

The control panel has switches and knobs for control.

The switches are identified as follows:

- STORM/DOME
- DOME.

The knobs are identified as follows:

- OVERHD CONSOLE
- GLARESHIELD
- CTR CONSOLE, FWD
- CTR CONSOLE, AFT.

It gives control for the areas that follow:

- Overhead console
- Glareshield panel
- Forward center console
- Aft center console
- Pilot flight panel
- Co-pilot flight panel.

Refer to Figure 33-6. Side Console Lighting Panel.

The side console lighting panels are installed in the pilot and co-pilot side consoles. The control panels also have switches and knobs for control.

The knobs are identified as follows:

- PILOTS FLT PNL
- COPILOTS FLT PANEL.

The switches are identified as follows:

- CIR BKR LIGHT
- W/S WIPER ICE DETECT.

The pilot's side console lighting panel gives control for the areas that follow:

- Pilot side console lighting panel
- Pilot index control panel.

The co-pilot's side console lighting panel provides control for the areas that follow:

- Co-pilot side console lighting panel
- Co-pilot index control panel
- Landing gear control panel
- GPWS/Hydraulic control panel.

CAUTION

DO NOT GET SCRATCHES ON THE FACING PANEL. LIGHT WILL SHOW THROUGH THE SCRATCHES. IT IS NOT EASY TO READ THE PANEL LEGEND DURING THE NIGHT IF THE PANEL HAS TOO MANY SCRATCHES. THIS CONDITION CAN CAUSE AN ACCIDENT IF YOU OPERATE THE AIRCRAFT DURING THE NIGHT.

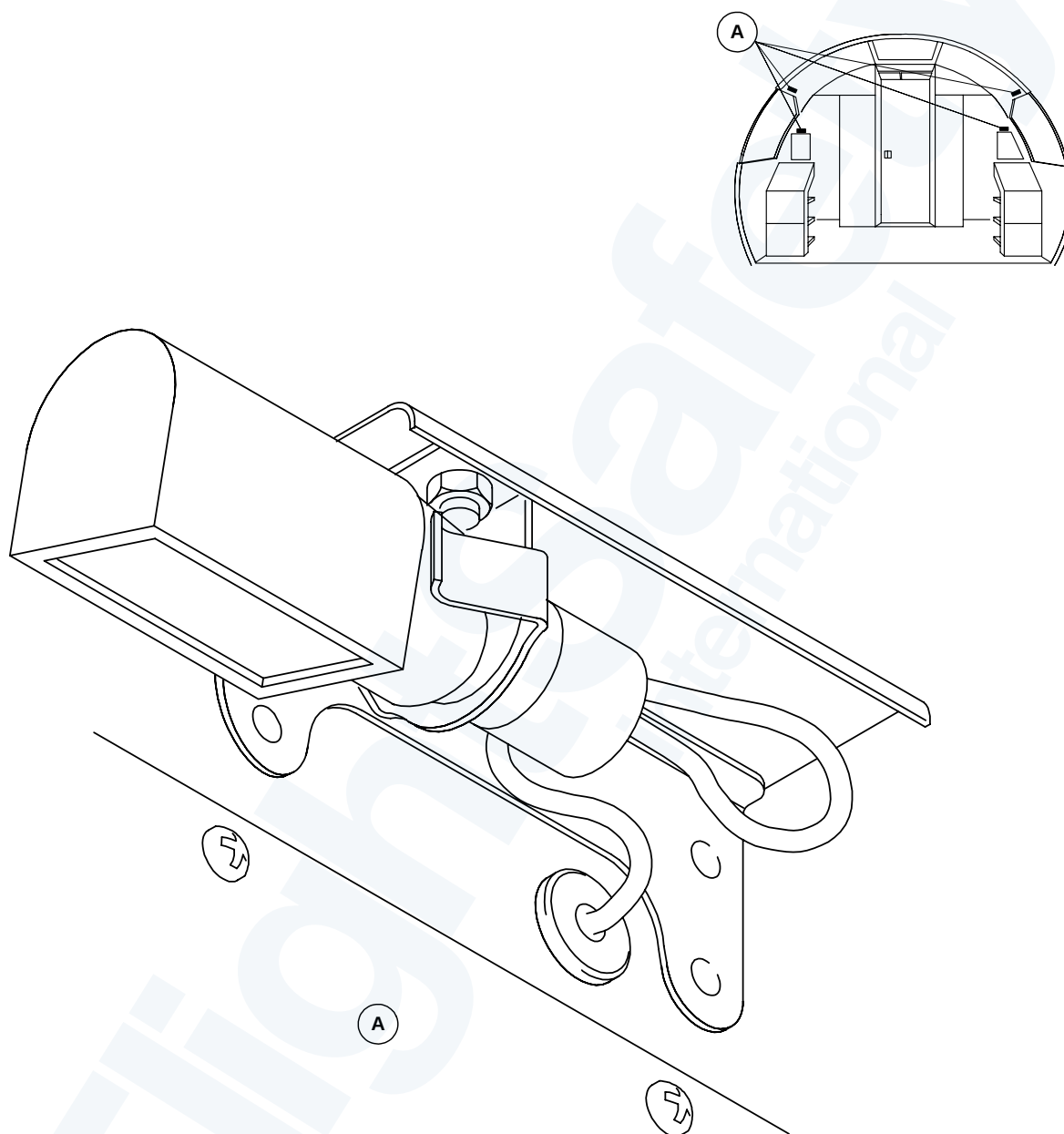
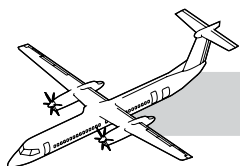
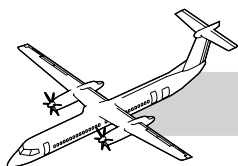


Figure 33-7. Circuit Breaker Panel Lights - Locations



Circuit Breaker Panel Lights

NOTES

Refer to:

- Figure 33-7. Circuit Breaker Panel Lights - Locations.
- Figure 33-8. Circuit Breaker Panel Lights on Side Consoles.

Illumination of the four circuit breaker panels is provided by four flood lights.

For controls, refer to Figure 33-8. Circuit Breaker Panel Lights on Side Consoles.

The illumination of the left DC circuit breaker panel and the avionics circuit breaker panel is controlled by a switch identified as CIRCUIT BREAKER PNL LTG. Power to the lights is supplied by the left essential DC bus.

Refer to Figure 33-9. Circuit Breaker Panel and Ice Detector Lights Schematic.

The illumination of the right DC circuit breaker panel, and the 115 VAC variable frequency circuit breaker panel is controlled by a switch identified as CIRCUIT BREAKER PNL LTG. Power to the lights is supplied by the right essential DC bus.

The illumination of the left ice detector is controlled by a switch identified as W/S WIPER ICE DETECT. Power to the light is supplied by the left essential bus.

The illumination of the right ice detector is controlled by a switch identified as W/S WIPER ICE DETECT. Power to the light is supplied by the right essential bus.

The circuit breaker panel light assemblies are installed above pilot and copilot circuit breaker panels.

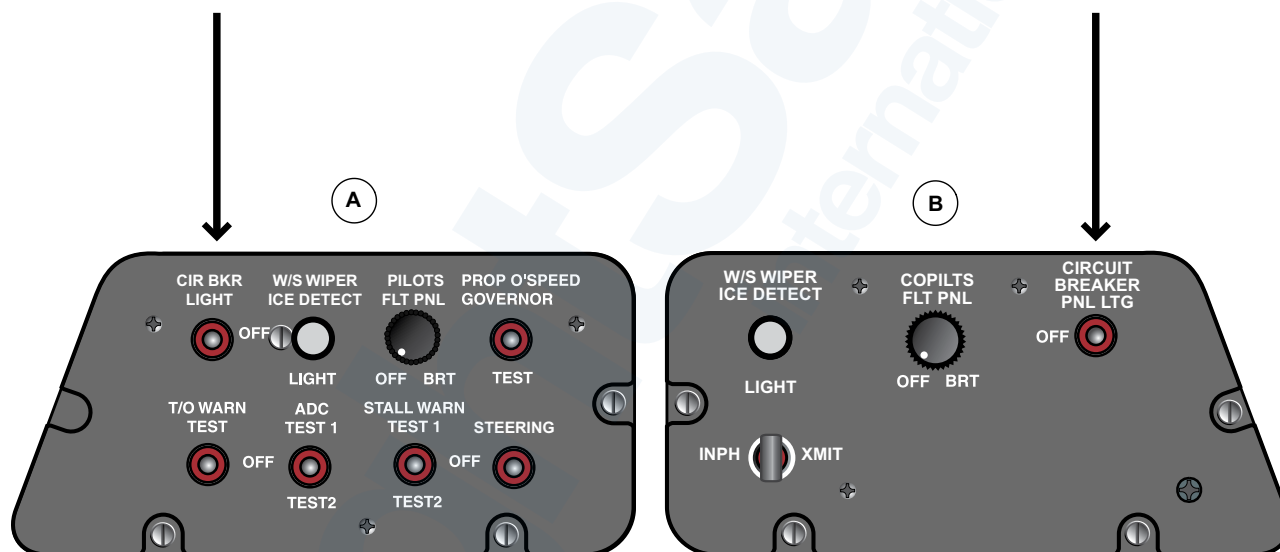
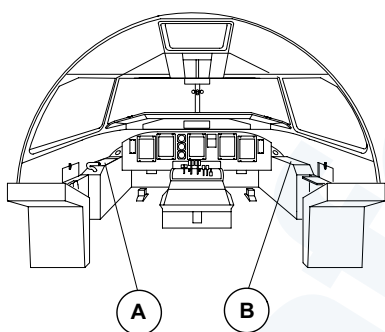
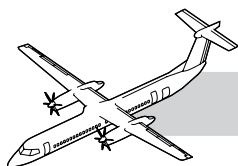


Figure 33-8. Circuit Breaker Panel Lights on Side Consoles

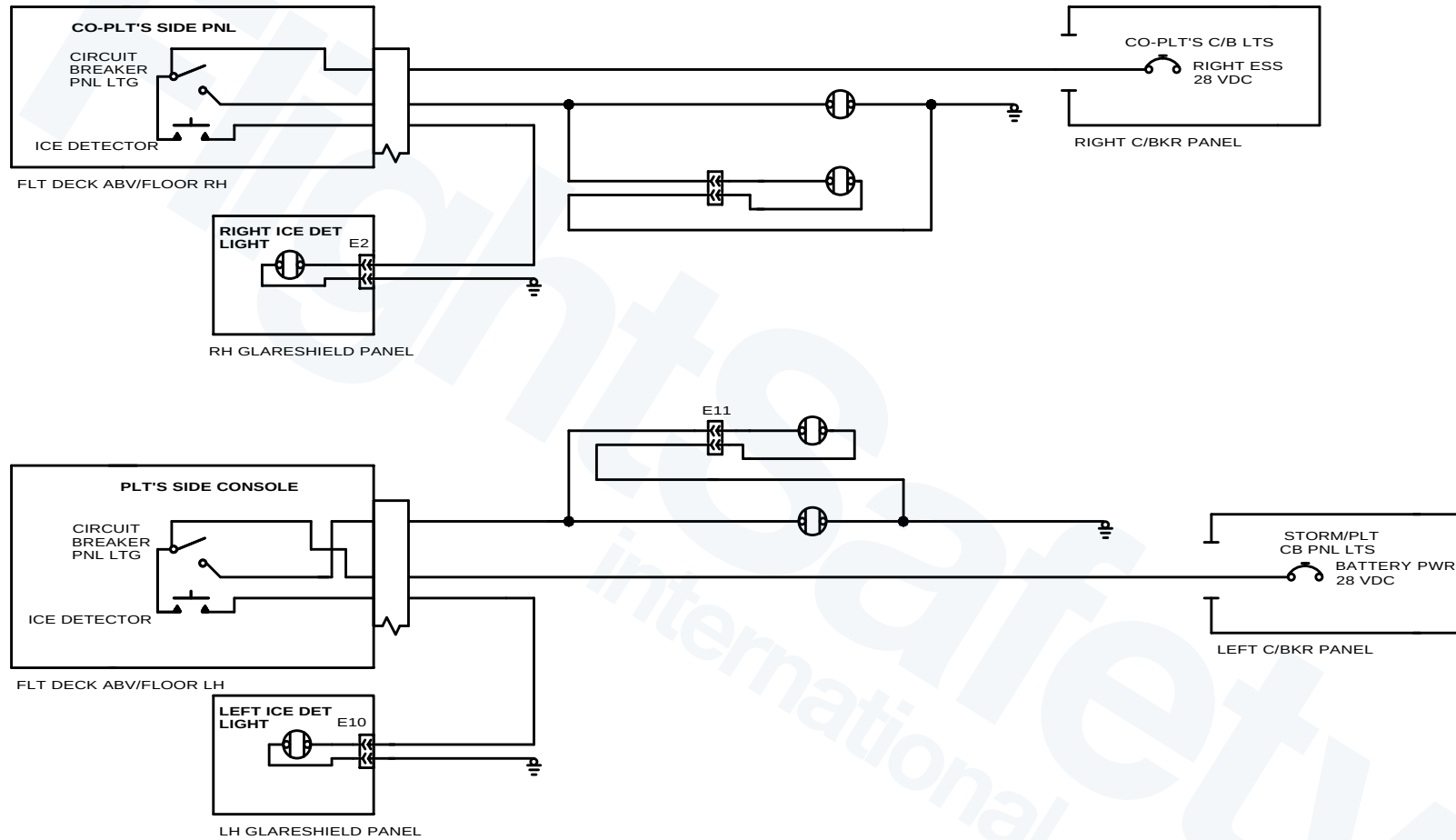
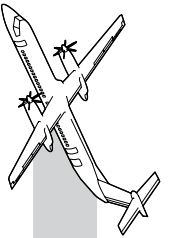


Figure 33-9. Circuit Breaker Panel and Ice Detector Lights Schematic

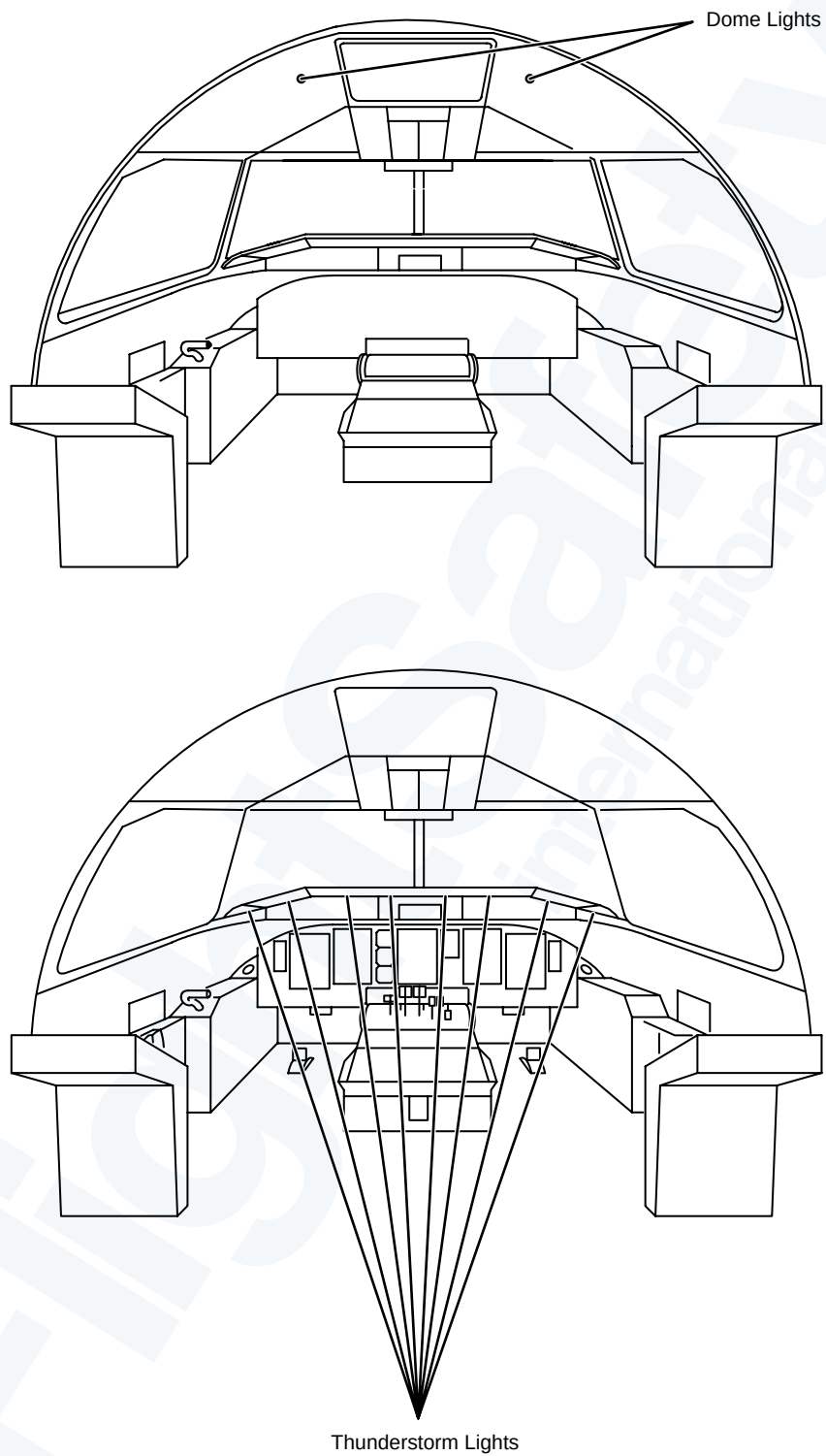
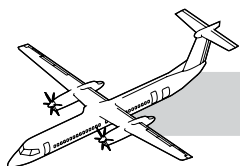
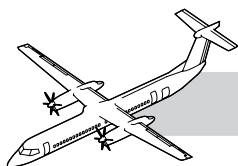


Figure 33-10. Dome and Thunderstorm Lights Location



33-12-00 DOME AND THUNDERSTORM LIGHTS

NOTES

INTRODUCTION

Refer to Figure 33-10. Dome and Thunderstorm Lights Location.

Illumination of the flight compartment is supplied by two dome and eight thunderstorm lights.

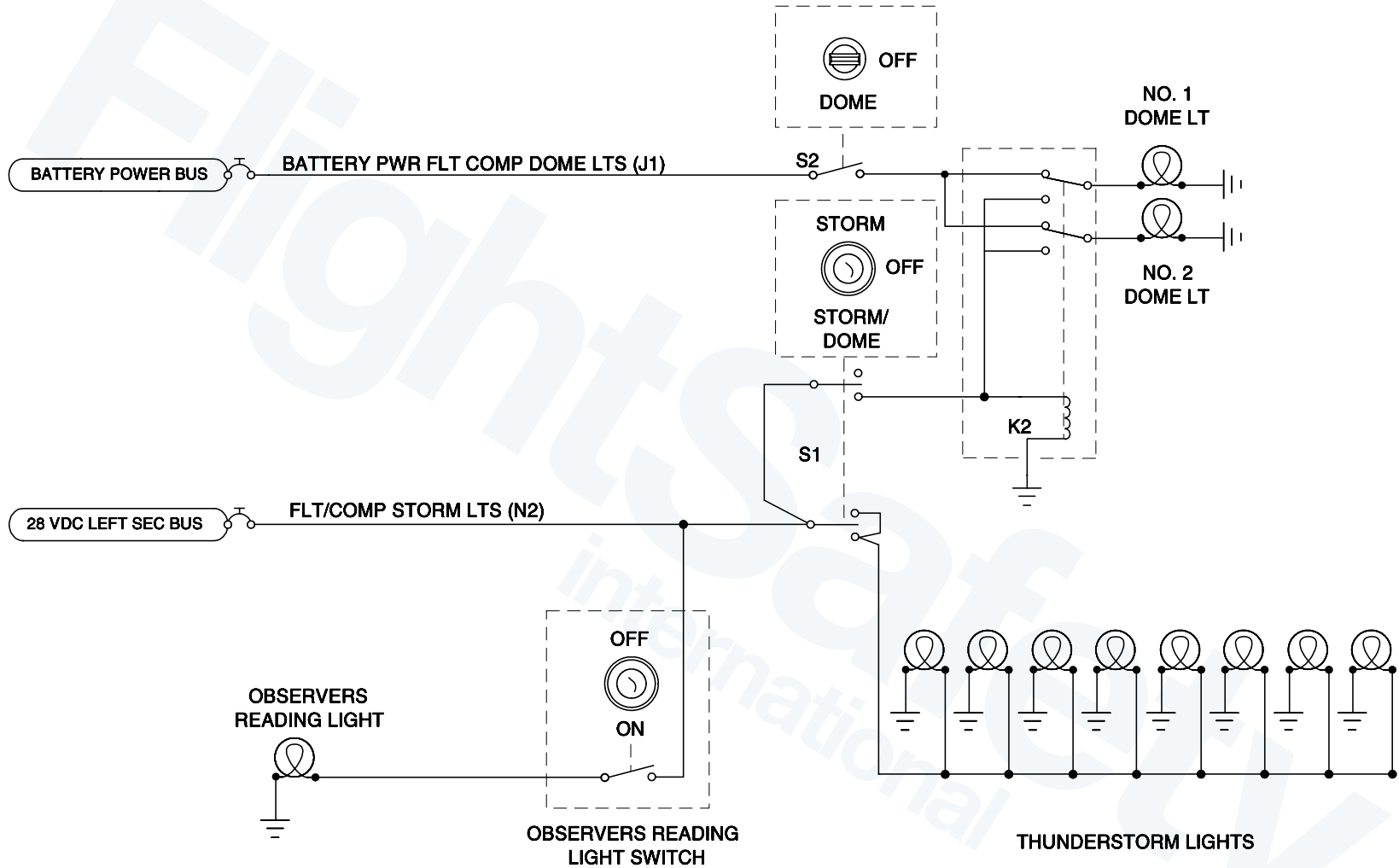
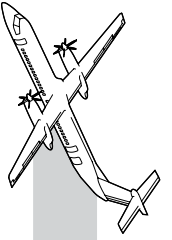
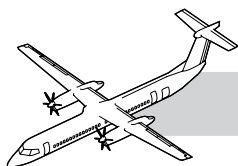


Figure 33-11. Storm/Dome Lights Wiring Schematic



GENERAL

NOTES

Refer to:

- Figure 33-11. Storm/Dome Lights Wiring Schematic.
- Figure 33-12. Panel Lighting Panel.

The dome lights are controlled by the STORM/DOME or DOME switches. The thunderstorm lights are controlled by the STORM/DOME switch.

Power to the dome lights is supplied from the left secondary DC bus or from the right battery bus. The source of power is determined by a relay.

When the switch is set to the STORM/DOME position, power is supplied from the Left Secondary Bus to the dome lights. When the other switch is set to the DOME position, power is supplied from the battery bus to the dome lights.

The thunderstorm lights can come on independently or with the dome lights. Power to the thunderstorm lights is supplied from the left battery.

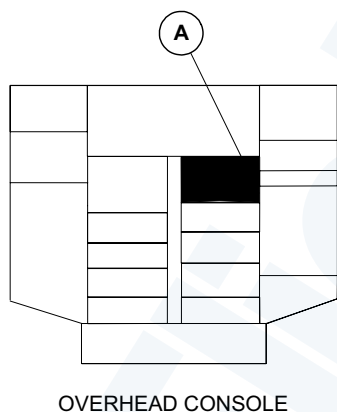
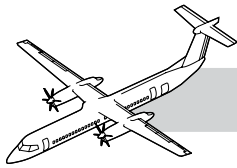


Figure 33-12. Panel Lighting Panel



COMPONENT DESCRIPTION

Dome Lights

Refer to Figure 33-13. Dome Lights.

Two dome lights are installed in the ceiling. One dome light is installed above the pilot's seat, and the other dome light is installed above the co-pilot's seat.

There is one lamp in each dome light.

CAUTION

YOU MUST USE A COTTON GLOVE TO REMOVE THE LAMP. IF YOU TOUCH THE LAMP WITH YOUR SKIN, YOU CAN CAUSE DAMAGE TO THE LAMP.

The dome lights are comprised of:

- Housing
- Lens
- Lamps.

Dome Light Lamp Removal

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Carefully push and turn the applicable dome light lens counterclockwise to disengage the dome light lens from the housing assembly.
- Carefully push in the lamp. Turn the lamp $\frac{1}{4}$ counterclockwise to remove.

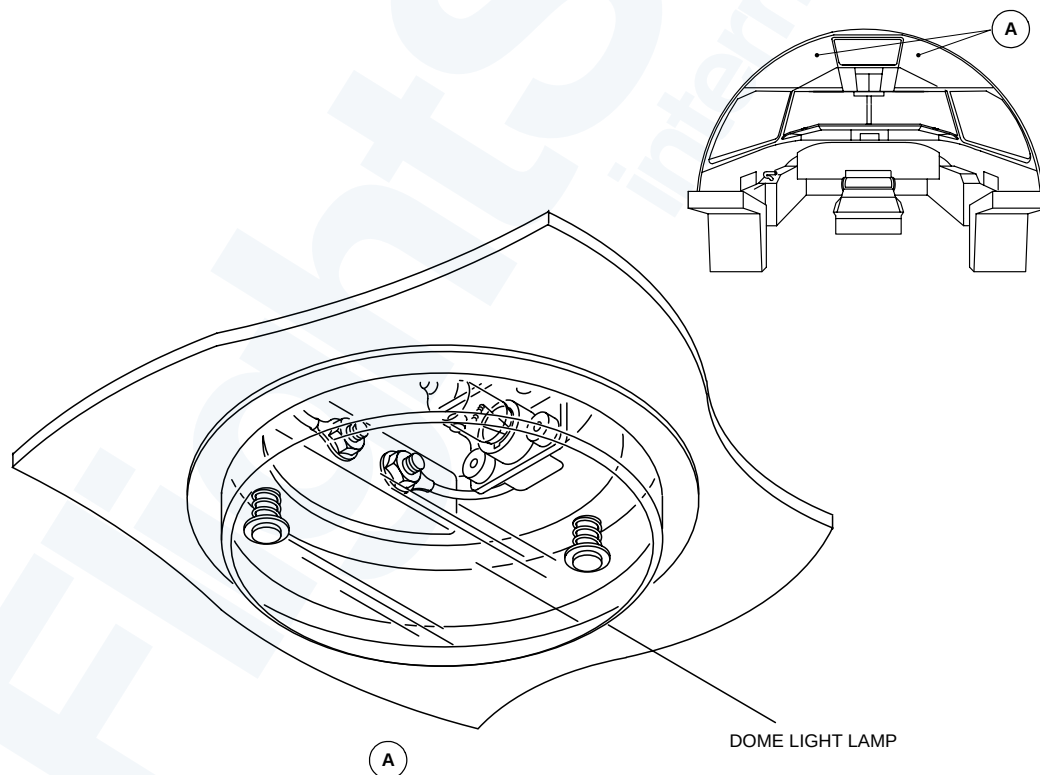
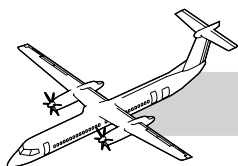


Figure 33-13. Dome Lights



Thunderstorm Lights

Refer to Figure 33-14. Thunderstorm Lights.

The thunderstorm lights are installed below the glareshield.

They are comprised of:

- Boot
- Housing
- Light bulb.

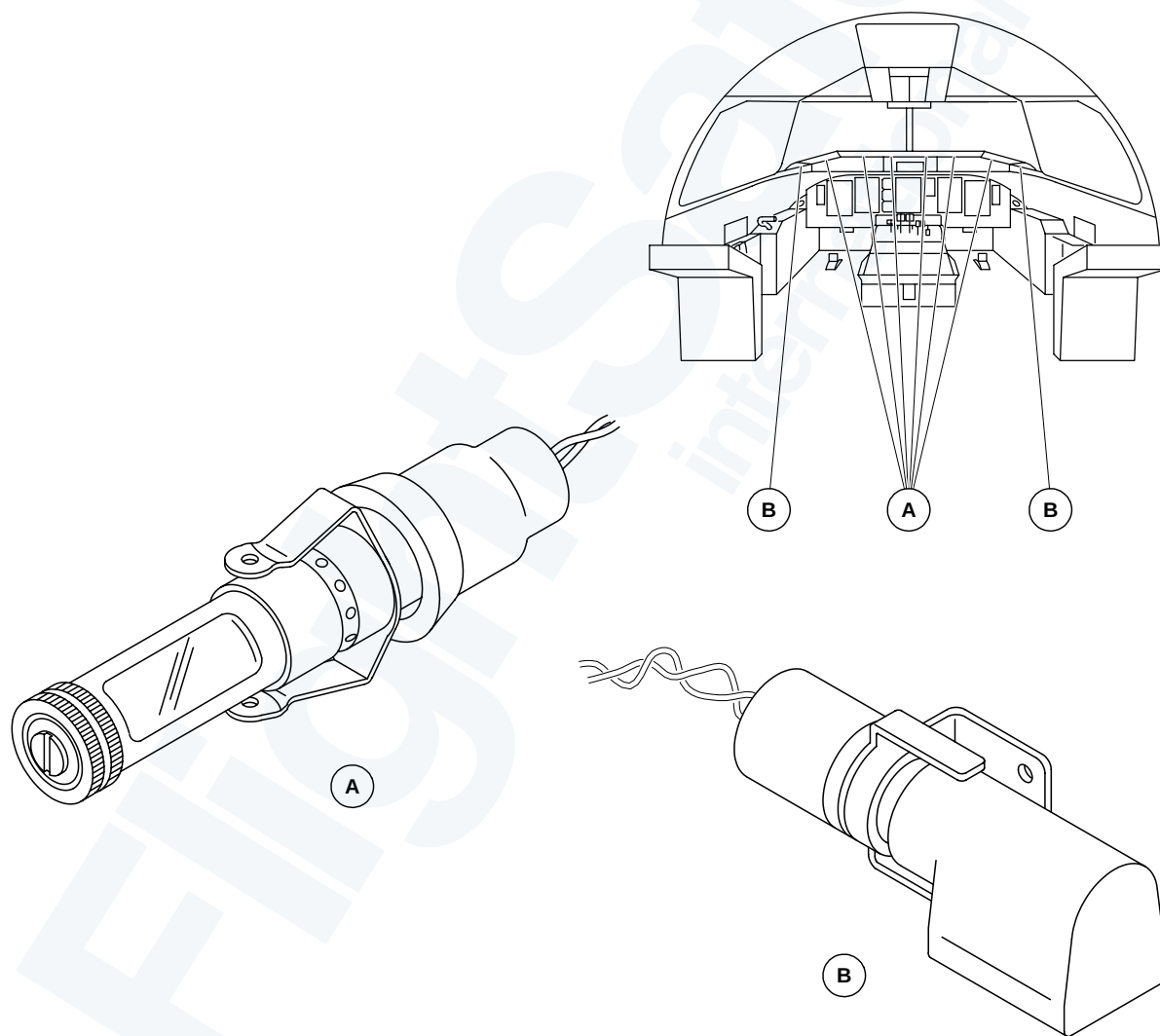


Figure 33-14. Thunderstorm Lights

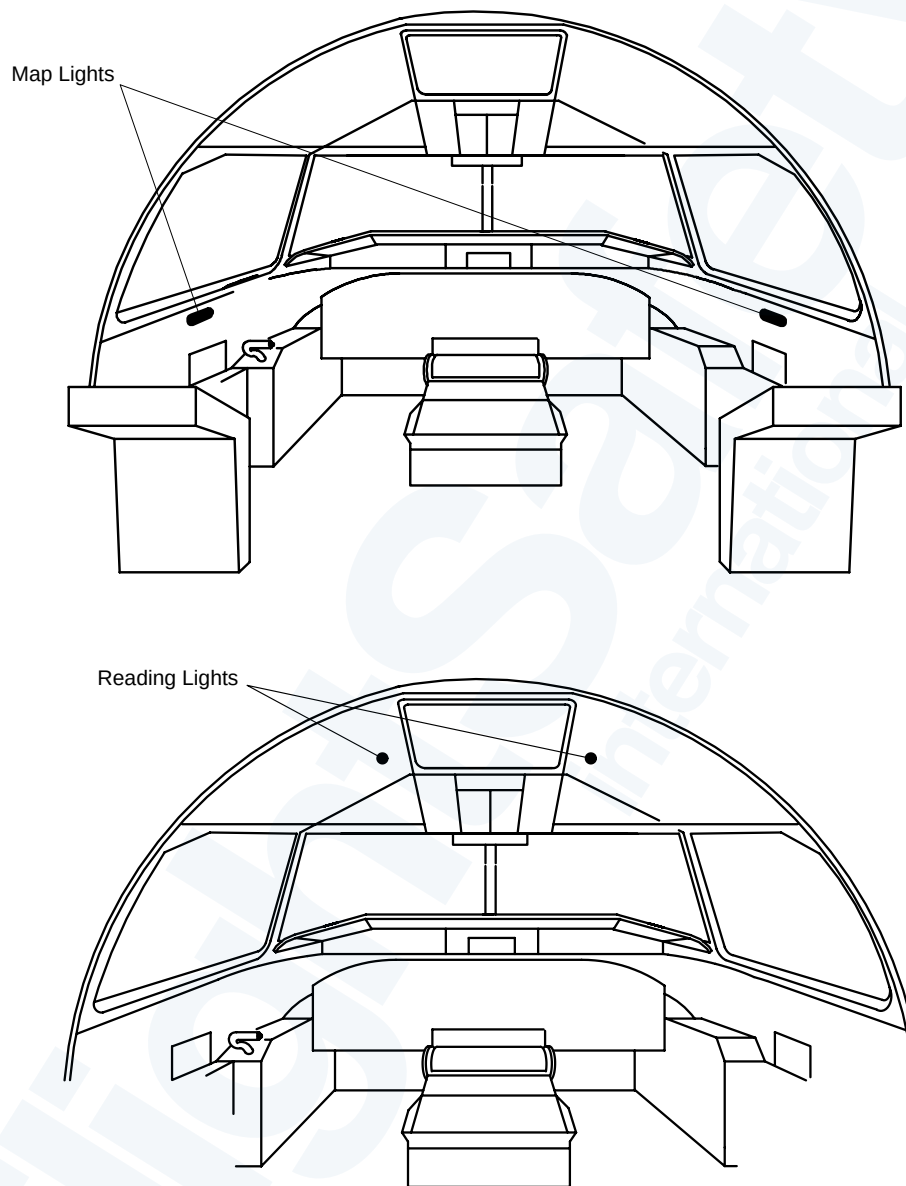
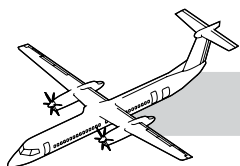
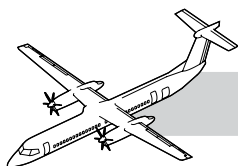


Figure 33-15. Maps and Reading Lights Location



33-13-00 MAP LIGHTS

NOTES

INTRODUCTION

Refer to Figure 33-15. Maps and Reading Lights Location.

Two map lights and two reading lights are supplied.

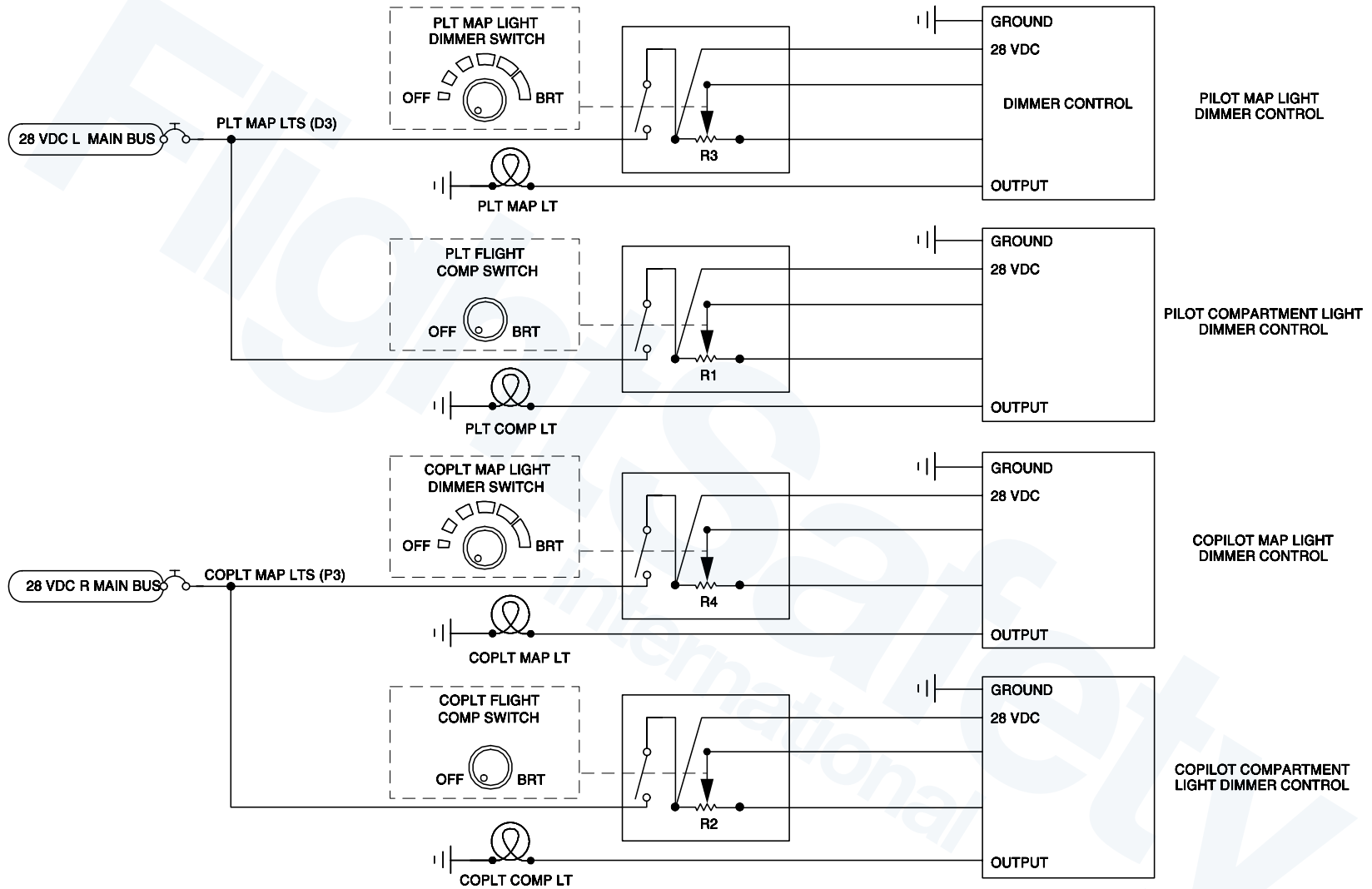
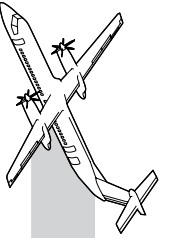
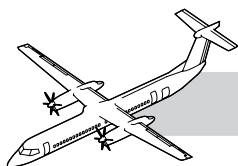


Figure 33-16. Map and Utilities Lights Wiring Schematic



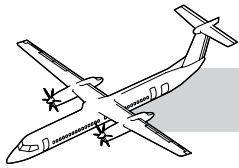


GENERAL

NOTES

Refer to Figure 33-16. Map and Utilities Lights Wiring Schematic. |

Variable resistance is supplied from a potentiometer on either side panel to the power supply to change the output voltage.



COMPONENT DESCRIPTION

5 VDC Power Supplies

Refer to Figure 33-17. 5 VDC Power Supplies.

Four 5 VDC power supplies are installed on the side of the avionic rack electrical assembly.

It changes the voltage at the output from 0 to 5.7 VDC.

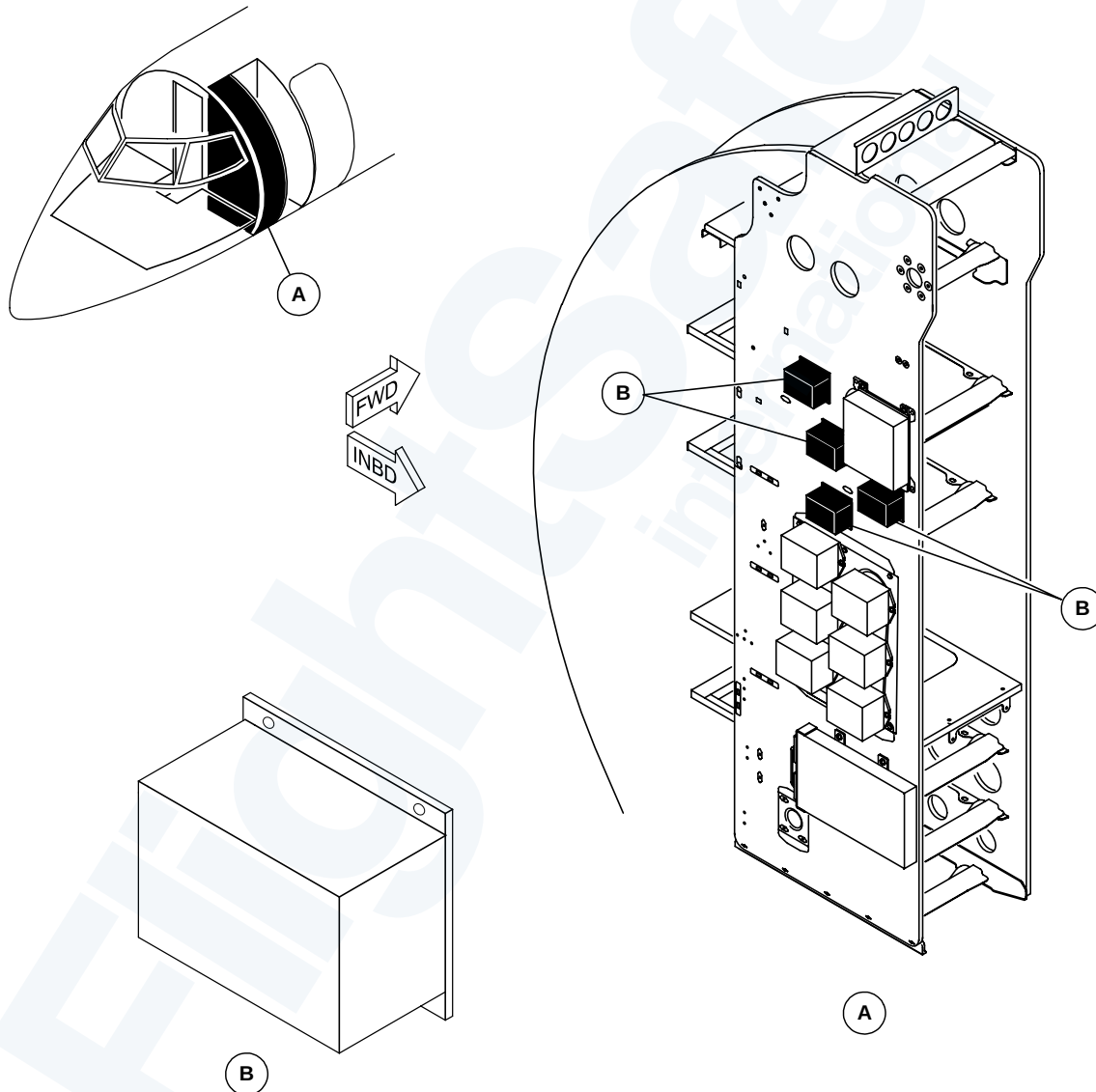
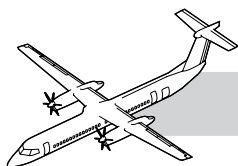


Figure 33-17. 5 VDC Power Supplies



Map Lights

Refer to Figure 33-18. Map Lights - Location.

The map lights are installed aft of the pilot and co-pilot side console.

They are comprised of bracket, lamp holder, lens cover and lamp.

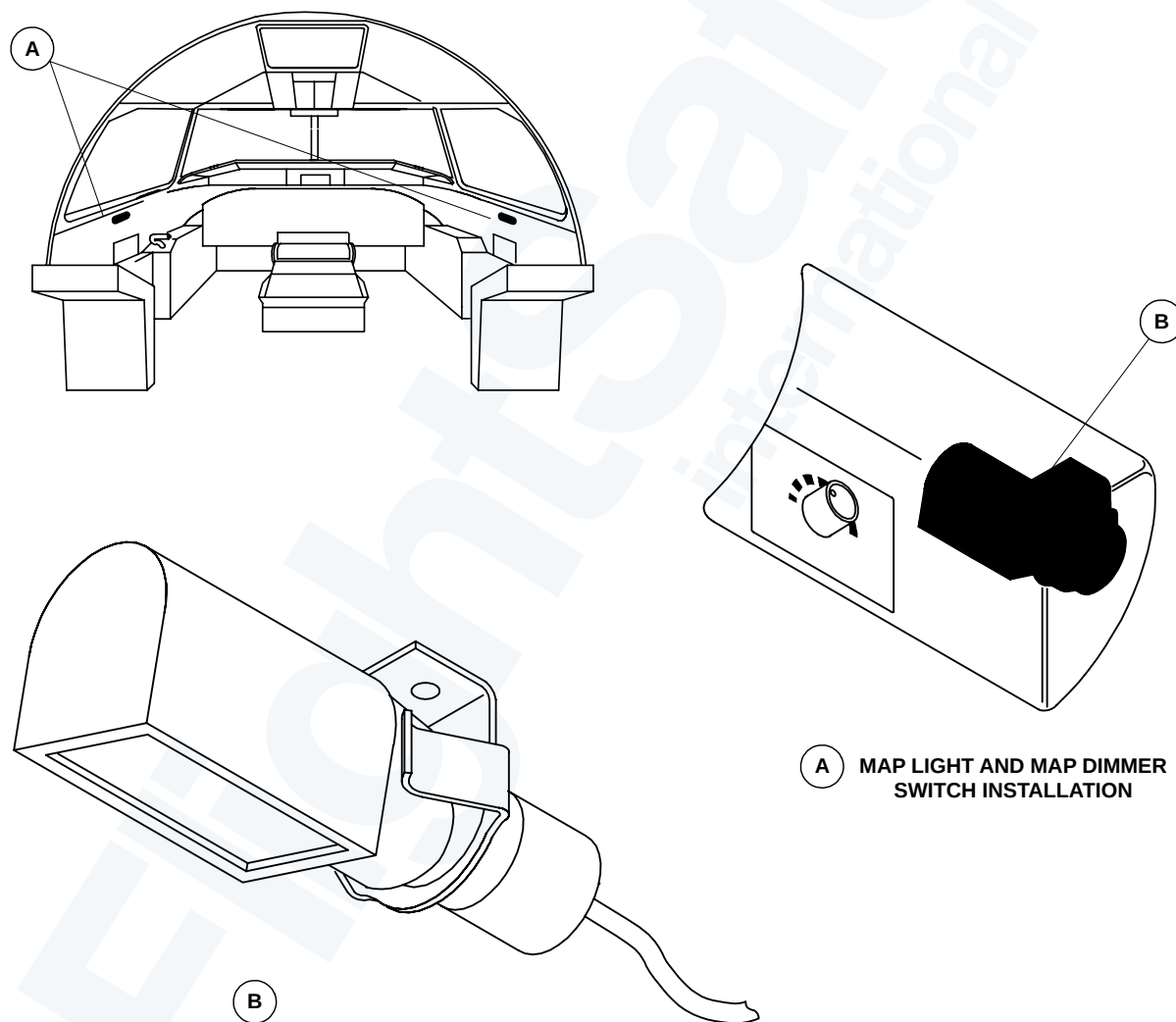
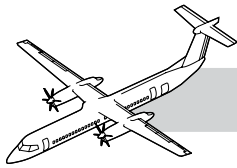


Figure 33-18. Map Lights - Location



Pilot and Co-Pilot Reading Lights

Refer to Figure 33-19. Pilot and Co-Pilot Reading Light.

The pilot and co-pilot reading lights are installed on the overhead console. They are comprised of a reading light and a spring clamp.

Windshield Wiper Ice Detector

The windshield wiper ice detector lights are installed on the top of the glareshield panel shroud. They have three sub-component - light assembly, hood and bulb.

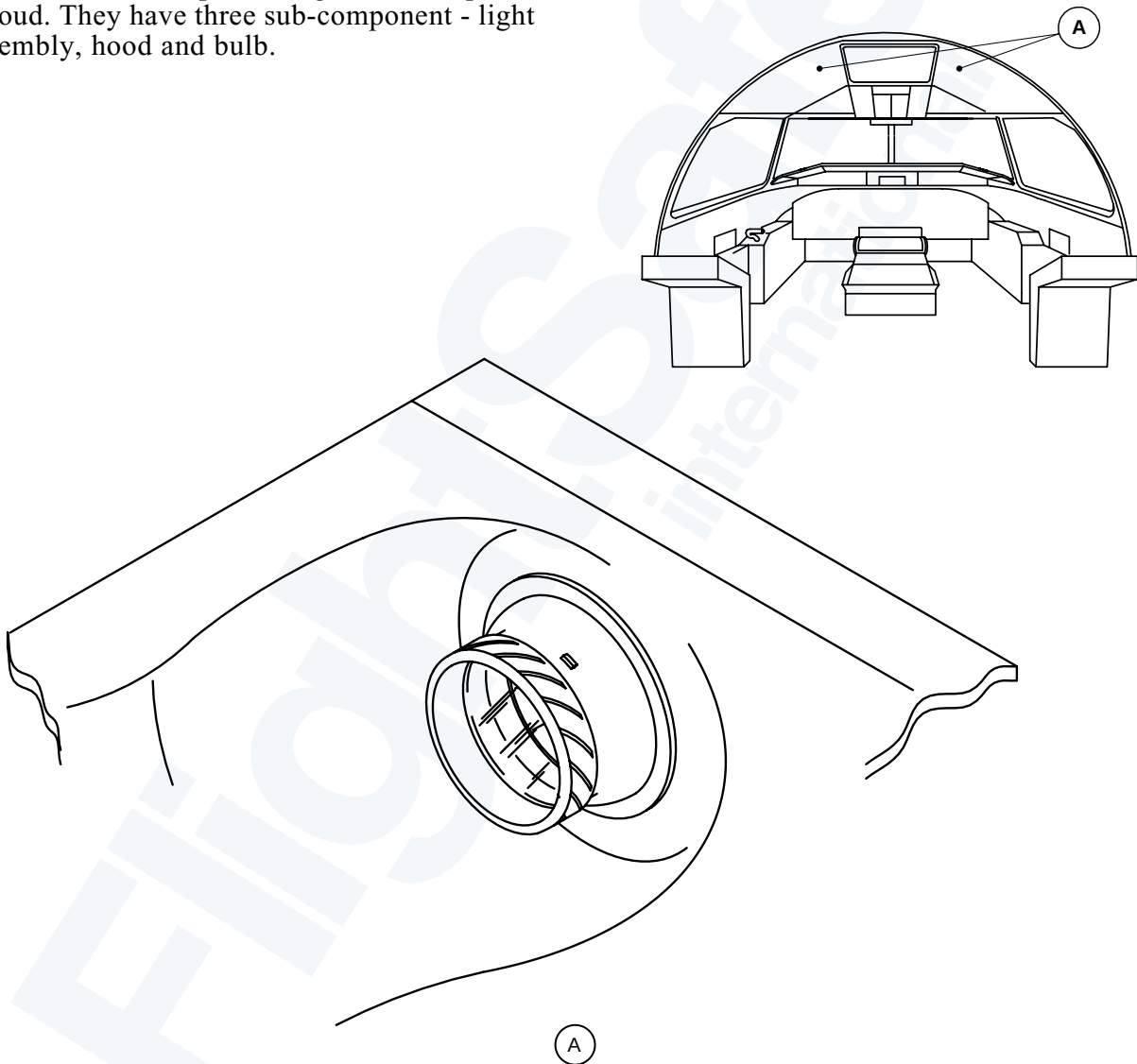
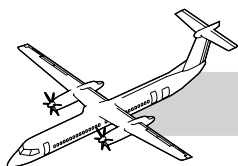


Figure 33-19. Pilot and Co-Pilot Reading Light



33-20-00 PASSENGER COMPARTMENT LIGHTING

NOTES

INTRODUCTION

The passenger compartment lights supply the general lighting requirements of the passengers, and crew in the passenger compartment area.

GENERAL

Passenger compartment lighting has the components that follow:

- Flight Attendant Panel
- Accent Lights
- Boarding Lights
- Cabin Overhead and Sidewall Lights
- Galley Lights
- Lavatory Lights
- Passenger Information Signs
- Reading Lights
- Wardrobe Lights.

Q400 Next Gen Interior - Passenger Cabin LED Lighting

With the Next Gen interior the passenger cabin, fluorescent and incandescent lamps were replaced with more reliable LED assemblies. This installation led to a reduction in ceiling panel size and resulted in an increased accessibility to emergency lighting battery packs 2 and 3. A new Flight Attendant Control Panel and accent lighting was also introduced. Incandescent light operate at approximately 3% efficiency giving up 97% of the consumed power in heat. LED lights operate at approximately 15% efficiency resulting in less heat and an estimated mean time between failures (MTBF) in excess of 200,000 hours.

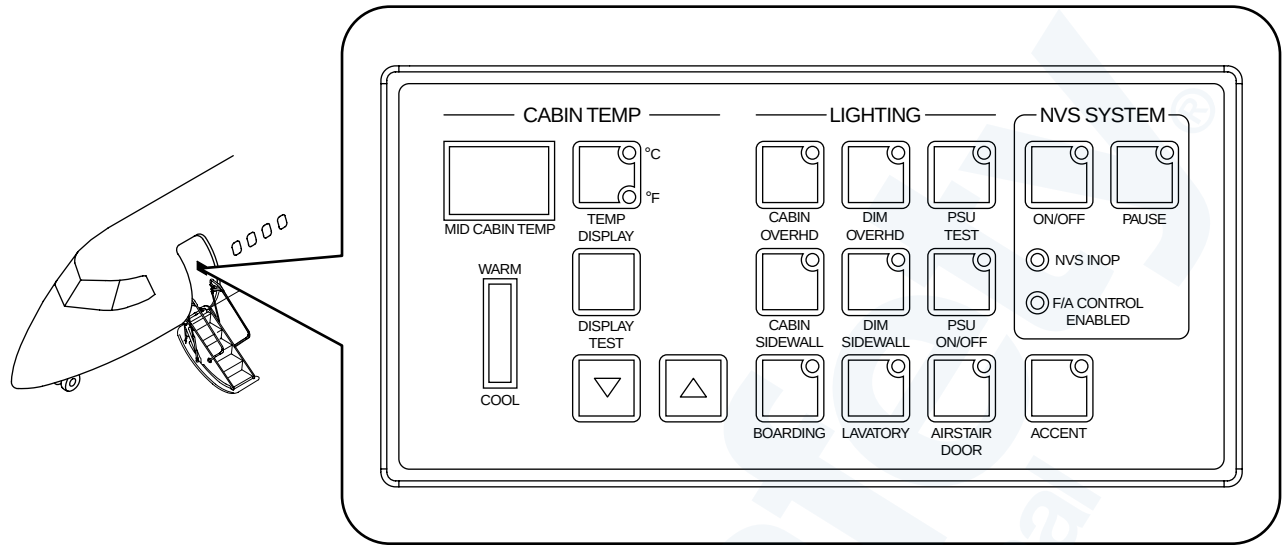
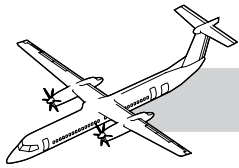
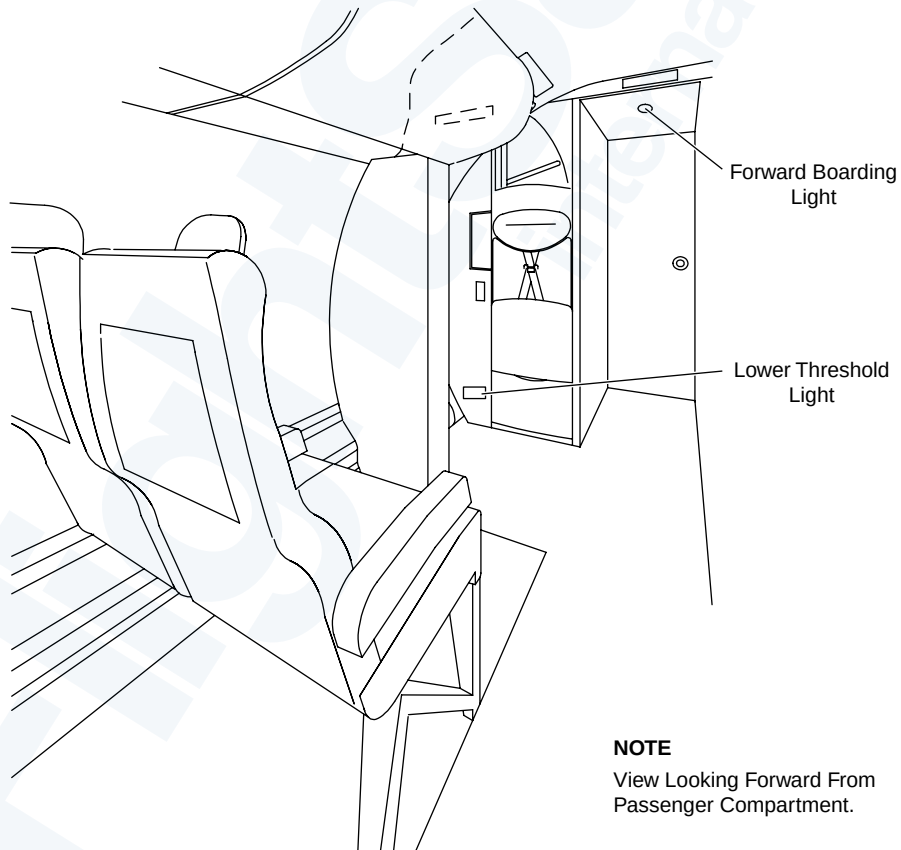


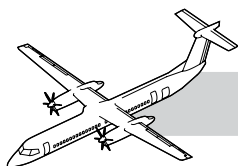
Figure 33-20. Flight Attendant Panel



NOTE

View Looking Forward From Passenger Compartment.

Figure 33-21. Boarding Lights Locator



Flight Attendant Panel

Refer to Figure 33-20. Flight Attendant Panel.

The redesigned Flight Attendant Panel (FAP) installed on the forward face of the left hand overhead bin end cap, is used to control passenger cabin systems. All functionality from the previously installed panel is maintained with the addition of a switch for Accent Lighting. The “Accent” lighting pushbutton is a membrane type on/off switch, which enables the routing of power to the ceiling Accent Lights. The red pushbutton LED will come on when the switch button is selected on and power to the lights is available.

Boarding Lights

Refer to:

- Figure 33-21. Boarding Lights Locator.
- Figure 33-22. Boarding Lights Switch.
- Figure 33-23. Boarding Lights Electrical Schematic.

The boarding lights supply adequate lighting. They help the passengers and crew in boarding the aircraft.

The boarding lights provide illumination of the passageway from the airstair door to the flight compartment. With the introduction of the Next Generation Interior, the lights that comprise the boarding lights system are LED based lights, and no longer utilize incandescent bulbs.

NOTE

Boarding light system functionality remains unchanged with the introduction of the LED lights.

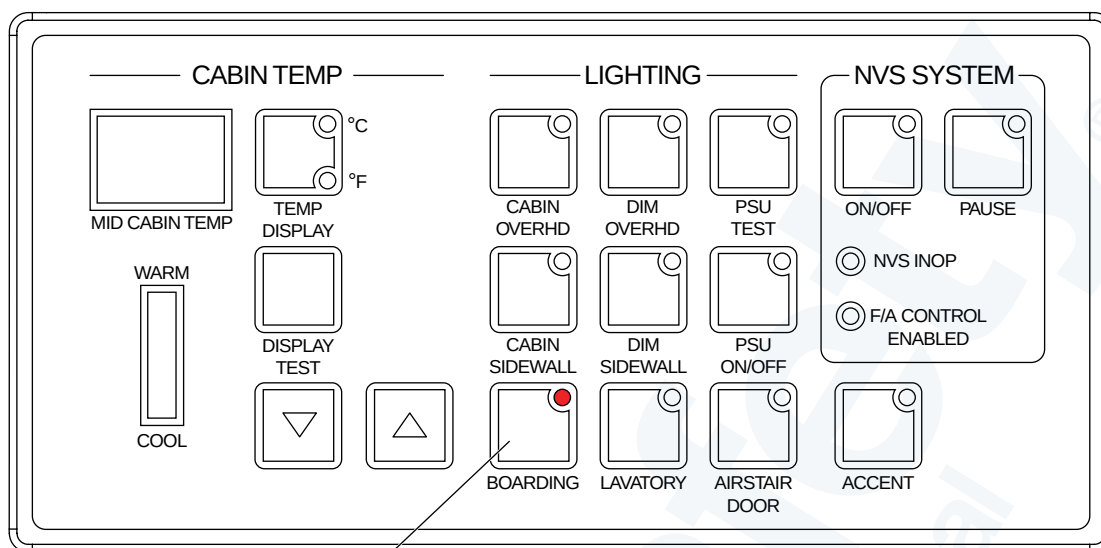
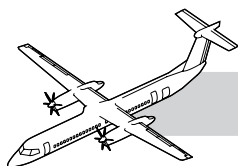
The lower threshold boarding light is located on the forward flight attendant's bulkhead, immediately above the floor level on the forward side of the forward passenger door entrance. This light gives illumination of the entrance floor, and when the forward passenger door is opened, the top step.

The forward boarding light is located in the ceiling, aft of the flight compartment door. This light gives illumination of the area between the flight compartment and the passenger compartment.

The Threshold light unit consists of a circuit board marker light, diffused lens, lens cover and housing assembly. The circuit board consists of 21 surface mounted LEDs. The circuit divides the LEDs into three parallel segments. Each segment consists of seven LEDs and a resistor connected in series. Thus, failure of any one LED segment will not cause failure to the remaining segments of the LED light strip unit.

Training Information Points

To avoid loss of battery power, the boarding lights must not be left on unless external DC power is available.



Boarding Lights Switch

Figure 33-22. Boarding Lights Switch

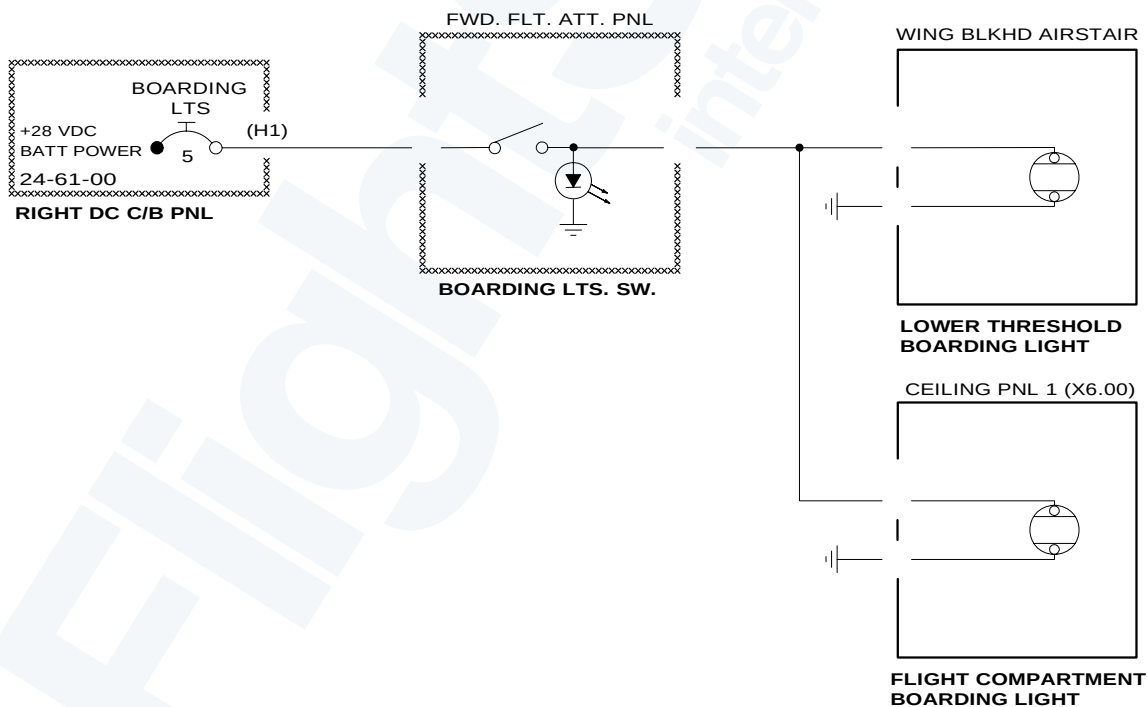
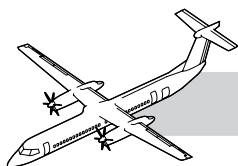


Figure 33-23. Boarding Lights Electrical Schematic



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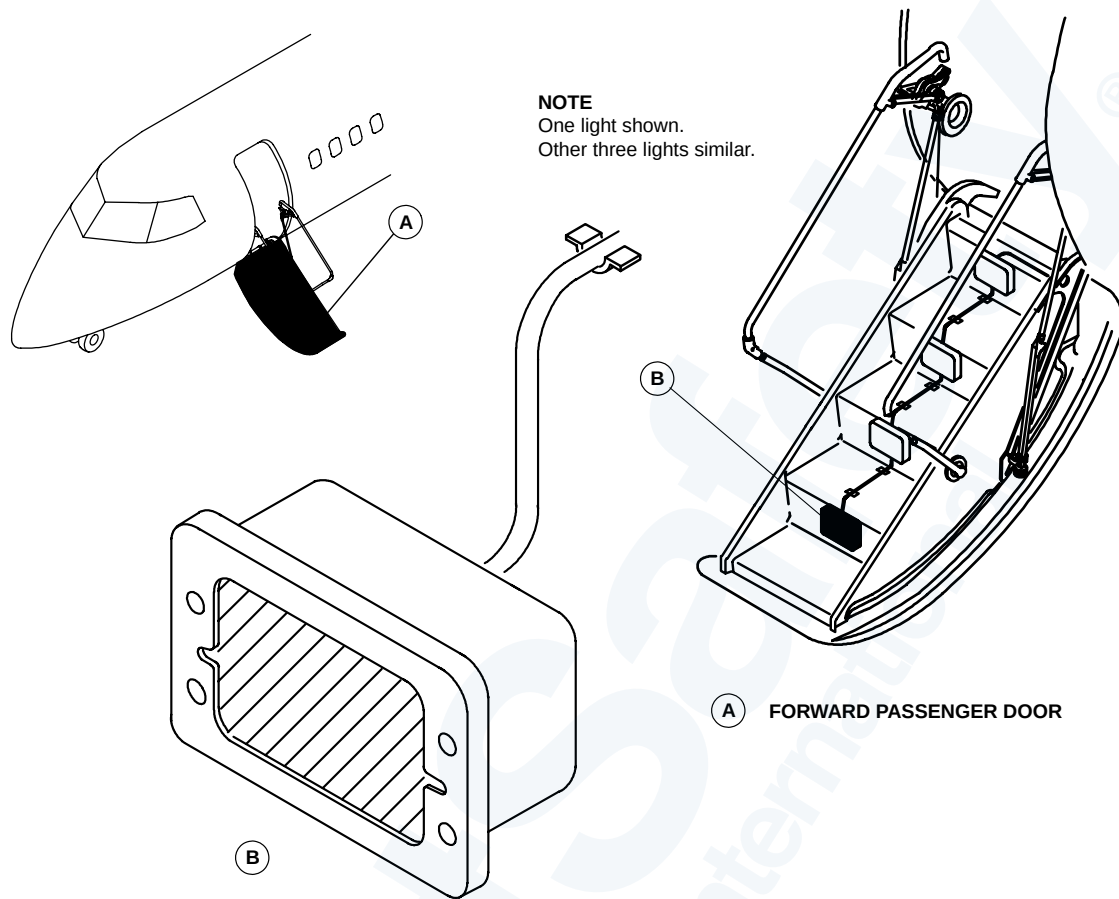
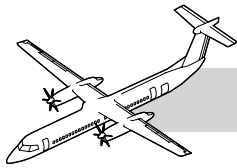


Figure 33-24. Airstair Door Lights Locator

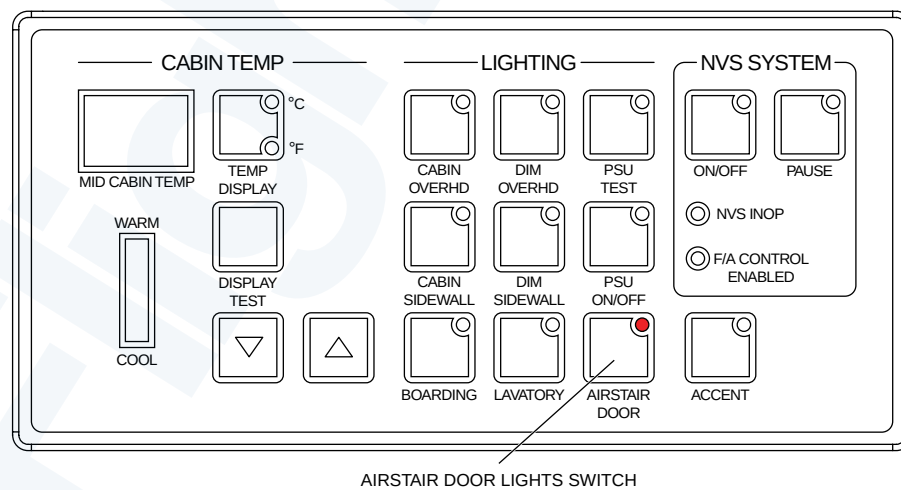
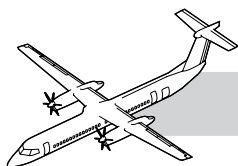


Figure 33-25. Airstair Door Lights Switch



Airstair Door Lights

Refer to:

- Figure 33-24. Airstair Door Lights Locator.
- Figure 33-25. Airstair Door Lights Switch.
- Figure 33-26. Airstair Door Lights Electrical Schematic.

The airstair door tread lights give illumination of the steps of the forward passenger door. Four airstair lights are installed in the top center part of the stair risers. A membrane touch pad switch identified as AIRSTAIR on the flight attendant panel controls the airstair lights. The airstair lights are energized by the left main bus.

MAINTENANCE PRACTICES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Airstair Boarding - Lights Lamp Replacement

- There are four of the airstair boarding light lamps. Each light lamp is installed in the top center part of one of the airstair risers. Carefully remove the sealant along the edge of the retainer assembly then loosen the screws that attach the retainer to the light assembly. Replace the lamp.
- Install the retainer assembly on the light assembly and apply the silicone sealant along the edge of the retainer assembly.

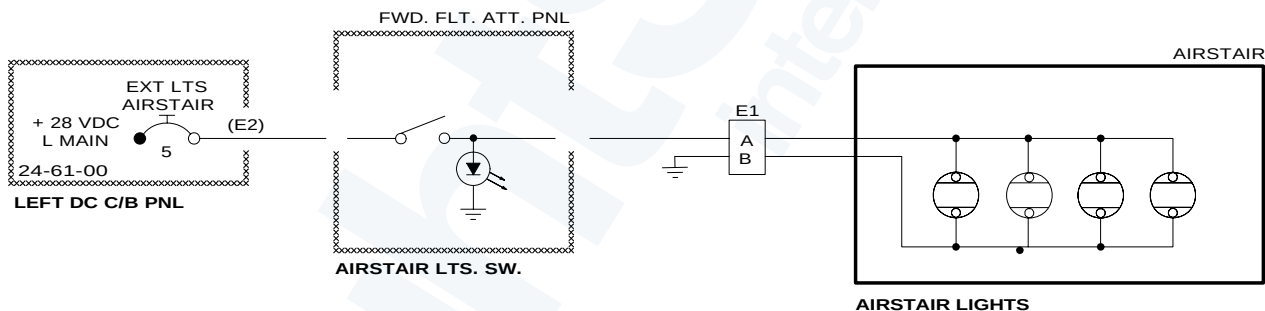


Figure 33-26. Airstair Door Lights Electrical Schematic

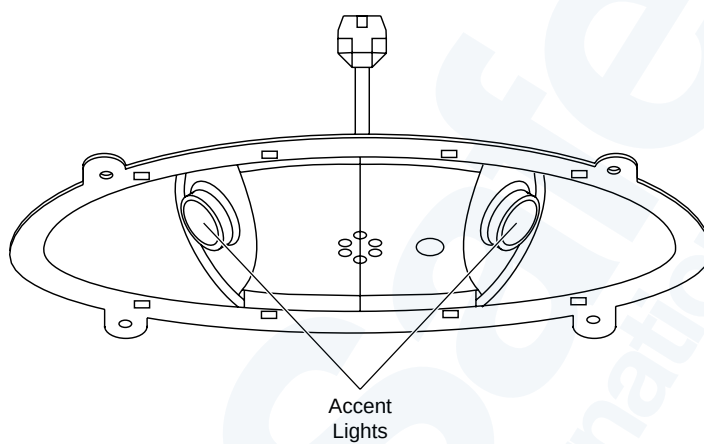
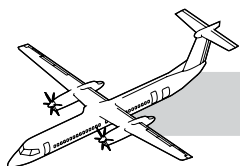
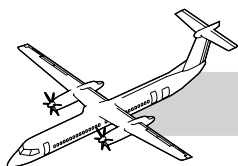


Figure 33-27. Accent Lights Locator



Accent Lights (Next Generation)

Refer to:

- Figure 33-27. Accent Lights Locator.
- Figure 33-28. Accent Lights Switches.
- Figure 33-29. Accent Lights Electrical Schematic.

Eleven accent light assemblies provide spotlight illumination to the overhead storage bins. When a bin is open, the bin interior is illuminated. These light assemblies are identical and are located along the center aisle of the passenger cabin. In the area from ceiling panel #10 to the wing-box a different part number is installed due to insufficient space.

There are four identical and interchangeable under wing accent light assemblies. The accent light assemblies include 2 spotlights, 1 emergency flood light and a zone indicator light. (The zone indicator light is inoperative on the Q400).

The accent lighting is controlled by the Accent Light pushbutton on the Flight Attendant Panel. This provides 28 VDC power from the CABIN ACCENT LTS CONT 28 VDC Right Secondary Bus to a relay which when energized allows power from CABIN ACCENT LTS PWR1 circuit breaker to the lights.

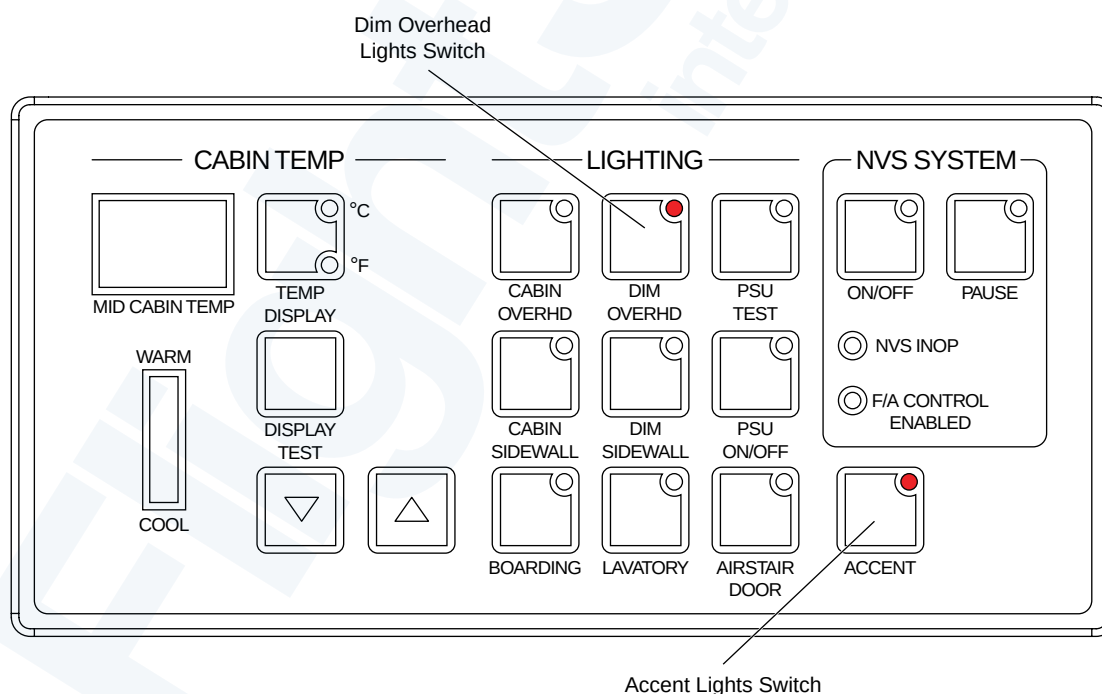


Figure 33-28. Accent Lights Switches

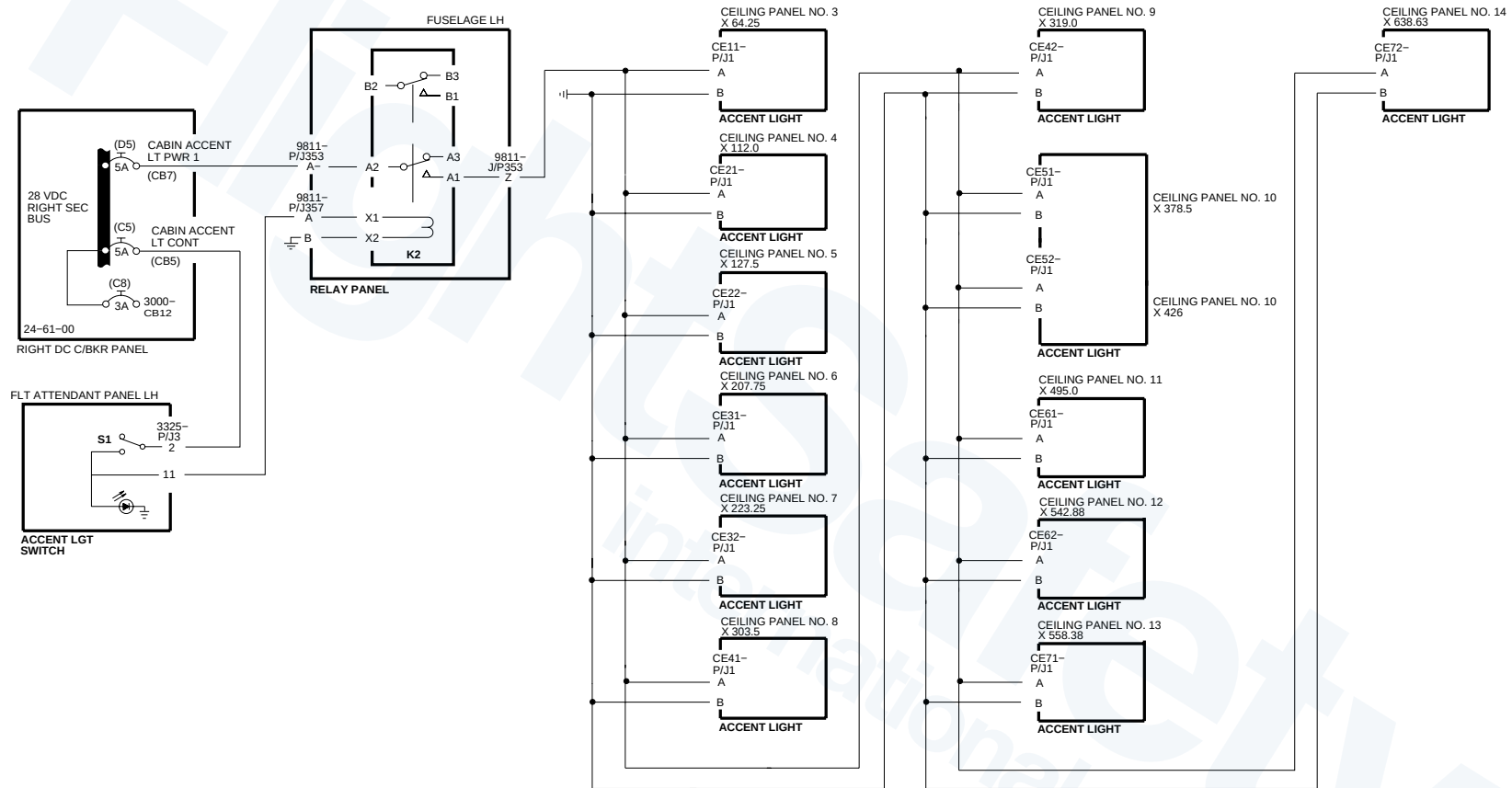
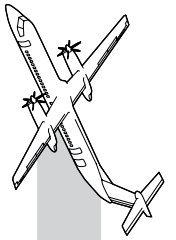
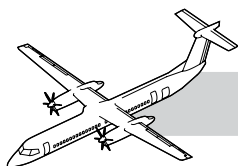


Figure 33-29. Accent Lights Electrical Schematic





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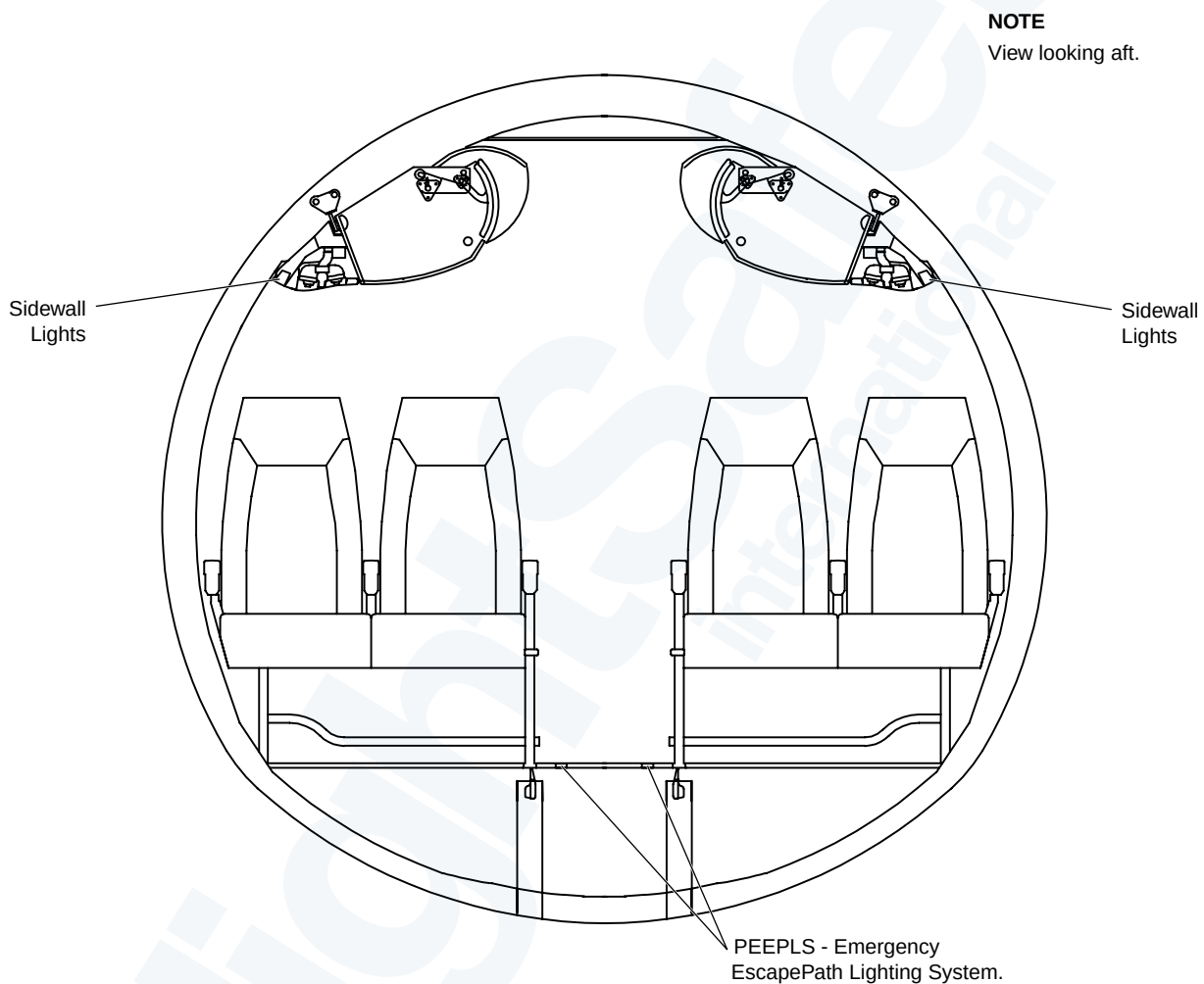
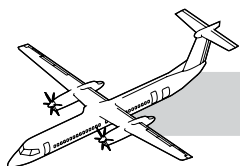
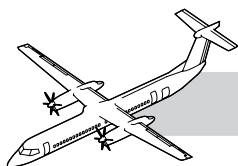


Figure 33-30. Sidewall Lights Locator



Sidewall Lights (Next Generation)

Refer to:

- Figure 33-30. Sidewall Lights Locator.
- Figure 33-31. Sidewall Lights Switches.
- Figure 33-32. Cabin Sidewall Lights Wiring Schematic.

Cabin sidewall lighting is supplied by lighting units located in the lower part of the Passenger Service Units (PSU), adjacent to the cabin sidewall. The sidewall lights, are installed to supply downwash lighting.

The overhead ceiling lights are replaced from the fluorescent type lamps to the LED type lights. The membrane touch pad switch identified as CABIN SIDE WALL on the flight attendant panel controls the cabin sidewall lights.

If installed, the DIM SIDE WALL membrane touch pad switch controls the brightness of the ceiling lights.

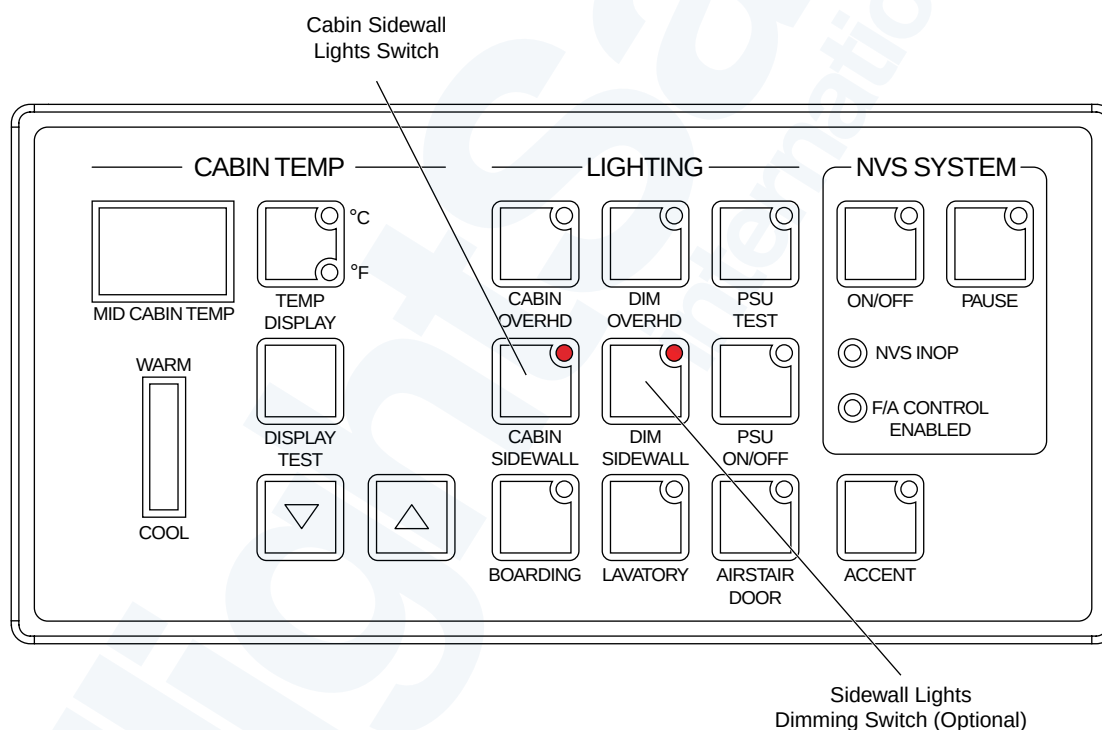


Figure 33-31. Sidewall Lights Switches

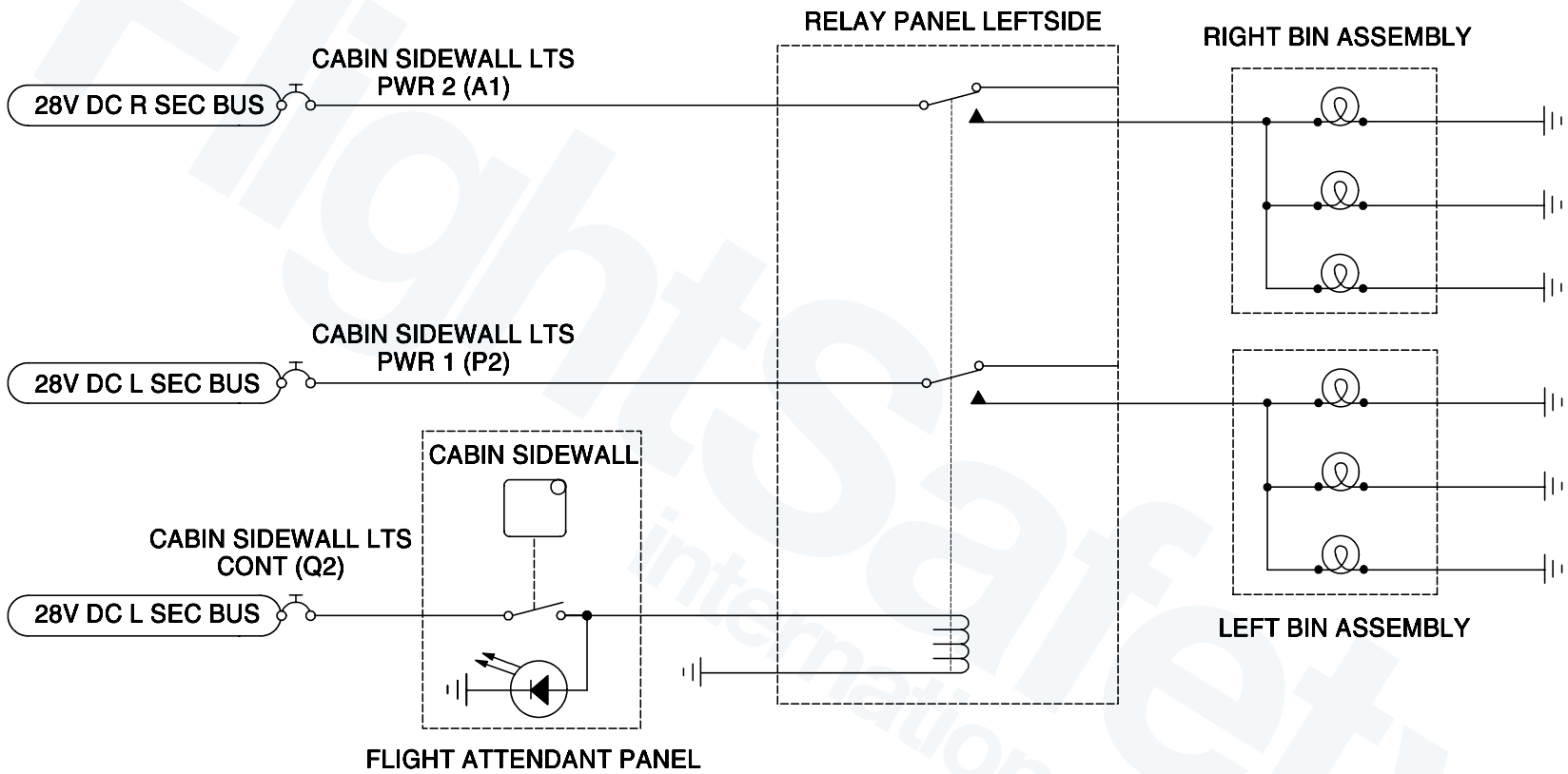
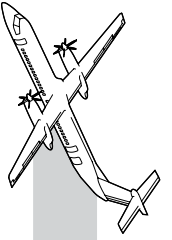
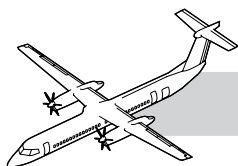


Figure 33-32. Cabin Sidewall Lights Wiring Schematic



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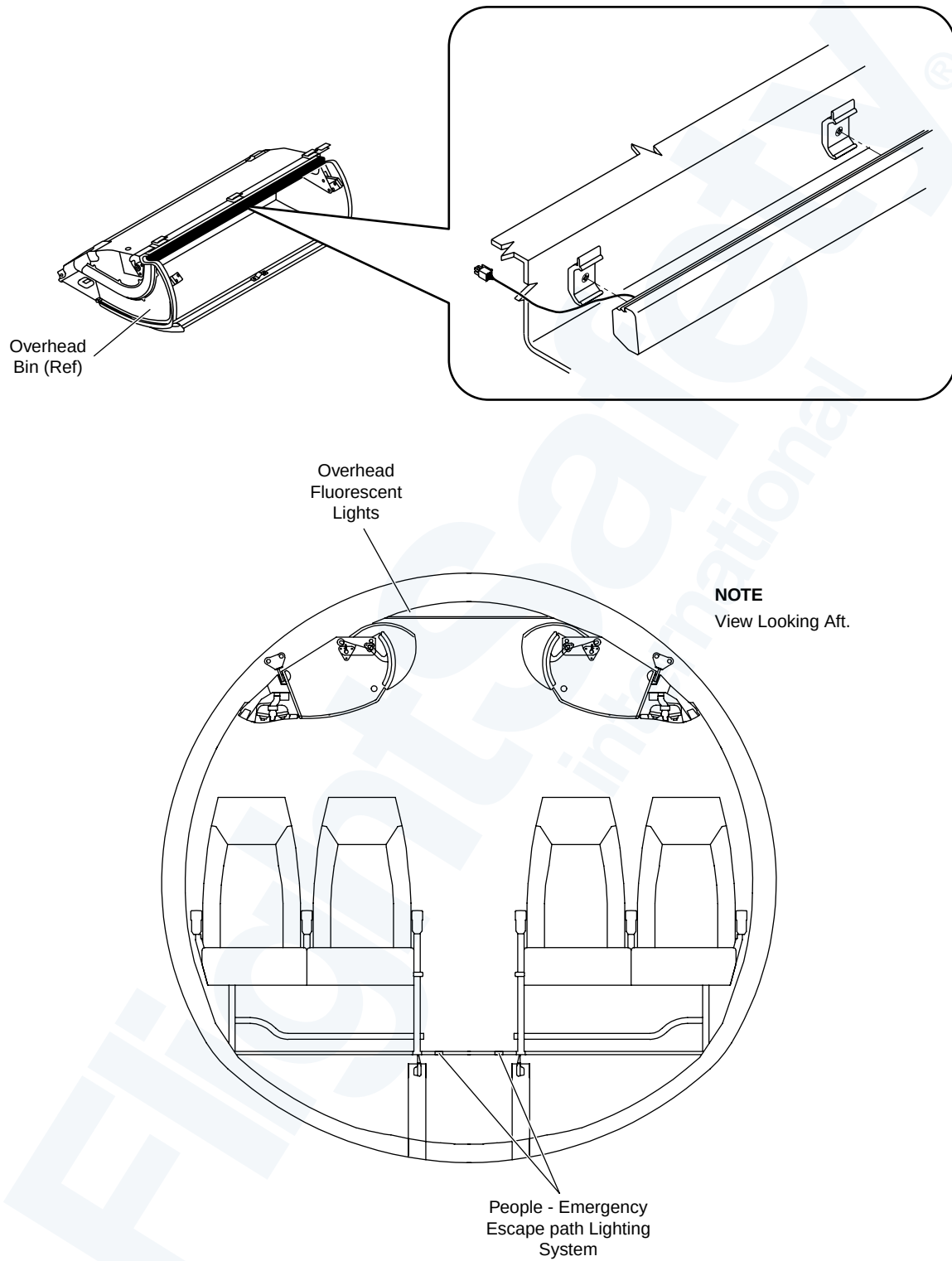
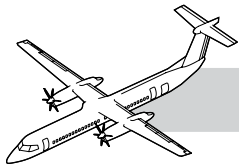
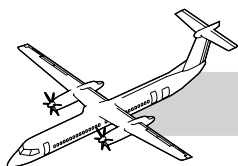


Figure 33-33. Cabin Overhead Lights Locator



Cabin Overhead Lights (Next Generation)

NOTES

Refer to:

- Figure 33-33. Cabin Overhead Lights Locator.
- Figure 33-34. Cabin Overhead Lights Switches.
- Figure 33-35. Cabin Overhead Lights Electrical Schematic (Sheet 1 of 2).
- Figure 33-36. Cabin Overhead Lights Electrical Schematic (Sheet 2 of 2).

The cabin overhead lights are the LED light strips and are installed on the top of the cabin overhead bins.

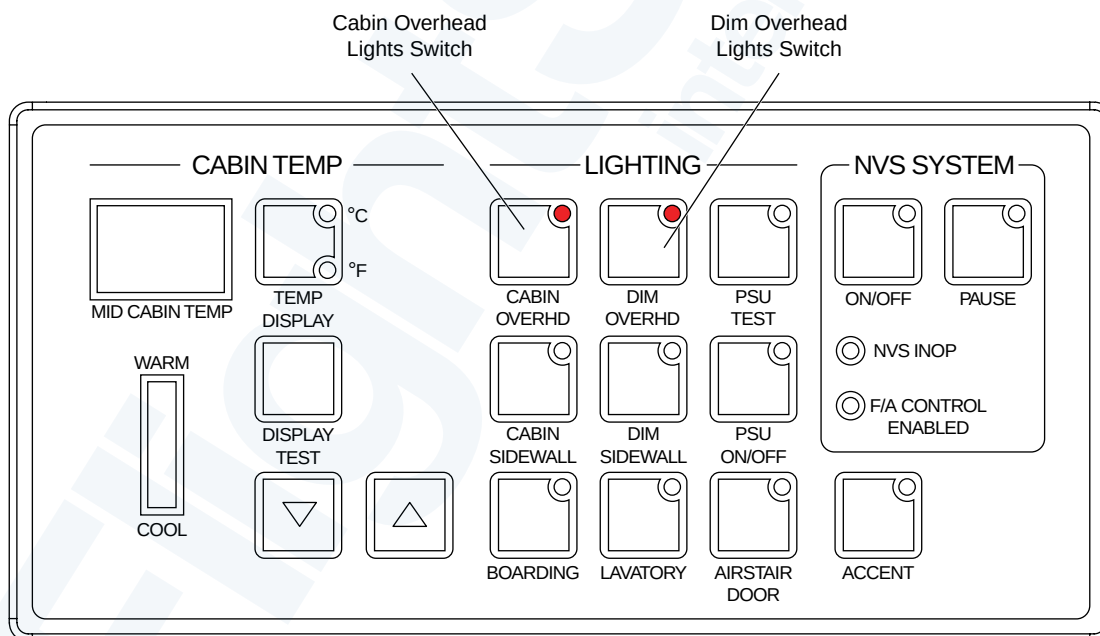


Figure 33-34. Cabin Overhead Lights Switches

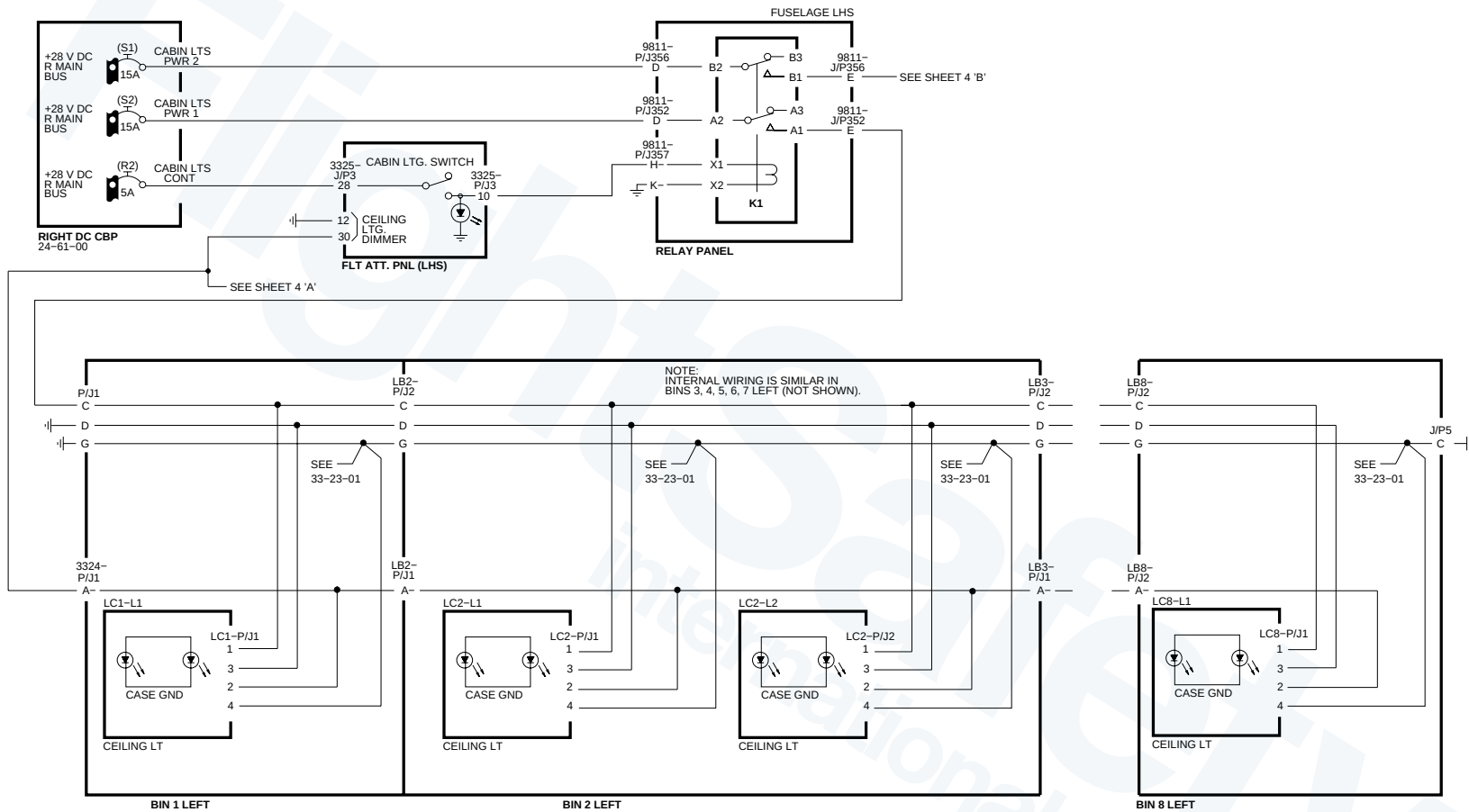
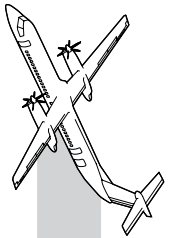


Figure 33-35. Cabin Overhead Lights Electrical Schematic (Sheet 1 of 2)



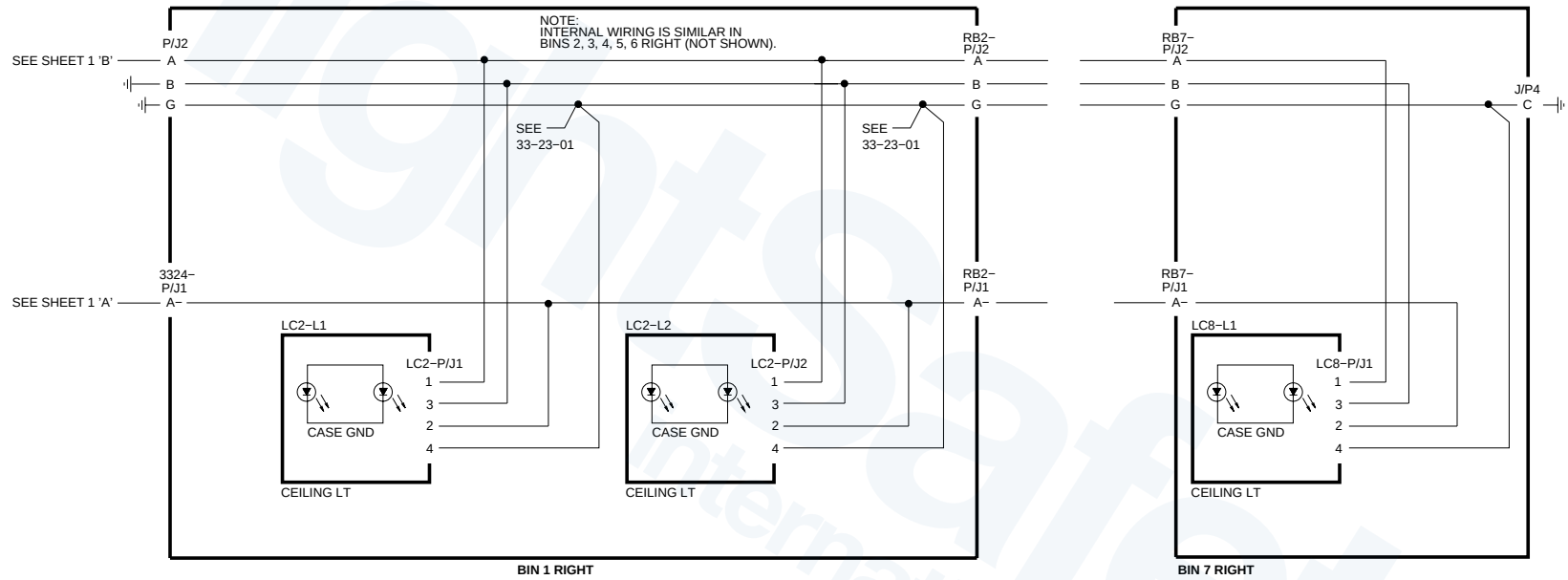
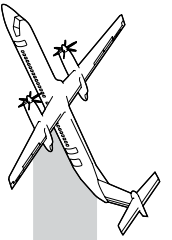
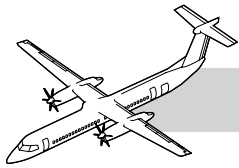


Figure 33-36. Cabin Overhead Lights Electrical Schematic (Sheet 2 of 2)



NOTES

- ① One passenger service unit shown.
- ② One passenger service unit installed per two seats; all units similar.

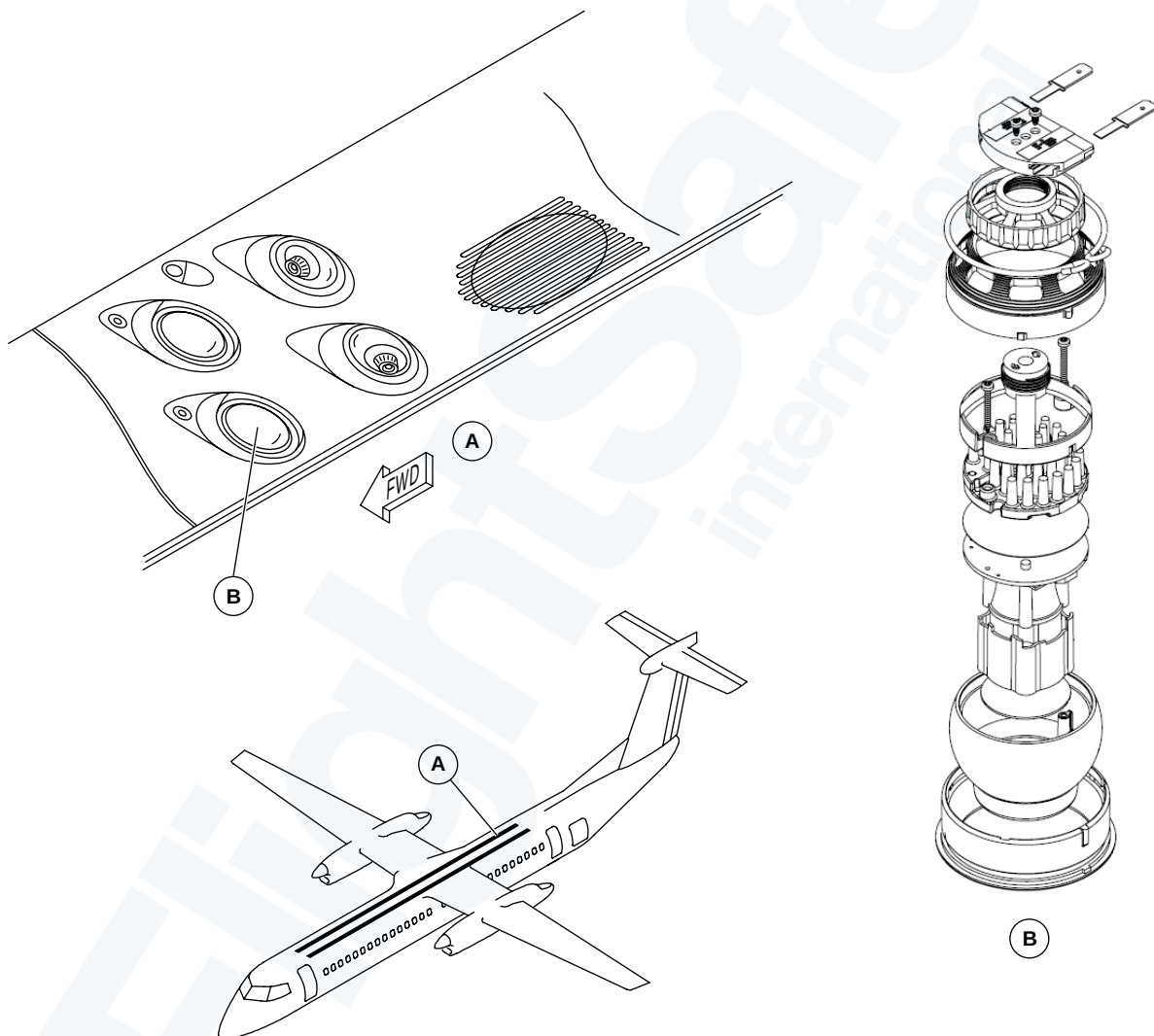
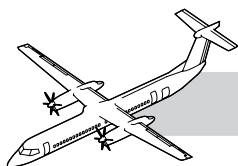


Figure 33-37. Reading Lights Locator



Reading Lights (Next Generation)

Refer to:

- Figure 33-37. Reading Lights Locator.
- Figure 33-38. PSU Lights Switches.
- Figure 33-39. Passenger Reading Lights Wiring Schematic.

Reading lights supply overhead lights for each passenger seat, forward and aft flight attendant. The passenger reading lights are installed in the Passenger Service Unit (PSU), at each seat location.

The forward and aft flight-attendant reading lights are installed on the ceiling panels, above each flight-attendant seat location. The PSU's are located at the bottom of the overhead storage bins.

Two switchlights, each located next to its reading light on the PSU, control the lights. The outboard reading light is provided for the window seat passenger and the inboard reading light for the aisle seat passenger.

The PSU ON/OFF pushbutton, on the forward-flight-attendant control panel, controls the passenger reading lights. When the pushbutton is pushed to the ON position, a red indicator light, on the pushbutton itself, comes on to indicate that the reading lights are operative. Each reading light can then be controlled by its individual switchlight on the PSU, at each seat location. When the pushbutton is pushed to the OFF position, the red indicator light goes off to indicate that the reading light system is inoperative.

The PSU TEST pushbutton, on the forward-flight-attendant control panel, is used to test all of the passenger reading lights and the flight-attendant call system. All of the passenger reading lights come on momentarily and any unserviceable reading light can be identified. When the PSU TEST pushbutton is pushed again, the red indicator light, on the pushbutton itself, goes off.

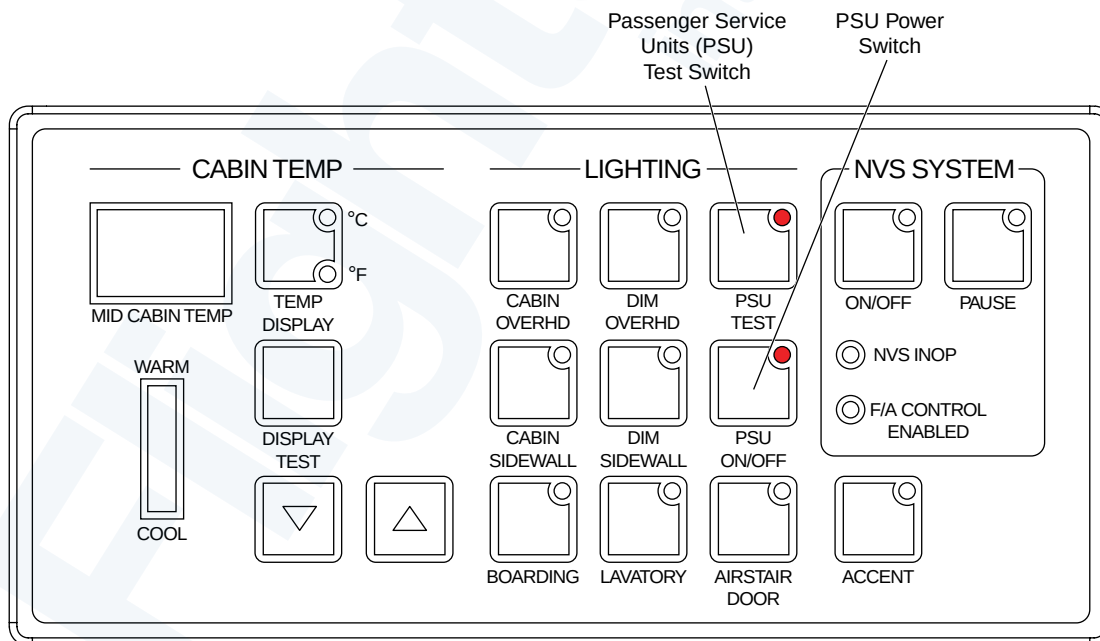


Figure 33-38. PSU Lights Switches

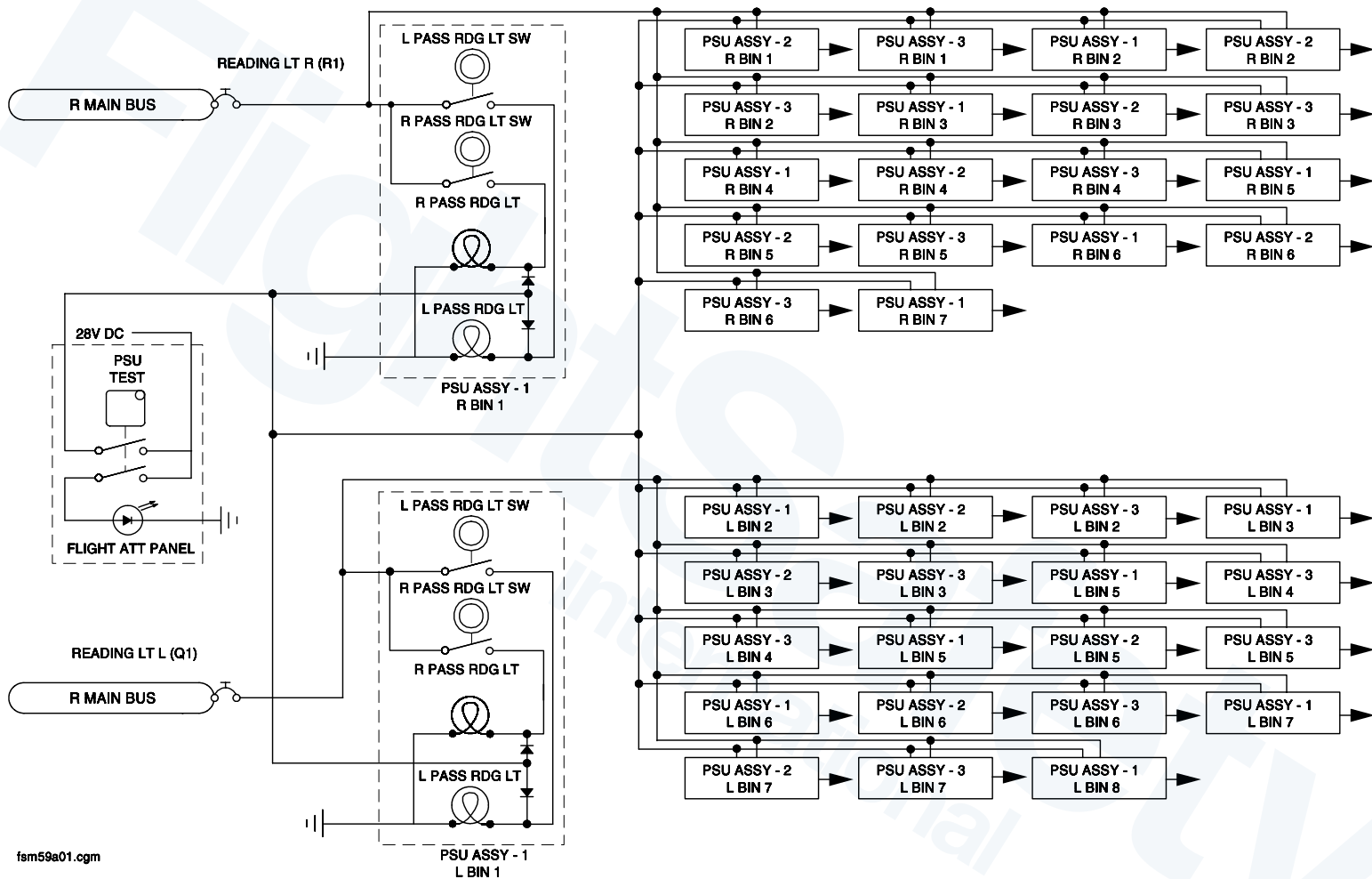
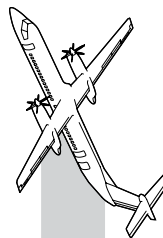
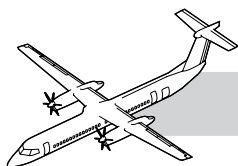


Figure 33-39. Passenger Reading Lights Wiring Schematic



MAINTENANCE PRACTICES

NOTES

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

Removal of the Reading Lights

- Release the lock on the clamp and adjust the light to get access to disconnect the electrical connectors from the terminal guard. Then release the spring clip from the reflector flange, to remove the light.

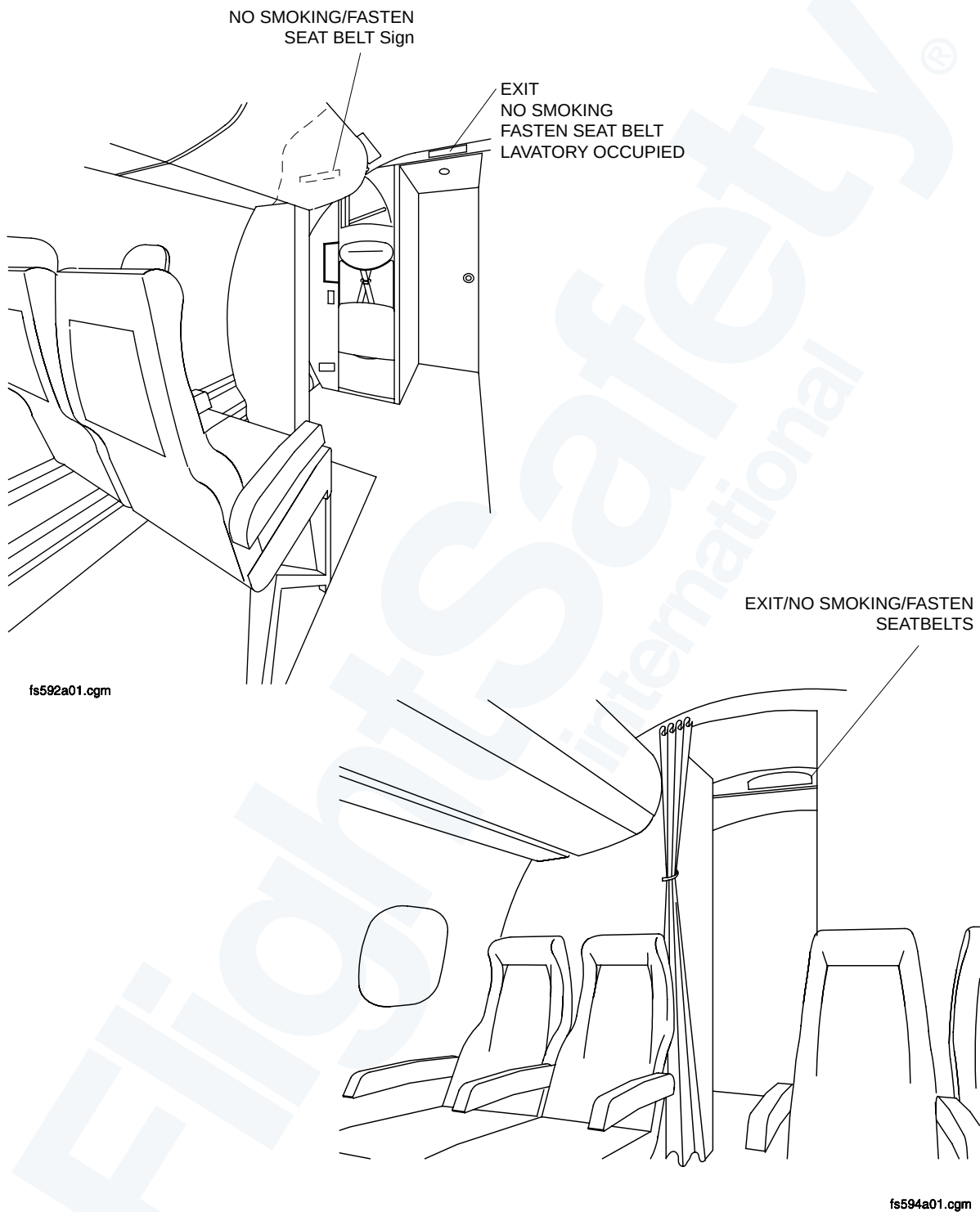
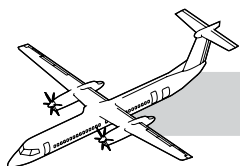
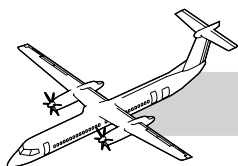


Figure 33-40. Information Signs Locator



Information Signs (Next Generation)

NOTES

Refer to Figure 33-42. Passenger Warning Lights Wiring Schematic.

The passenger warning lights are replaced from the fluorescent lamps to the LED lights. There are several passenger information signs located throughout the aircraft. Four individual information signs are installed in the combination sign unit located on the forward ceiling panel, above the flight compartment door as follows:

- Exit sign
- No smoking sign
- Fasten seat belt sign
- Lavatory occupied sign.

Three individual information signs installed in the two advisory light panels located on the centre ceiling panels, along the centre aisle as follows:

- No smoking sign
- Fasten seat belt sign
- Occupied lavatory sign.

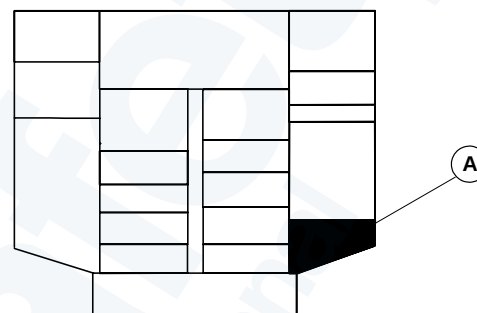
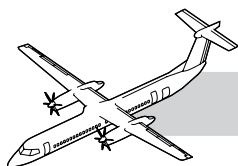
Four individual information signs installed advisory light panel located on the aft ceiling panel, above the cargo area as follows:

- Exit sign
- No smoking sign
- Fasten seat belt sign
- Lavatory occupied sign.

Two individual information signs installed in a combination sign unit located on the forward-flight-attendant control panel, on the end cap assembly, aft of the passenger door as follows:

- No smoking sign
- Fasten seat belt sign.

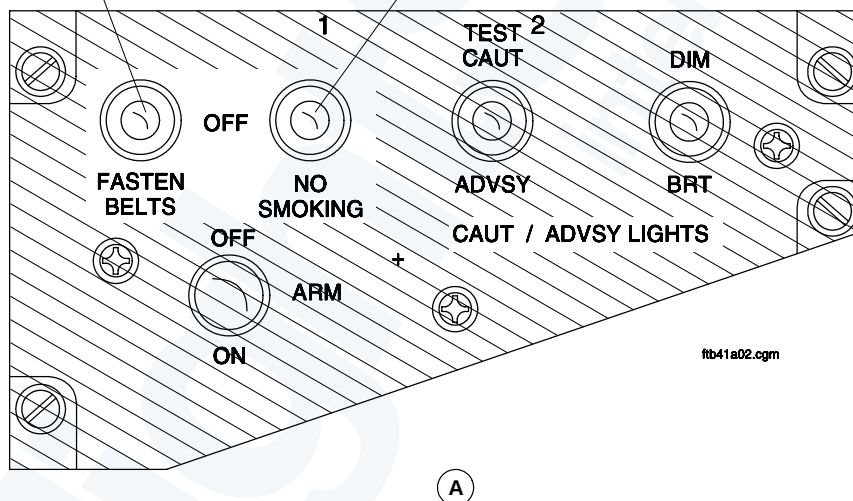
A single return-to-seat sign installed in the PSU located in the lavatory.



OVERHEAD CONSOLE

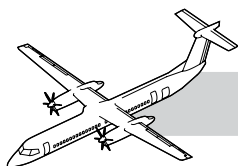
FASTEN BELTS Lights
Toggle Switch

NO SMOKING Lights
Toggle Switch



ftb41a02.cgm

Figure 33-41. Information Signs Controls - Passenger Warning Panel



Refer to Figure 33-41. Information Signs Controls - Passenger Warning Panel.

The passenger information signs are controlled from the passenger warning panel on the lower right side of the overhead console in the flight compartment. The NO SMOKING switch turns on all the NO SMOKING signs and symbols. The FASTEN BELTS switch turns on the FASTEN SEAT BELT signs and symbols and the RETURN TO SEAT sign in the lavatory.

The NO SMOKING signs also come on when the landing gear is selected to the down position.

TROUBLESHOOTING

PSU and Information Signs Operational Test

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

NOTE

The information signs include the no-smoking, fasten-seat-belt, return-to-seat and lavatory-occupied signs.

It is recommended to have two people to complete these tasks.

- The passenger reading lights and related information signs are tested by pushing and hold the PSU TEST pushbutton on the forward-flight-attendant control panel. Make sure that the indicator lights (red) come on and that at each seat location the two reading lights and the flight-attendant service-call annunciator-light (red) come on momentarily. Also the three lights on the advisory light panels (yellow) come on momentarily. Make sure that the reading lights and the flight-attendant service-call button function at each seat position.
- The lavatory lights and related information signs are tested by pushing and release the flight-attendant call button on the PSU, in the lavatory. Make sure that its annunciator light (red) and three lights (amber) on the aft, mid and forward advisory light panels, comes on and that a chime is heard. Push and release the flight-attendant call button again. Make sure that the annunciator light (red) and the three lights (amber) on the aft, mid and forward advisory light panels, go off. Open the lavatory door and make sure that the lights in the lavatory go dim. Close and lock the lavatory door and make sure that the lights inside the lavatory brighten and that the LAVATORY OCCUPIED sign comes on.
- To test the operation of the no-smoking information signs, set the NO SMOKING switch on the Advisory Signs/Passenger Warning panel, on overhead console to the NO SMOKING position. Make sure that the NO SMOKING light comes on, at each seat location. Also that a low tone chime is heard.
- To test the operation of the fasten-seat-belt and return-to-seat information signs, set the FASTEN BELTS switch on the Advisory Signs/Passenger Warning panel, on the overhead console to the FASTEN BELTS position. Make sure that the fasten-belt sign light comes on, at all the seat positions and lavatory. Also that a low tone chime is heard.

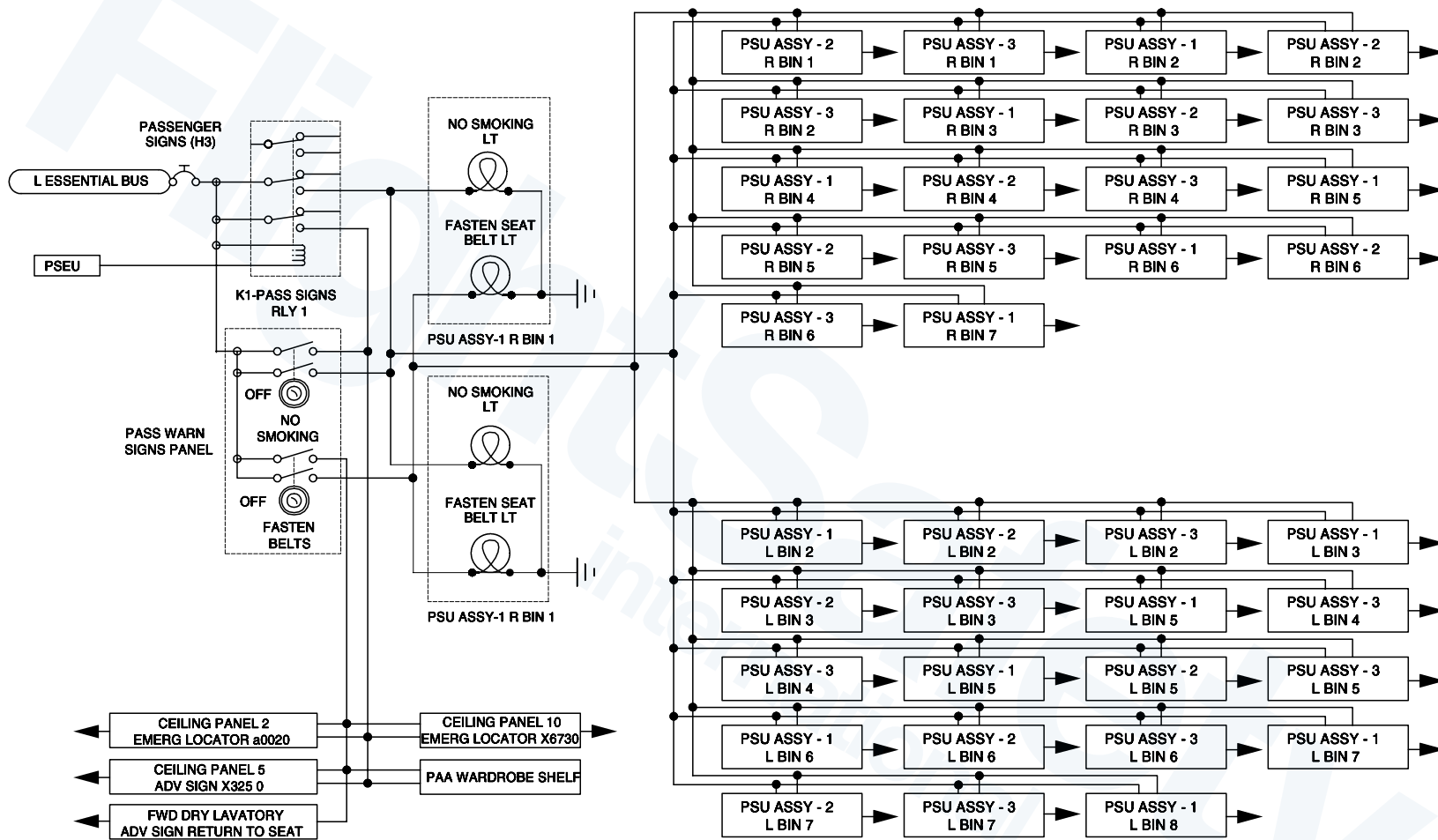
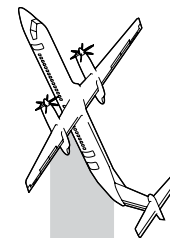
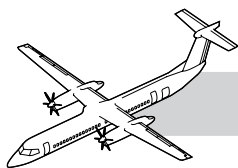
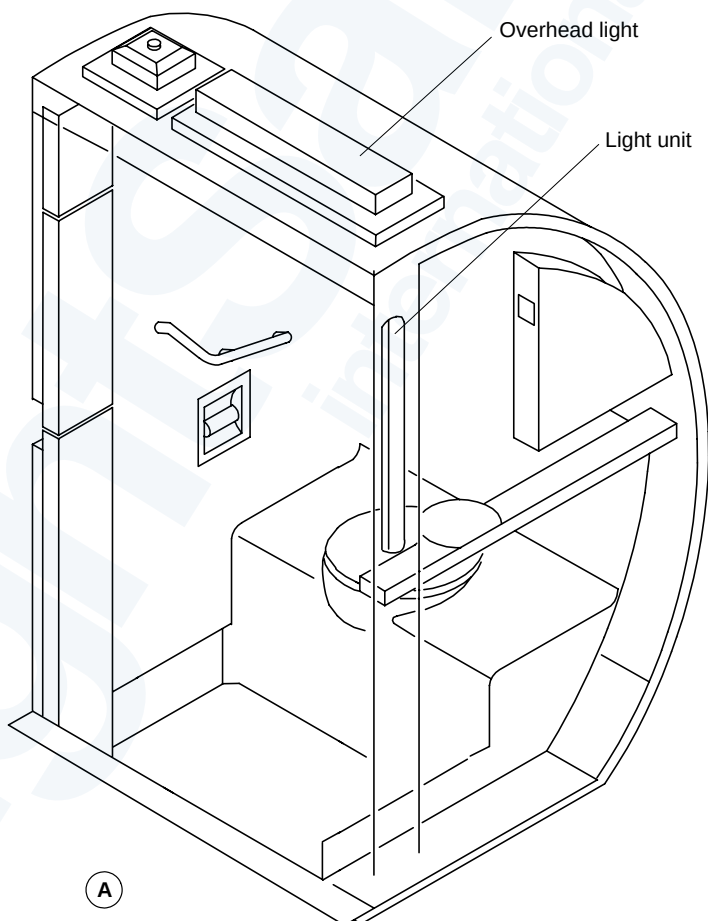
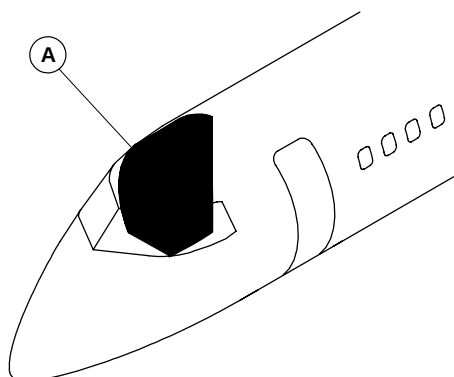
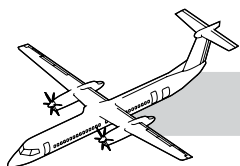


Figure 33-42. Passenger Warning Lights Wiring Schematic



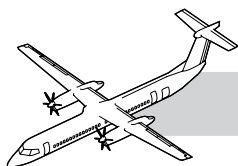


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Figure 33-43. Lavatory Lights Locator



Lavatory Lights (Next Generation)

Refer to:

- Figure 33-43. Lavatory Lights Locator.
- Figure 33-44. Lavatory Lights Switch.
- Figure 33-45. Lavatory Lights Wiring Schematic.

On the forward-flight-attendant control panel when the LAVATORY push-button is pushed, a red indicator light and the LAVATORY sign comes on.

When there is an Aft Lavatory installed the LED lights function the same.

Lavatory lights supply illumination of the lavatory.

Lavatory LED lights supply either full or dim illumination to the forward lavatory.

The lavatory door latch micro-switch control the operation of the lights. When the lavatory door latch is in the unlocked position, power is supplied to the lavatory lights for continuous dim illumination. When the lavatory door latch is in the locked position, the micro-switch is activated and the lights provide full illumination (bright).

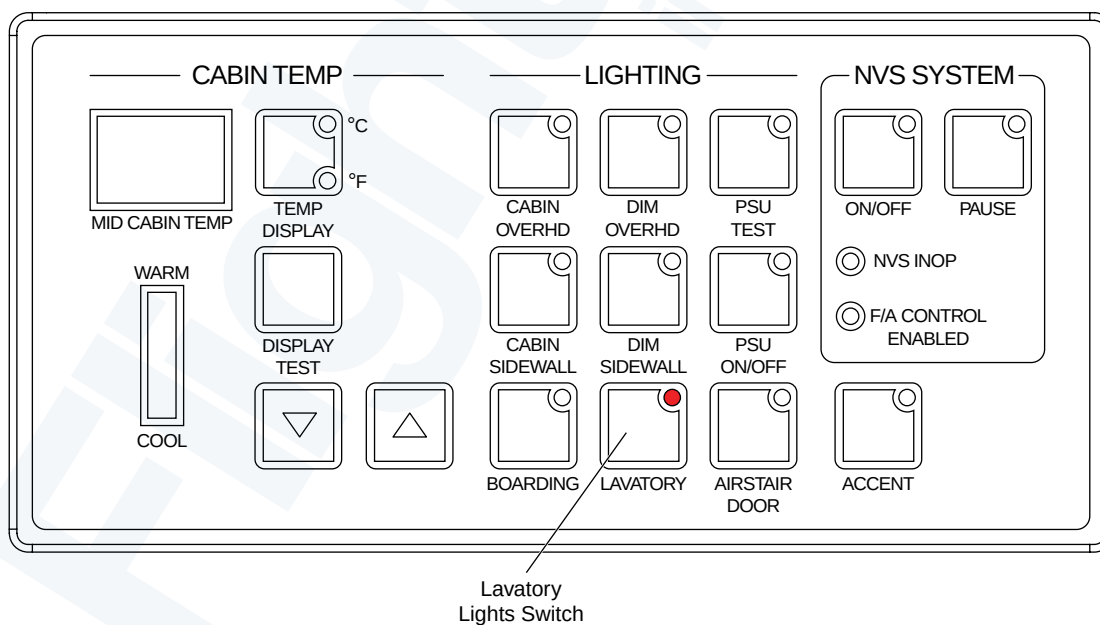


Figure 33-44. Lavatory Lights Switch

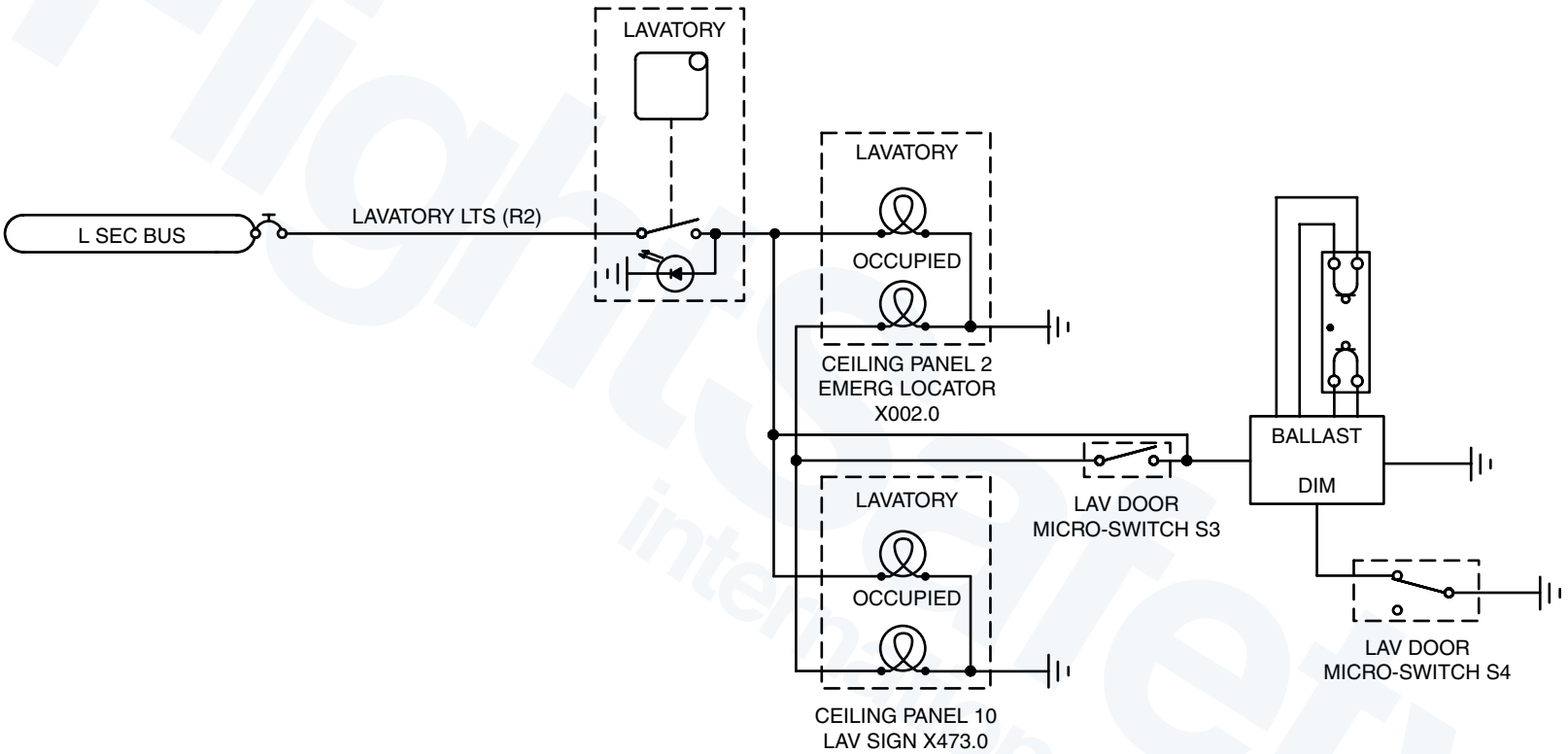
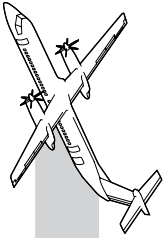
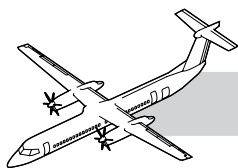


Figure 33-45. Lavatory Lights Wiring Schematic



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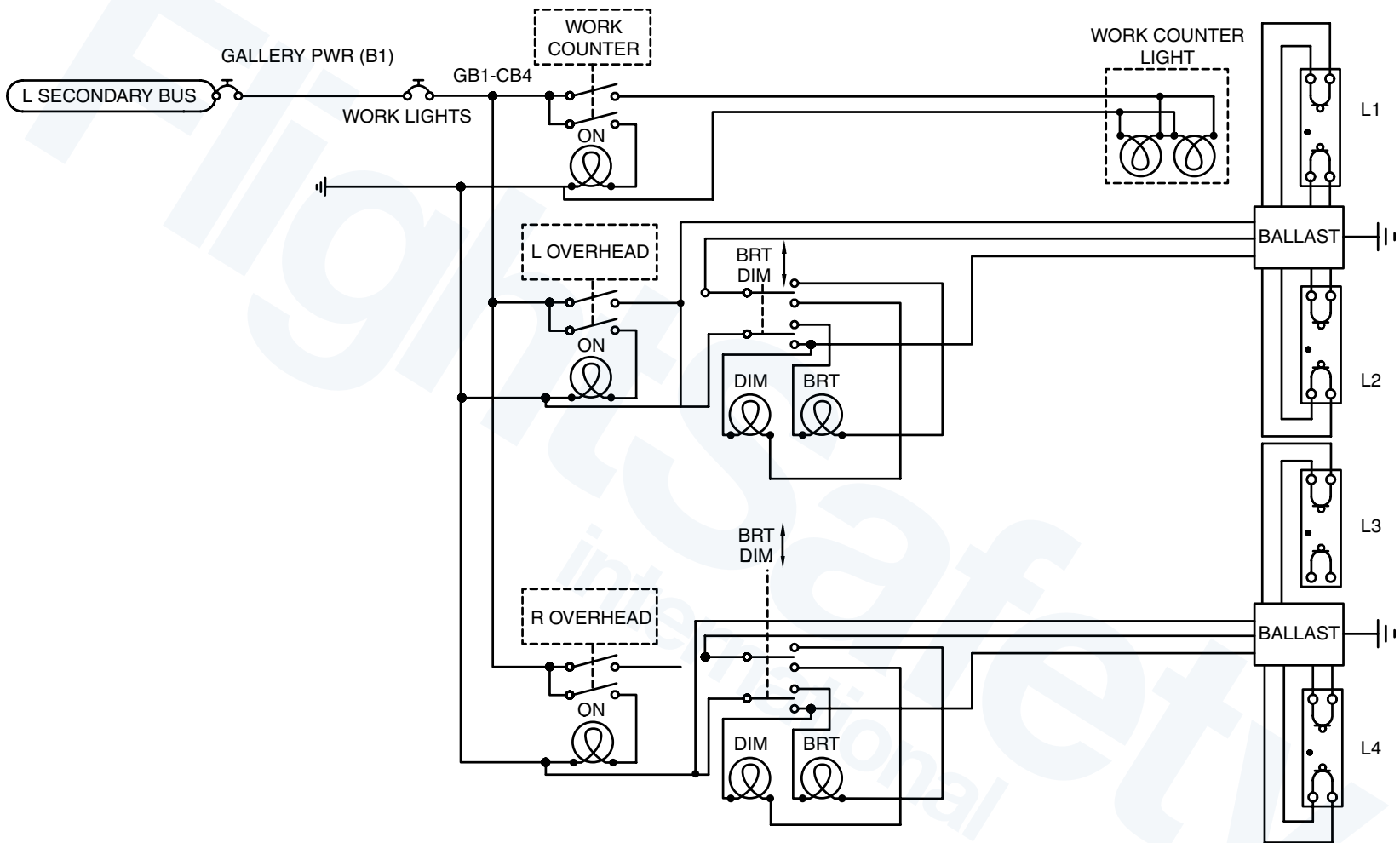
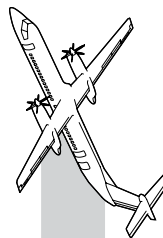
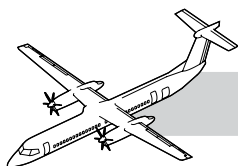


Figure 33-46. Galley Lights Wiring Schematic





Galley Lights (Next Generation)

NOTES

Introduction

Illumination for the aft passenger door and service door is supplied by the galley lights.

Refer to Figure 33-46. Galley Lights Wiring Schematic.

Two lights supply overhead illumination to the galley. The lights are installed in the aft-flight attendant area, on the left and right ceiling panel. Each lighting unit has a lamp fixture replaced with the LED lights.

Push-buttons, located on the galley control panel, near the top of the galley, control the operation of these lights. The galley has also a worklight to supply illumination to its countertop area. Power is supplied through the GALLEY PWR circuit breaker (20A) at location B1 on the R SECONDARY DC bus.

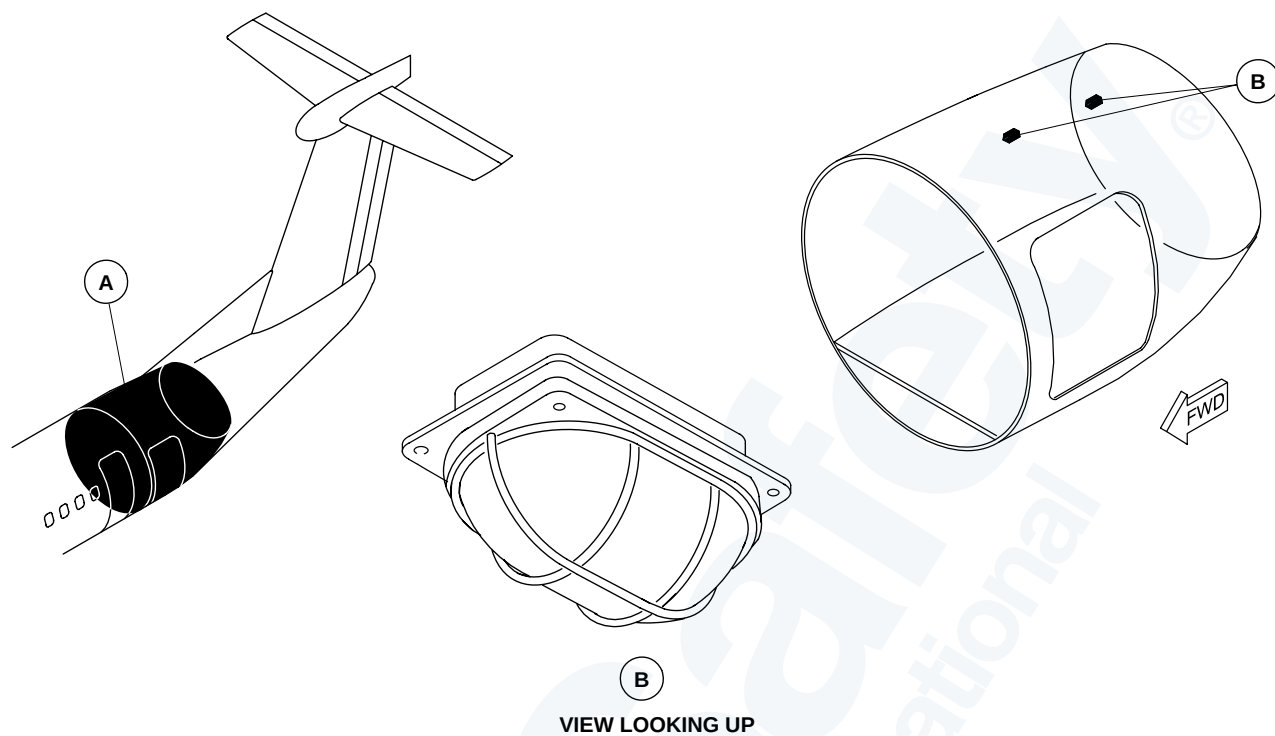
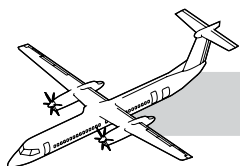


Figure 33-47. Aft Baggage Compartment Lights

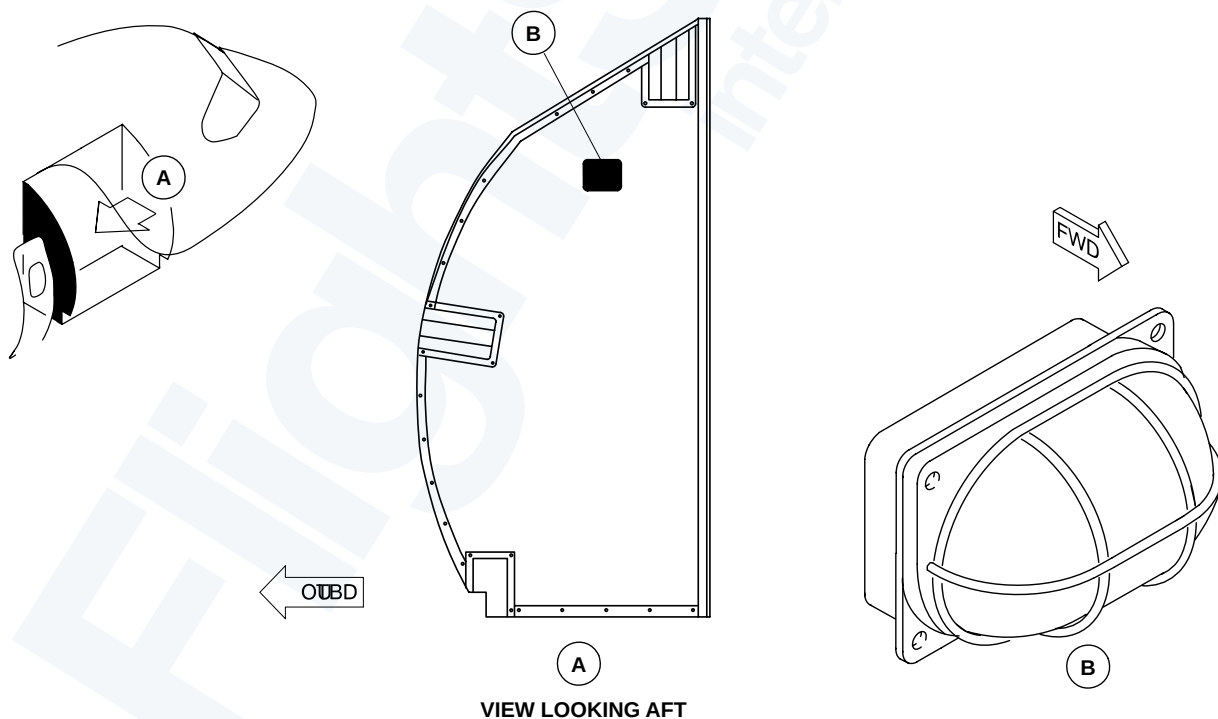
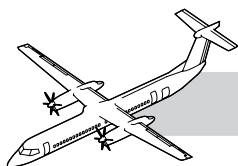


Figure 33-48. Forward Baggage Compartment Lights



33-30-00 BAGGAGE AND SERVICE COMPARTMENT LIGHTS (NEXT GEN)

NOTE

INTRODUCTION

The baggage compartment lights supply illumination for loading and unloading baggage.

GENERAL

Refer to:

- Figure 33-47. Aft Baggage Compartment Lights.
- Figure 33-48. Forward Baggage Compartment Lights.

The aircraft has three lights for the baggage areas. One is located on the aft wall of the forward baggage compartment. The other two are in the ceiling of the aft baggage compartment.

The wardrobe light is controlled by a door activated micro-switch.

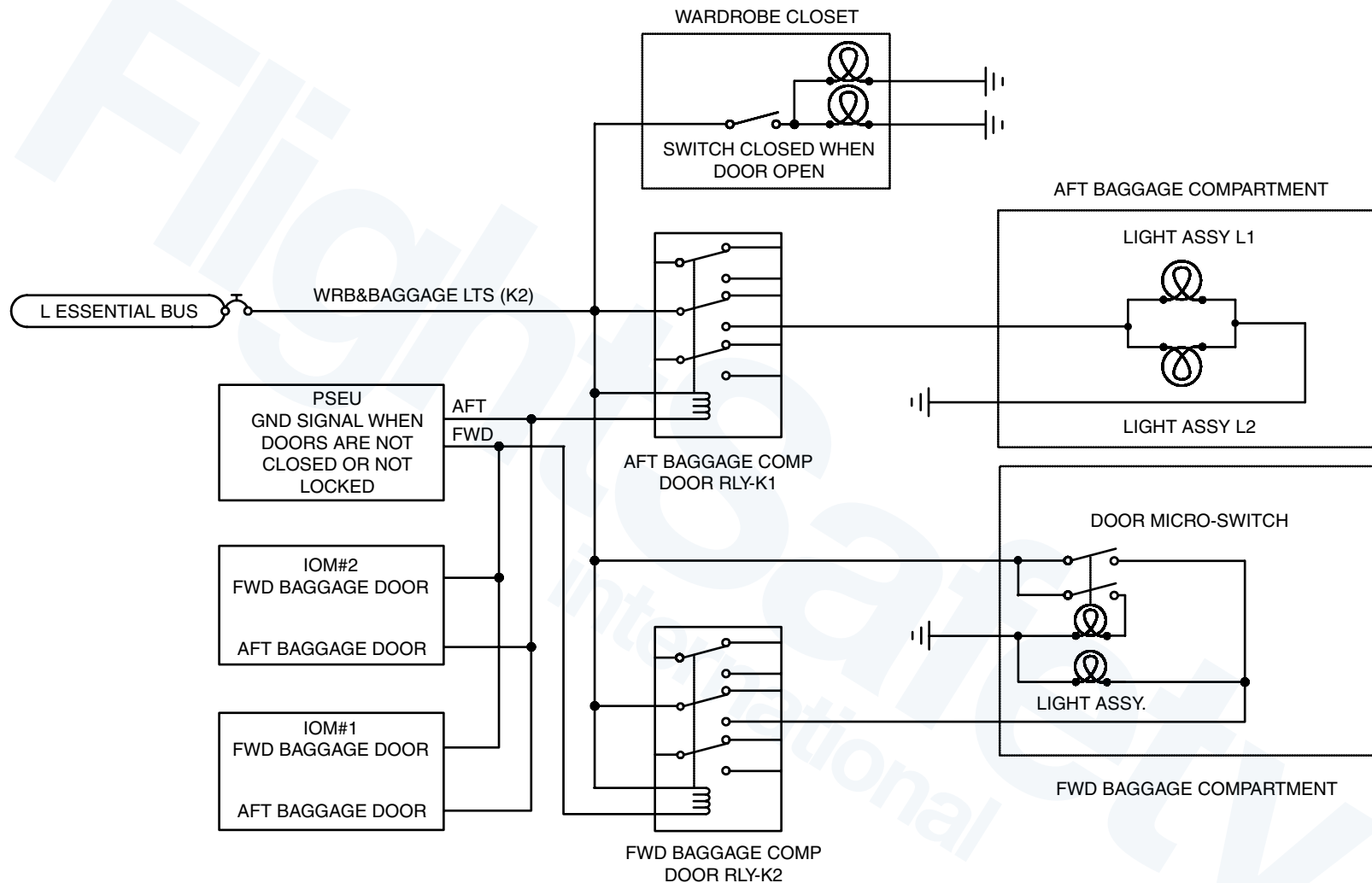
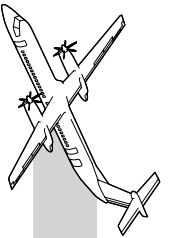
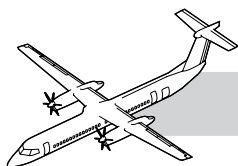


Figure 33-49. Baggage Compartment Lights Schematic





Refer to Figure 33-49. Baggage Compartment Lights Schematic.

NOTE

The aft baggage compartment lights are controlled by relay 2-K4 in Relay Junction Box (RJB) No.2. When the PSEU senses the aft baggage door is closed and locked, a ground discrete is supplied to the relay to remove power to the light assemblies.

The forward baggage compartment light is controlled by relay 2-K4 in RJB No.1. When the PSEU senses the forward baggage door is closed and locked, a ground discrete is supplied to the relay to remove power to the light assembly.

The forward baggage compartment light will also illuminate through the setting of an internal baggage door controlled micro-switch.

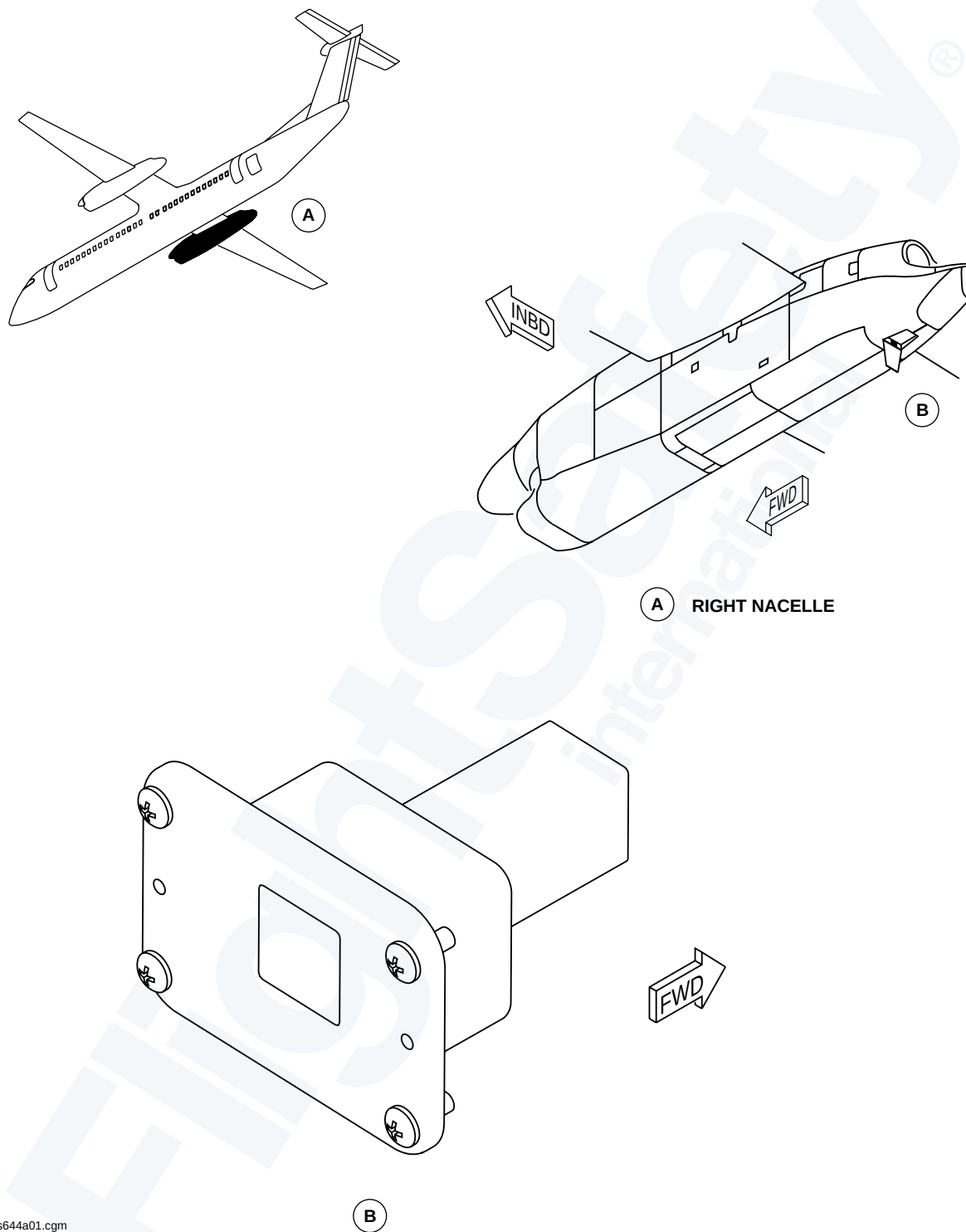
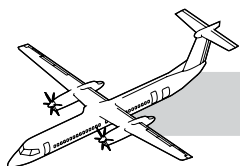
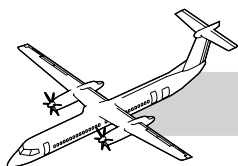


Figure 33-50. Refuel Panel Service Light



COMPONENT DESCRIPTION

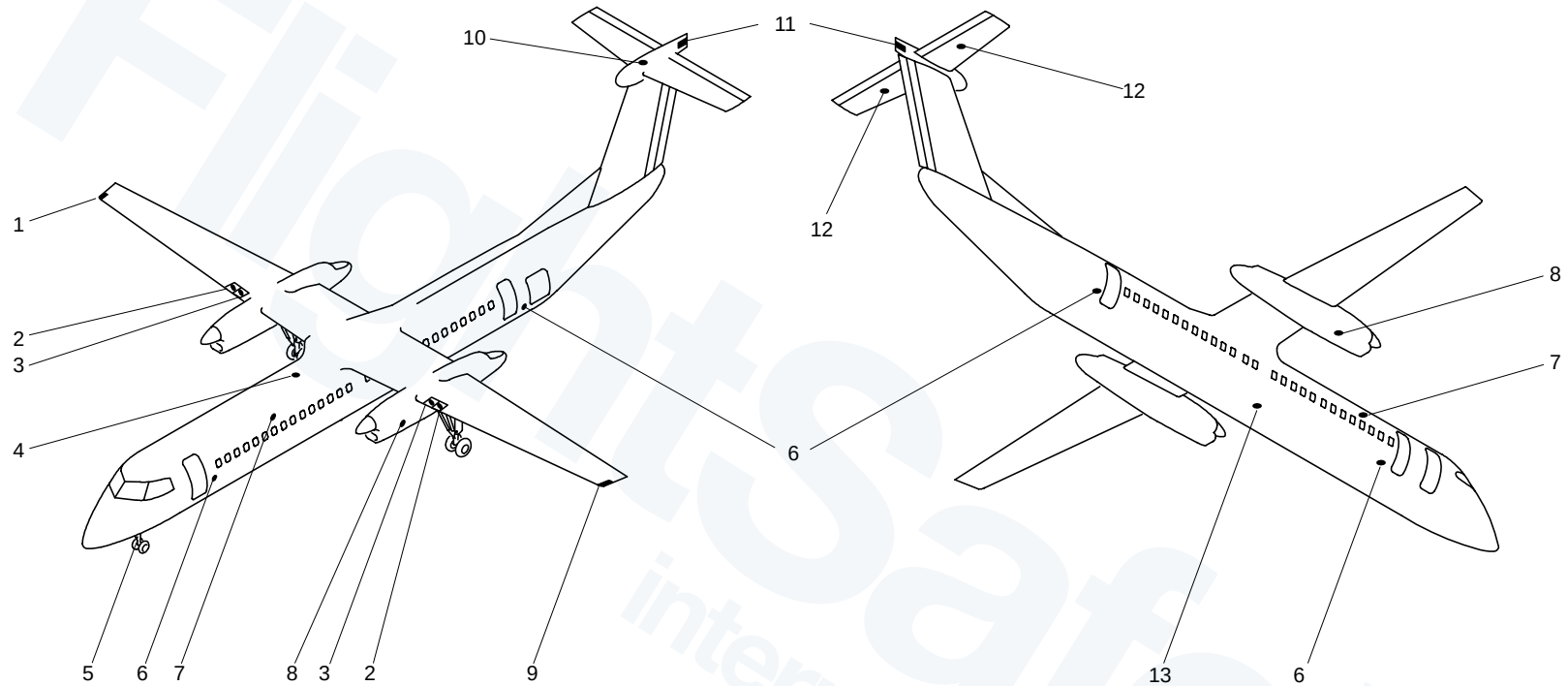
NOTES

Refuel Panel Service Light

Refer to Figure 33-50. Refuel Panel Service Light.

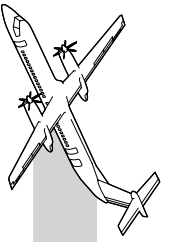
The aircraft has a fixed refuel panel service light, installed on the right nacelle, refuel/defuel access panel to supply lighting to this area. The light comes on automatically whenever the access door is opened.

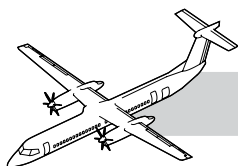
The service light has a rectangular lens, with simple housing and a reflector light, operated from a 28 VDC power source.

**LEGEND**

- | | |
|-----------------------------------|----------------------------------|
| 1. Right Position Lights (Green) | 8. Wing Inspection Lights |
| 2. Approach Lights | 9. Left Position Lights (Red) |
| 3. Flare Lights | 10. Upper Anti-Collision Light |
| 4. Ground Recognition Light (red) | 11. Tail Position Light |
| 5. Taxi Light | 12. Tail Logo Lights |
| 6. Egress Lights | 13. Lower Anti-Collision Lights. |
| 7. Engine Inspection Lights | |

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Figure 33-51. Exterior Lights



33-40-00 EXTERIOR LIGHTS

NOTE

INTRODUCTION

The exterior lights supply illumination for the crew, passengers, ground personnel, and aircraft identification. The aircraft has external lights installed at different locations on the fuselage, wings, empennage, engine cowlings, and landing gear.

GENERAL

Refer to Figure 33-51. Exterior Lights.

The exterior lights system has the lights that follow:

- Landing and Taxi
- Position
- Inspection
- Anti-Collision and Ground Recognition
- Logo.

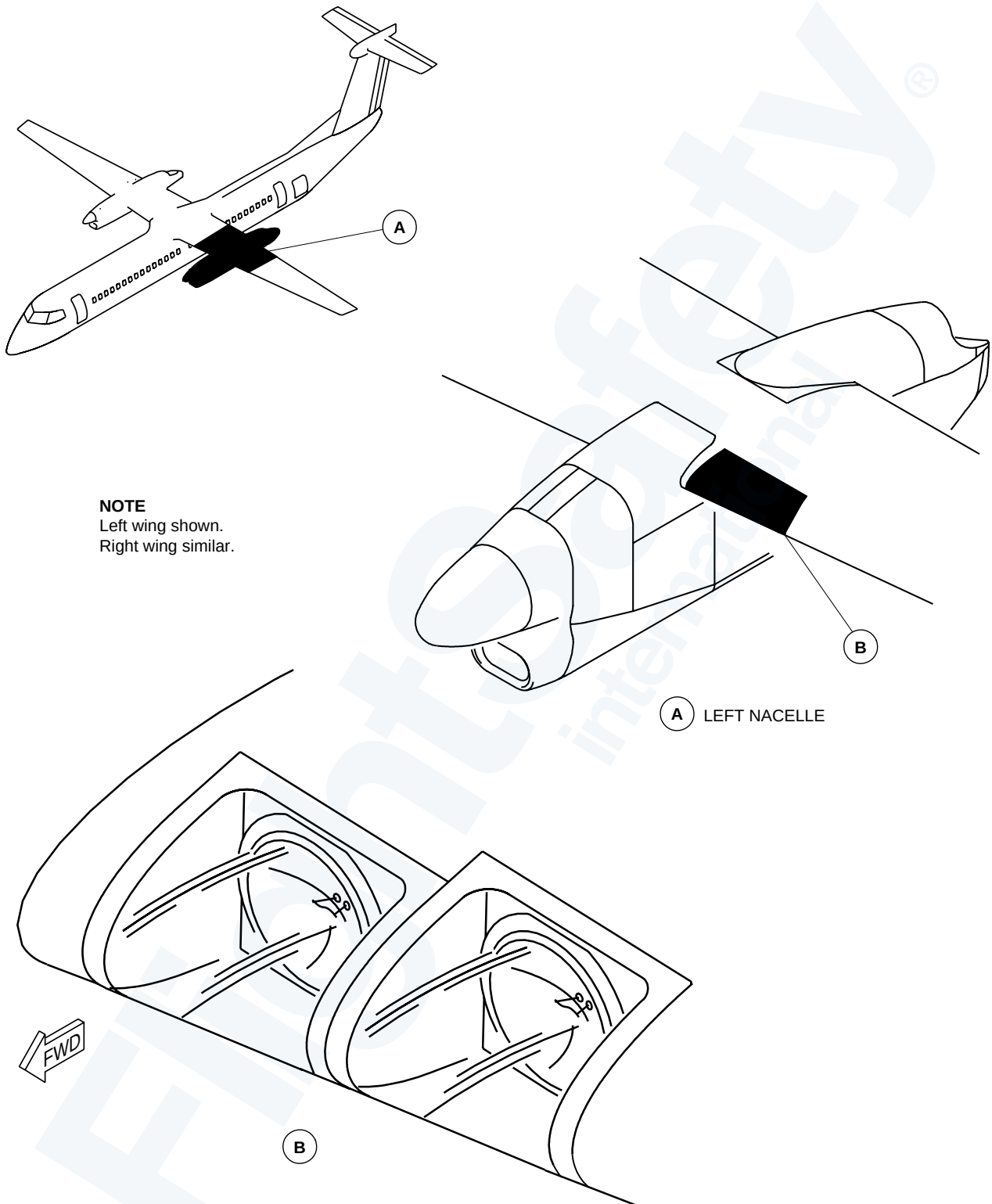
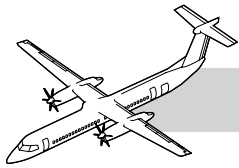
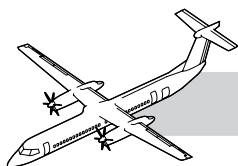


Figure 33-52. Landing Lights - Locator



COMPONENT DESCRIPTION

NOTES

Landing Taxi and Flare Lights

Refer to Figure 33-52. Landing Lights - Locator.

Two lights are installed on the leading edge of each wing, adjacent to, and outboard of the engine nacelles. They supply landing surface illumination during the landing approach.

The landing lights are controlled by independent switches located on the overhead EXTERIOR LIGHTS control panel.

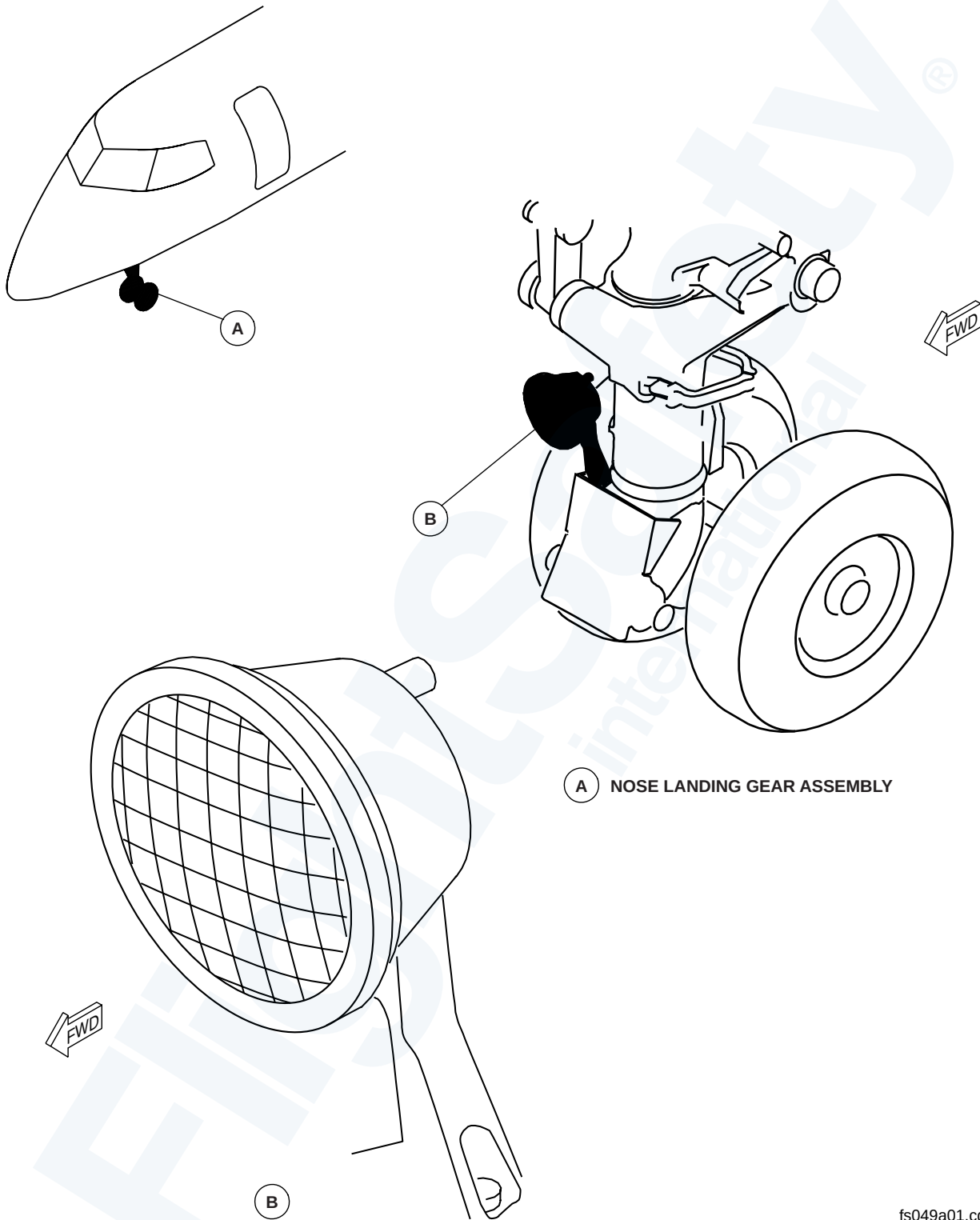
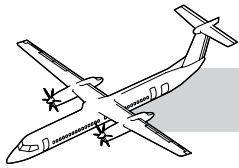
Each landing light assembly has a flare light and an approach light.

The outboard, or approach landing lights, are aimed forward in the horizontal plane with a toe-in of 1.3° relative to the aircraft centerline. They supply runway illumination during approach.

The inboard, or flare landing lights, aim directly forward parallel to the aircraft centerline. They are aimed downward from the horizontal at an angle of 4° so that, during the aircraft pitch up flare maneuver, the runway remains illuminated.

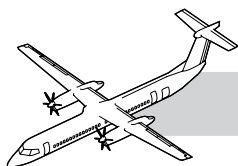
CAUTION

DO NOT KEEP THE LANDING LIGHTS ON FOR MORE THAN ONE MINUTE WHILE THE AIRCRAFT IS ON THE GROUND. IF YOU DO, THEY CAN GET TOO HOT AND CAUSE DAMAGE.



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Figure 33-53. Taxi Lights - Locator



Refer to:

- Figure 33-53. Taxi Lights - Locator.
- Figure 33-54. Landing and Taxi Light Switches.

A light is attached to the nose landing gear strut. It supplies illumination of the ground ahead of the aircraft, when taxiing.

For controls and indication, refer to Figure 33-54.

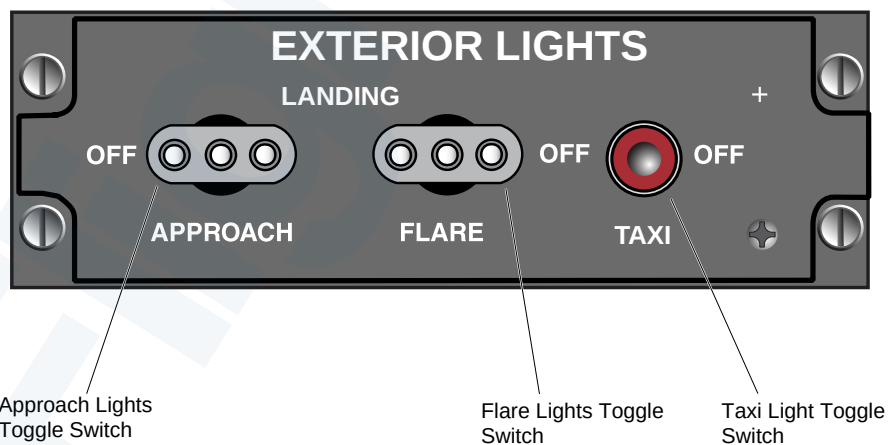
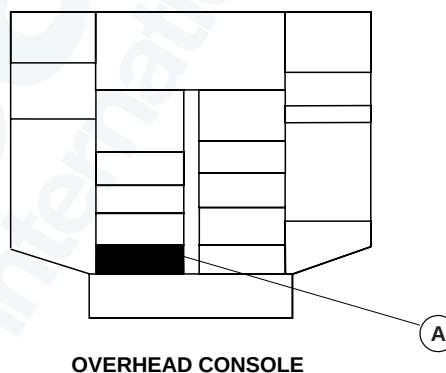


Figure 33-54. Landing and Taxi Light Switches

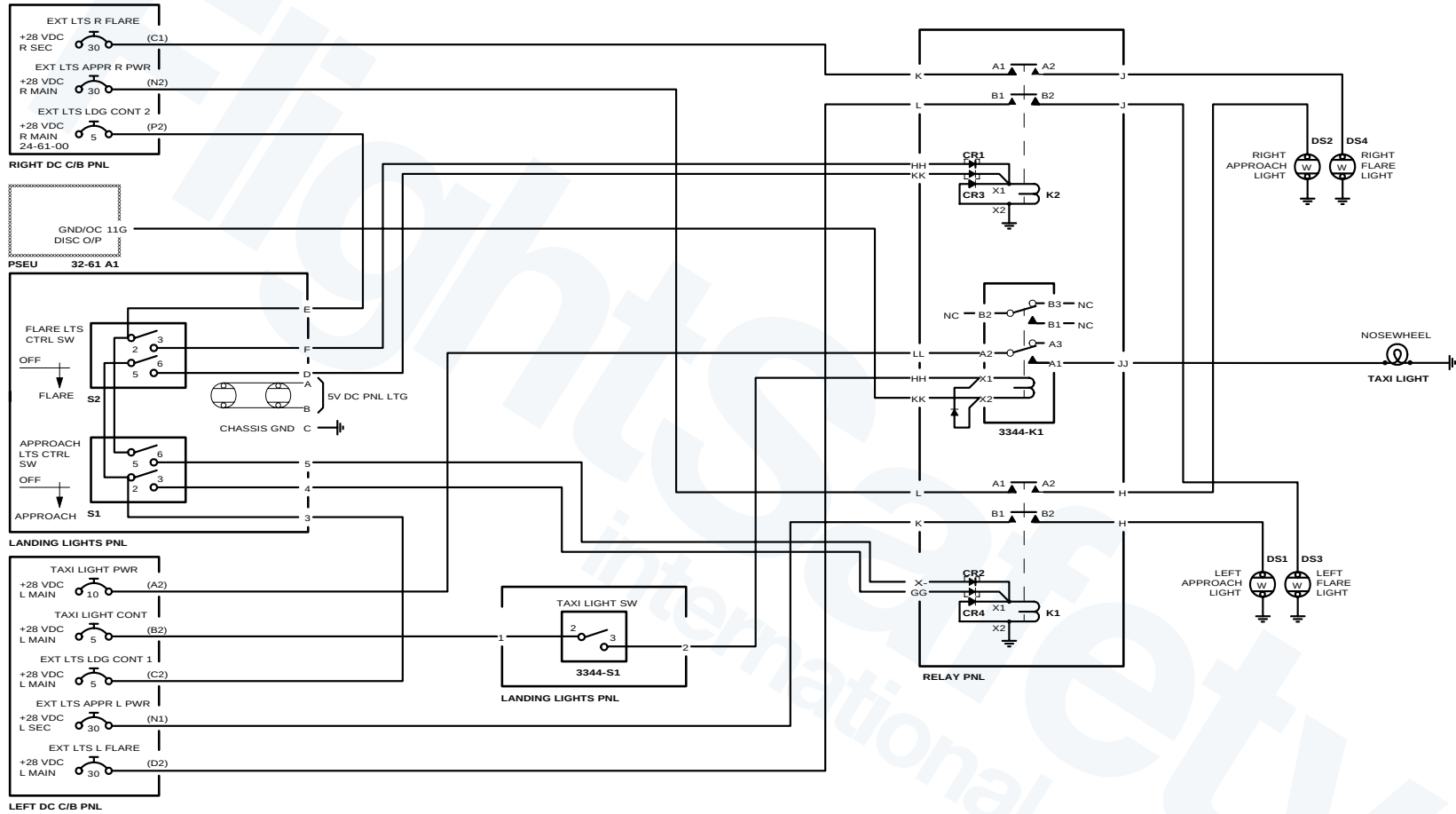
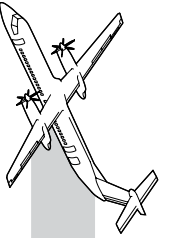
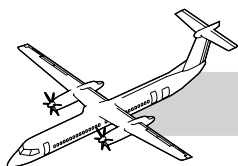


Figure 33-55. Approach Lights and Flare Schematic





Refer to Figure 33-55. Approach Lights and Flare Schematic.

The approach and flare lights draw a heavy current and are therefore isolated by contactors K1 and K2 located in the LH fuselage relay panel. Control of these contactors is from the 28 VDC main bus EXT LDG CONT 2 circuit breaker (P2).

Electrical power is supplied through relays from the main and secondary buses to the approach flare and taxi lights.

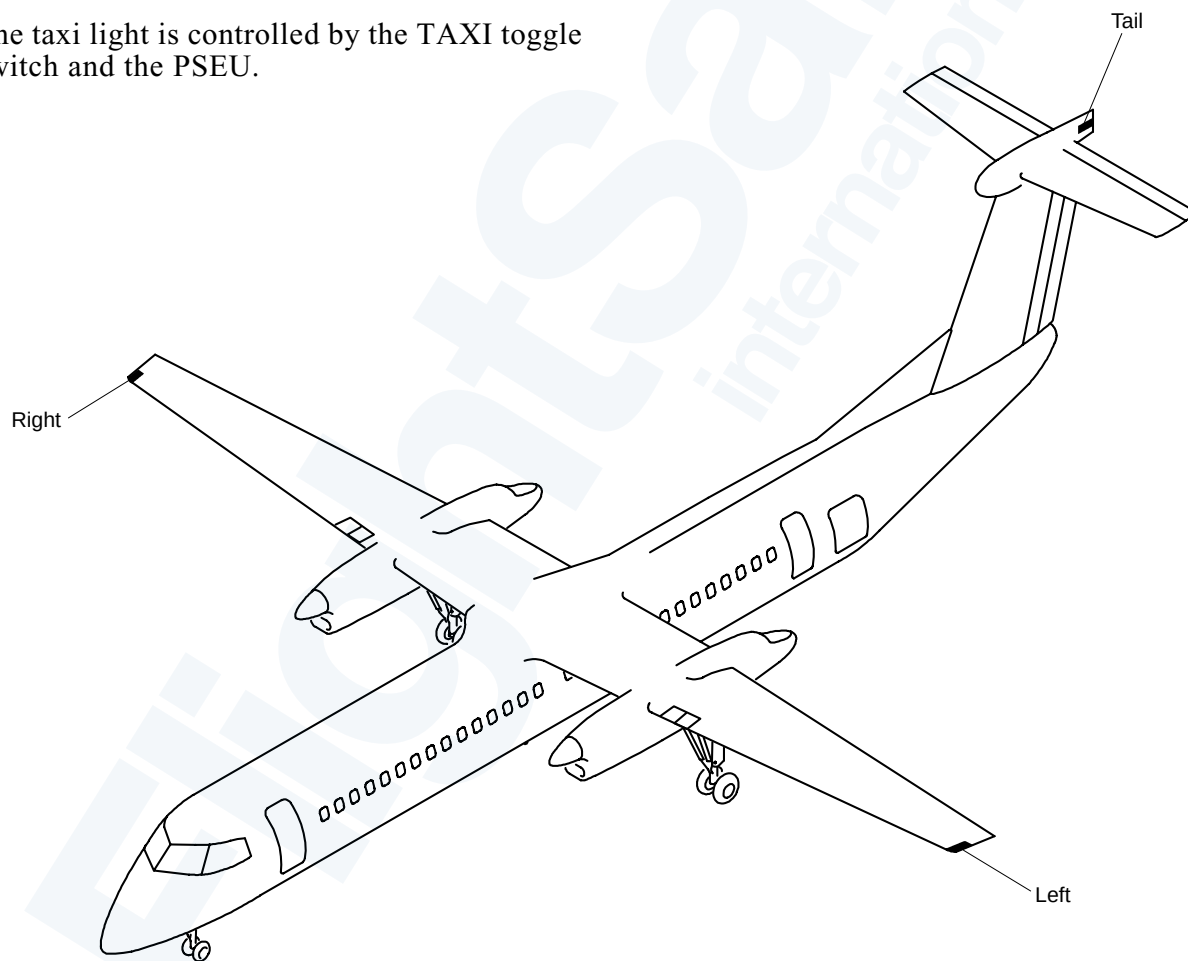
The relays are controlled by the APPROACH and FLARE toggle switches on the EXTERIOR LIGHTS control panel.

The taxi light is controlled by the TAXI toggle switch and the PSEU.

Navigation Lights

Refer to Figure 33-56. Navigation Lights.

Two position lights are installed on the left wing, right wing, and tail.



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Figure 33-56. Navigation Lights

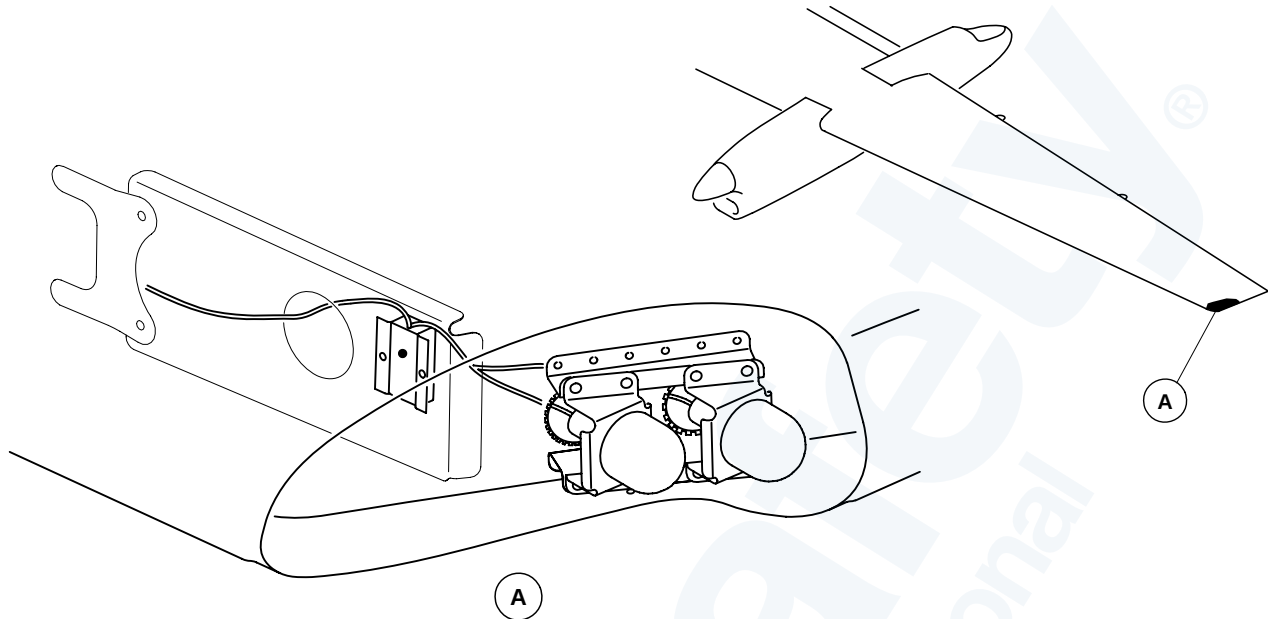
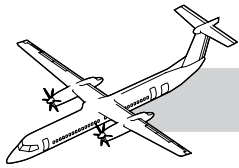


Figure 33-57. Electronic Switch Wing Position Light Assembly

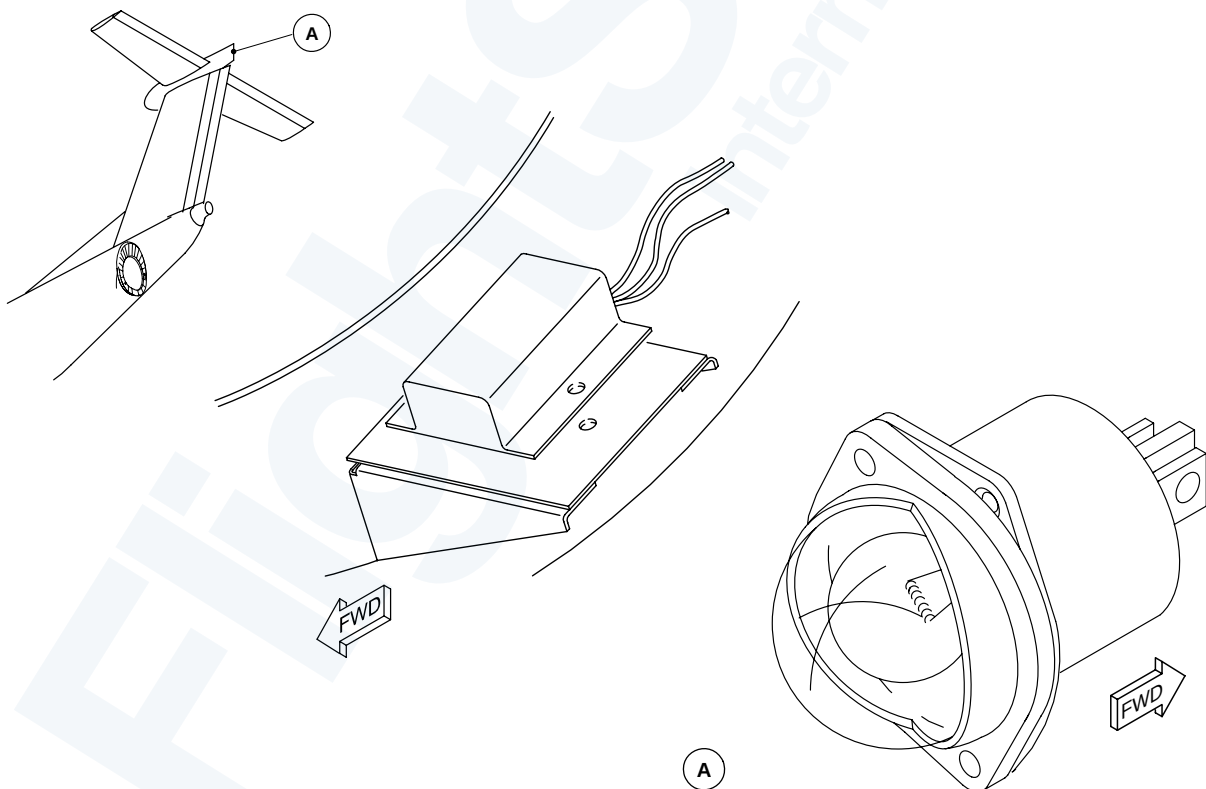
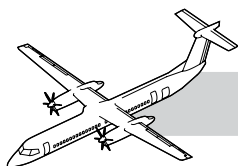


Figure 33-58. Tail Position Light Assembly



Refer to Figure 33-57. Electronic Switch Wing Position Light Assembly.

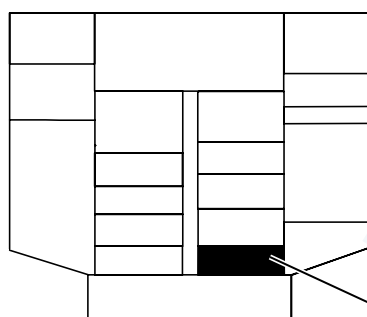
An electronic switch is installed in each wing tip.

Refer to Figure 33-58. Tail Position Light Assembly.

An electronic switch is also installed in the aft fuselage bay.

Refer to Figure 33-59. Controls - POSN-OFF Switch.

The wing and tail position lights are controlled by the same switch, labeled POSN-OFF, on the overhead EXTERIOR LIGHTS control panel.



OVERHEAD CONSOLE



A

Position Lights Toggle Switch

Figure 33-59. Controls - POSN-OFF Switch

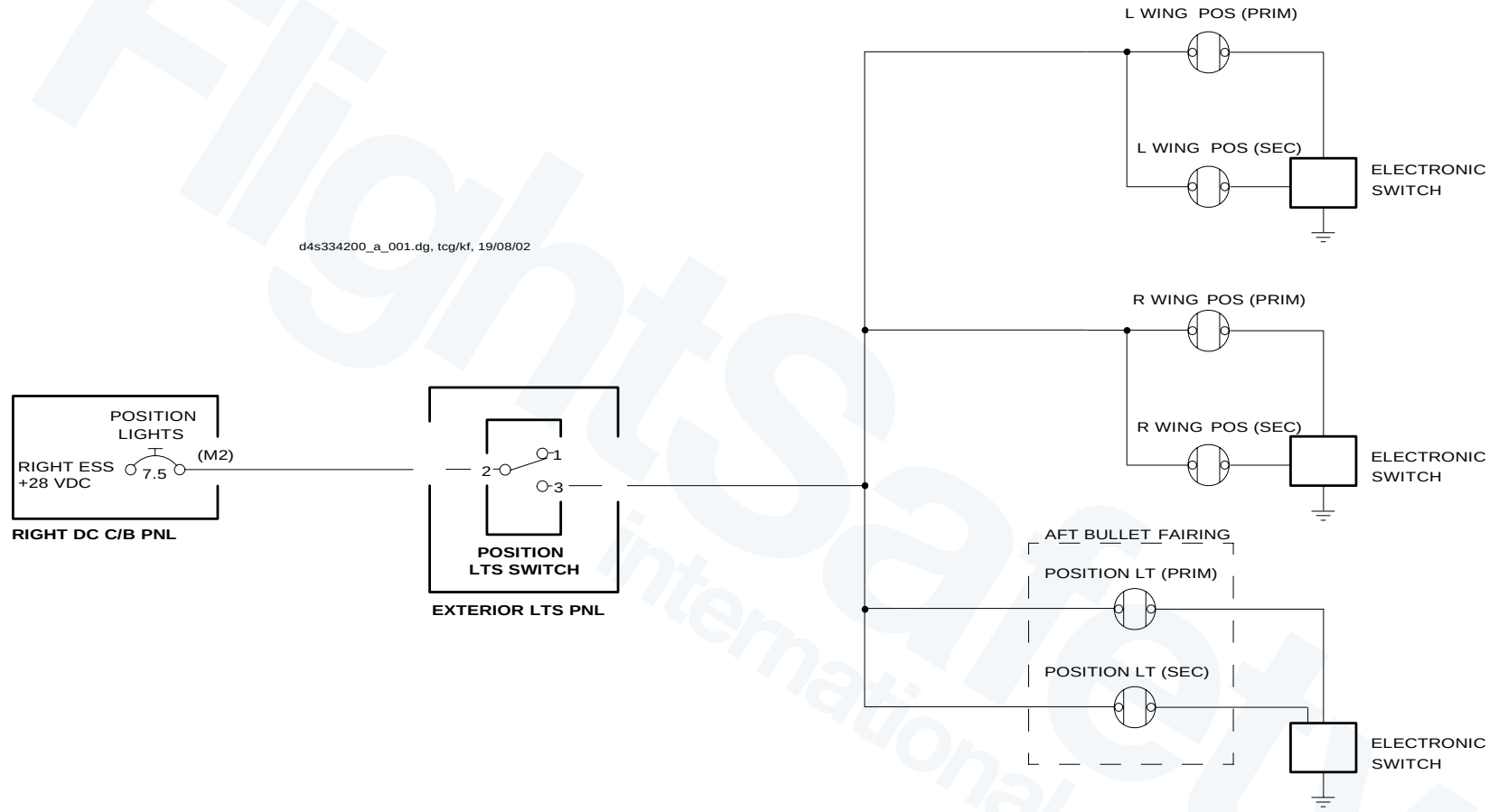
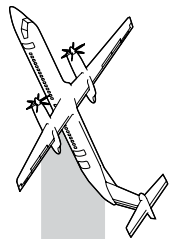
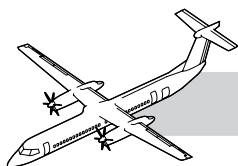


Figure 33-60. Position Light Schematic

33 LIGHTING



Position Lights

NOTES

Refer to Figure 33-60. Position Light Schematic.

When the switch is set to POSN, all position lights come on for 1 seconds to check the secondary lights. After 1 seconds, the secondary lights will go off.

If a primary light fails, the electronic switch will automatically cause the secondary light to come on.

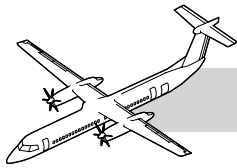
WARNING

BE CAREFUL WHEN YOU REPLACE THE HALOGEN LAMP. THE HALOGEN LAMP IS PRESSURIZED AND CAN EXPLODE DURING REPLACEMENT.

Navigation Lights Operational Test

The following is an abbreviated description of the maintenance practice and is intended for training purposes only. For a more detailed description of the practice, refer to the task in the Bombardier AMM PSM 1-84-2.

- Select the POSN switch on the EXTERIOR LIGHTS control panel to POSN. Make sure that the primary (forward) and secondary (aft) lights, on the wing tips and tail, come on. After one second, all of the secondary lights, on the wing tips and tail, will go off.
- Set the POSN switch to the OFF position. Make sure that all of the primary lights, on the wing tips and tail, go off.



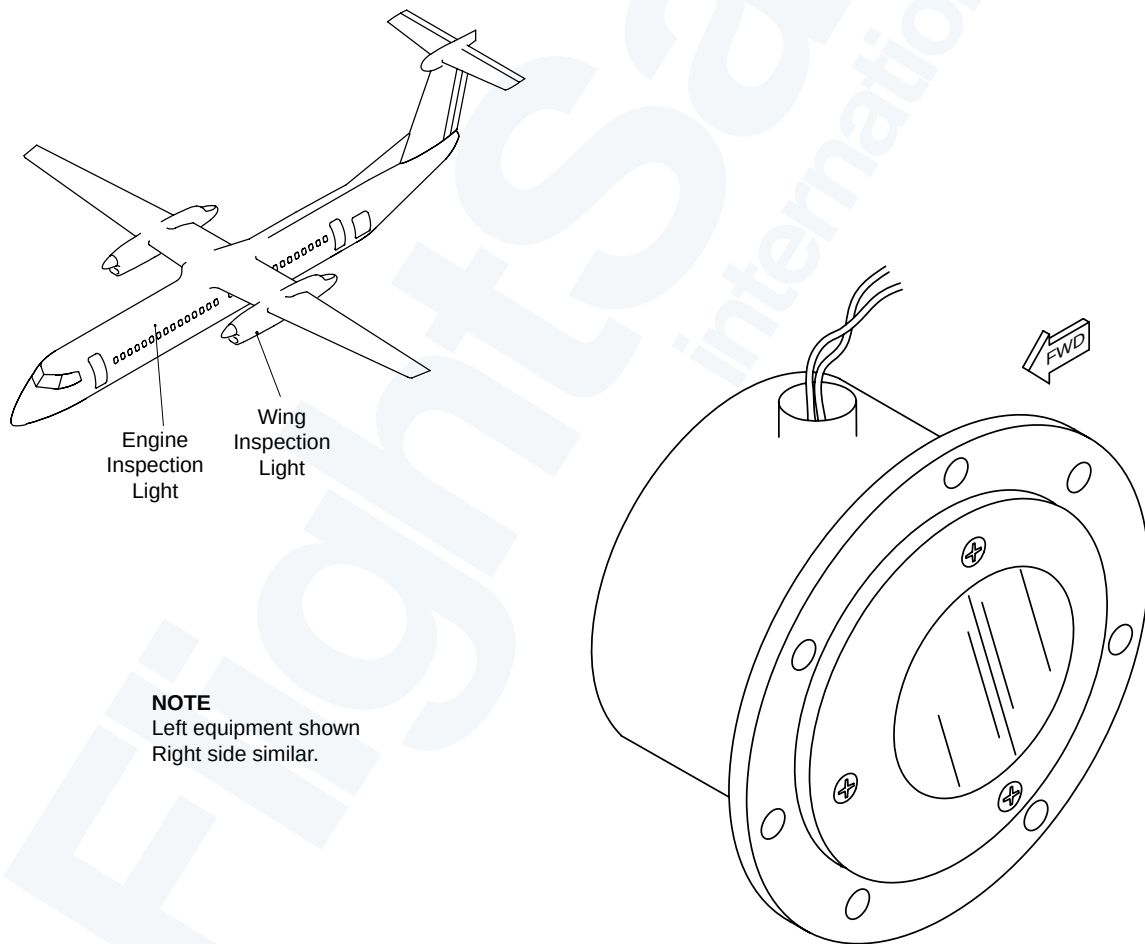
Inspection Lights

Refer to Figure 33-61. Inspection Lights Detail.

The inspection lights give illumination for visual inspection of ice formation on the wing leading edges and engine intakes.

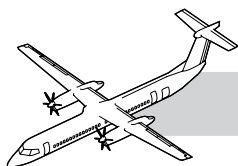
One light is located on each side of the fuselage for lighting the propeller spinners, engine intakes, and the wing leading edges between the nacelle and fuselage.

One light is located on the outboard side of each nacelle lower cowl, for illumination of the outboard wing leading edges.



NOTE
Left equipment shown
Right side similar.

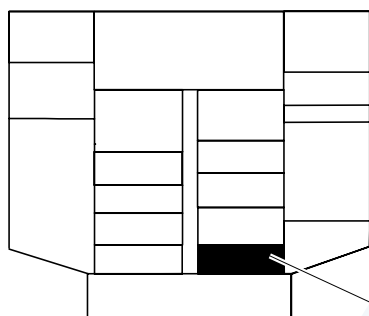
Figure 33-61. Inspection Lights Detail



Refer to:

- Figure 33-62. Wing Inspection Switch - Locator.
- Figure 33-63. Wing Inspection Light Schematic.

Electrical power is supplied through a toggle switch from the secondary buses to the inspection lights.



OVERHEAD CONSOLE



Wing Inspection Lights
Toggle Switch

Figure 33-62. Wing Inspection Switch - Locator

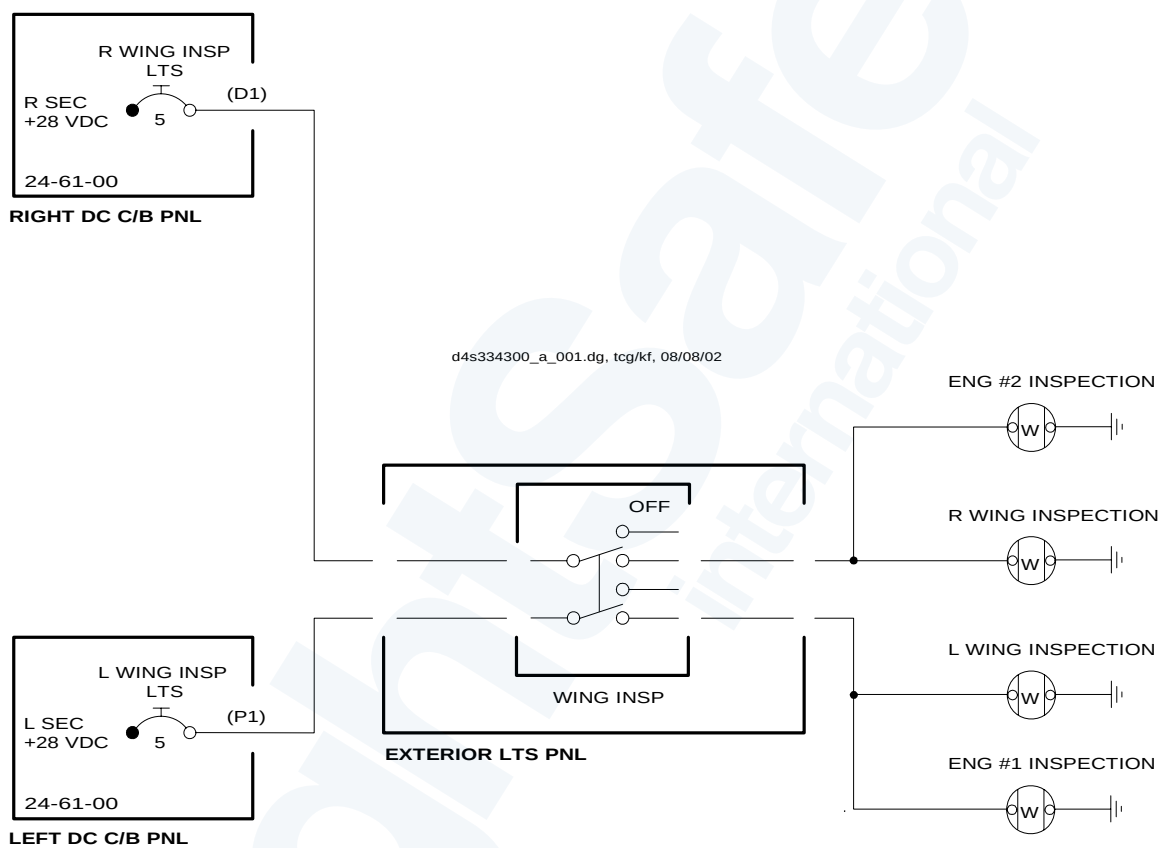
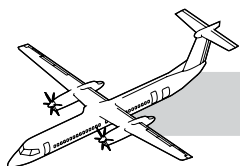
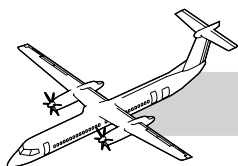


Figure 33-63. Wing Inspection Light Schematic



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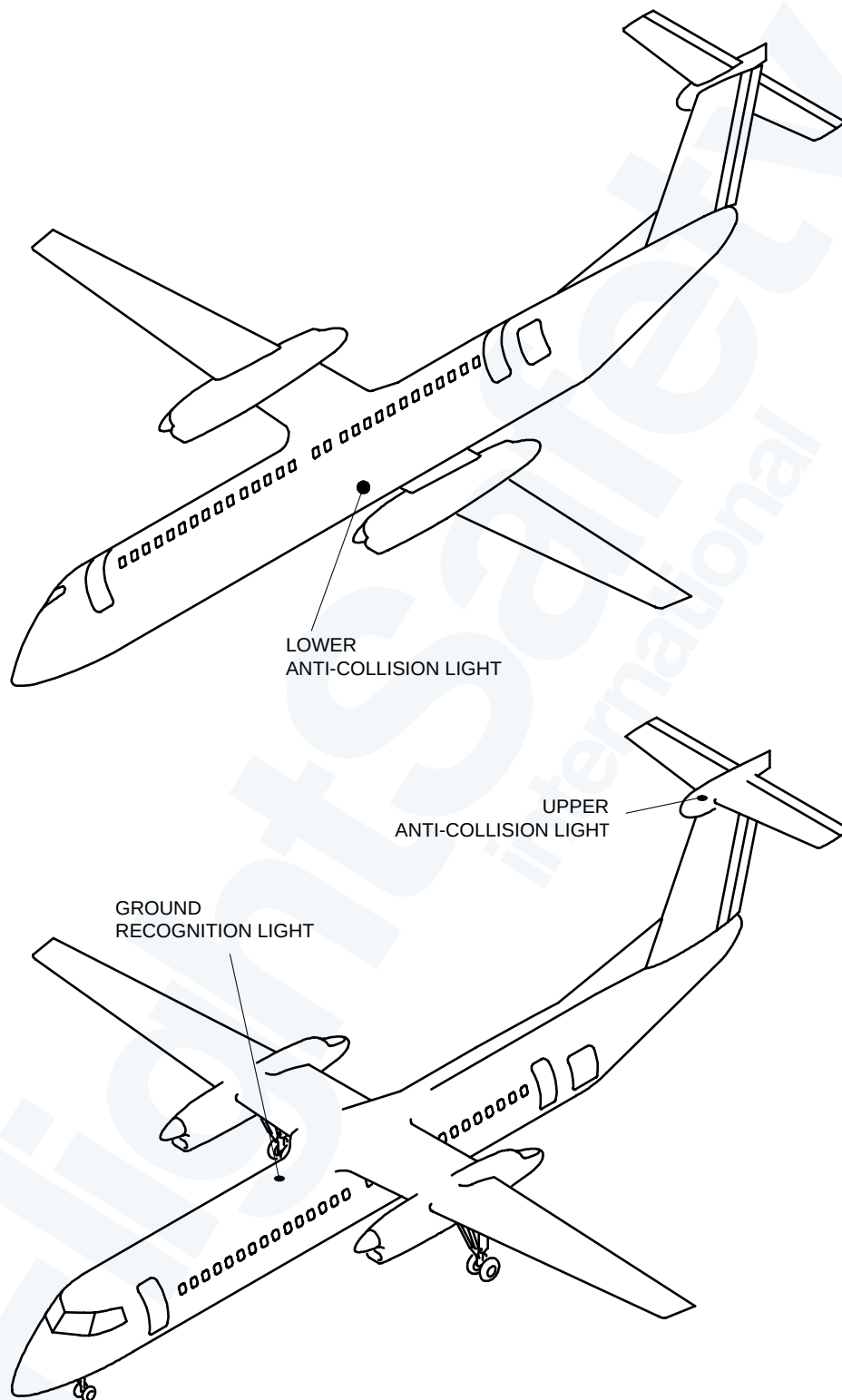
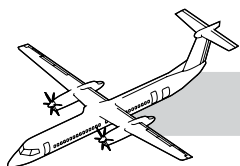
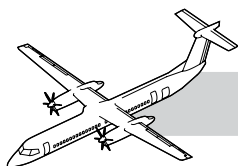


Figure 33-64. Anti-Collision Lights and Recognition Lights - Exterior Locator



Anti-Collision Lights and Recognition Lights

Refer to:

- Figure 33-64. Anti-Collision Lights and Recognition Lights - Exterior Locator.
- Figure 33-65. Anti-Collision and Recognition Light.

Two white anti-collision lights, located on the fuselage, flash to make the aircraft more visible. When the aircraft is on the ground, a red recognition light is set on, to show that the engines or APU are about to start or are operating.

The upper anti-collision light assembly is located on the top of the vertical stabilizer. The lower anti-collision light is located on the lower center line of the fuselage. The red recognition light is located on the upper forward fuselage.

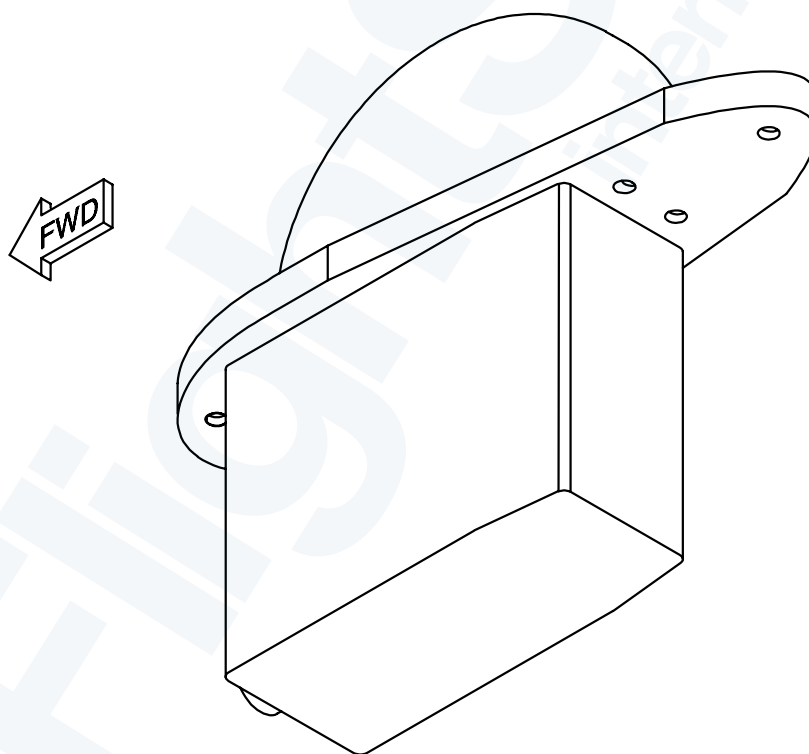
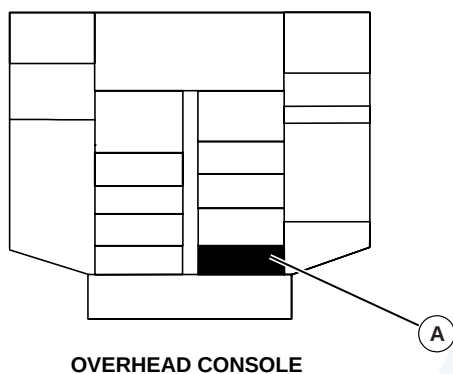
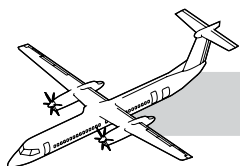
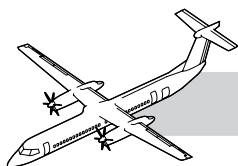


Figure 33-65. Anti-Collision and Recognition Light



Anti-Collision Toggle
Light Switch

Figure 33-66. Anti-Collision Lights Switch



Refer to Figure 33-66. Anti-Collision Lights Switch.

NOTES

The white anti-collision, and red recognition lights are controlled by a switch located on the EXTERIOR LIGHTS control panel.

The anti-collision light switch is set to the WHITE A/COL position to operate both white anti-collision lights. It is set to the RED position to operate the red recognition light.

CAUTION

DO NOT TOUCH THE STROBE LIGHTS WITH YOUR FINGERS. IF YOU DO THIS , OIL FROM YOUR FINGERS CAN DAMAGE THE LIGHTS.

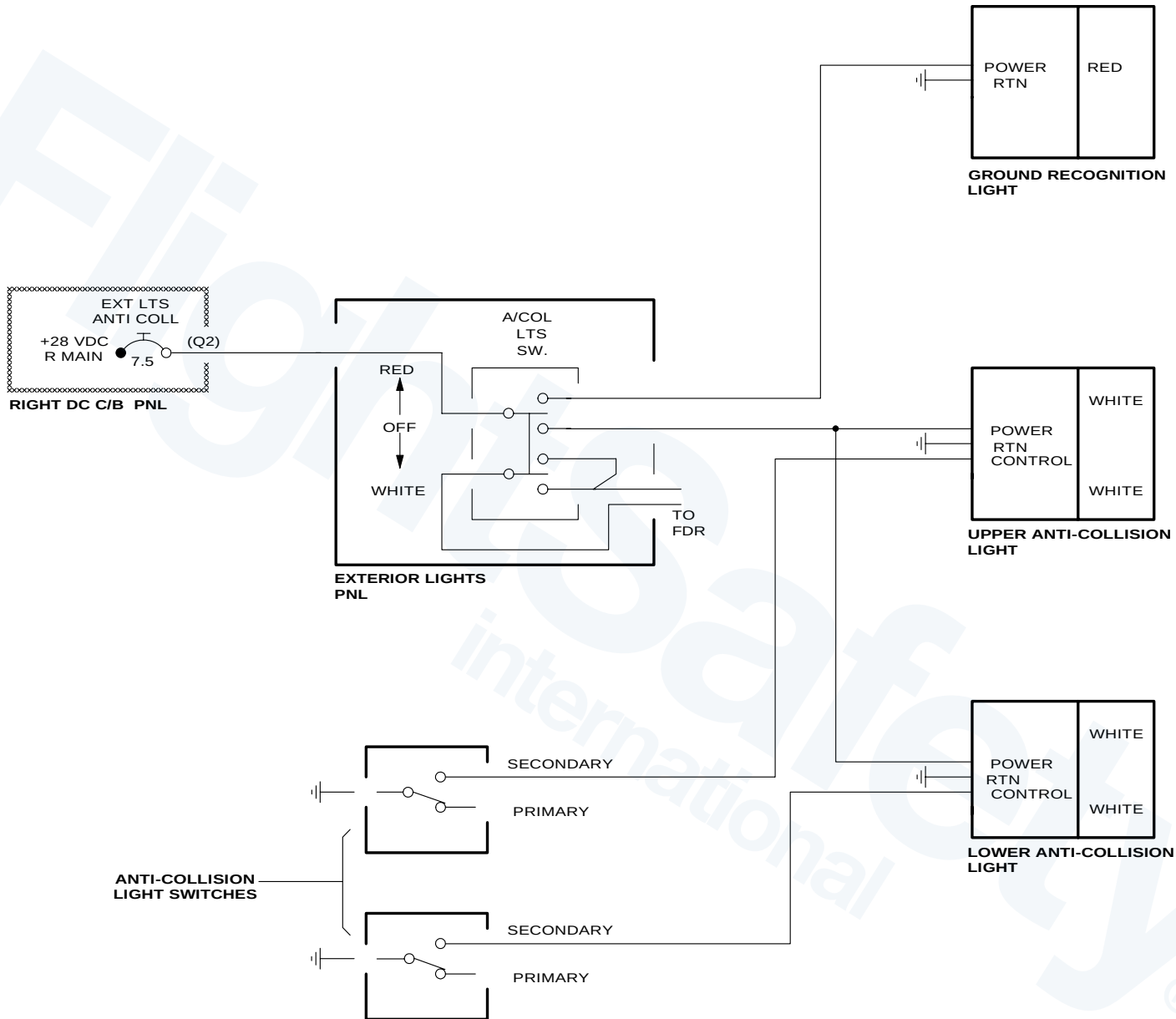
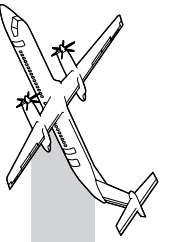
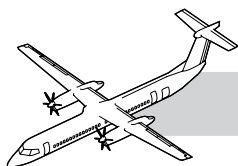


Figure 33-67. Anti-Collision Light Schematic



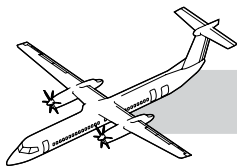


Refer to Figure 33-67. Anti-Collision Light Schematic.

NOTES

The right main bus supplies electrical power through a toggle switch to the lights.

The toggle switch also controls the operation of the flight data recorder (FDR).



Refer to Figure 33-68. Anti-Collision Light Selector.

The white anti-collision lights and the recognition light operate at 45 flashes per minute.

The white anti-collision light assembly have two anti-collision strobe type light elements.

If the primary element has malfunctioned, a secondary element selection is made using a toggle switch located in the aft fuselage equipment bay.

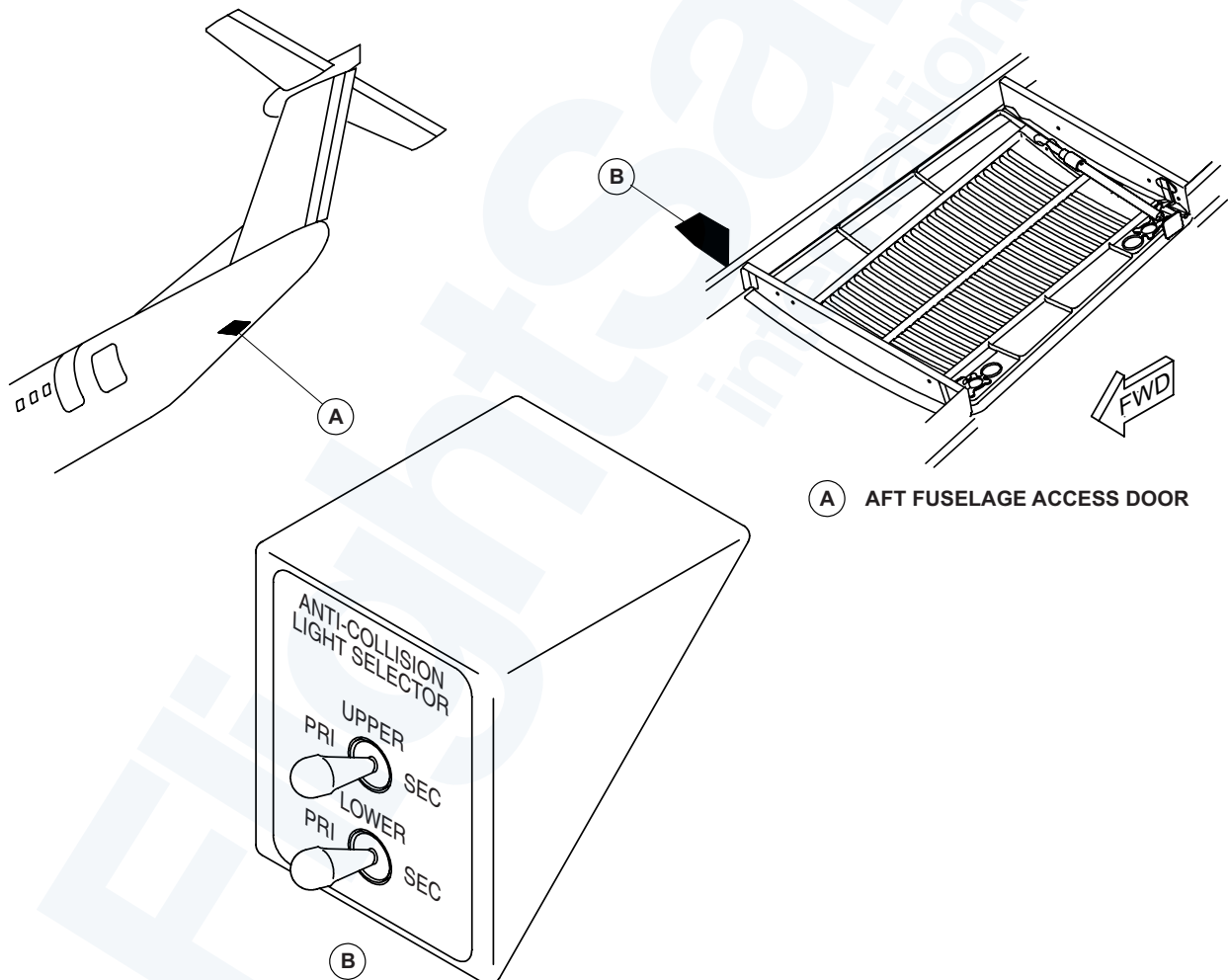
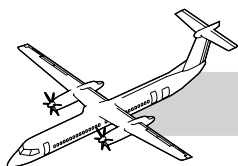


Figure 33-68. Anti-Collision Light Selector



Logo Lights

Refer to Figure 33-69. Logo Lights.

The logo lights provide illumination of the aircraft tail.

The two optional logo lights are installed flush with the bottom of the horizontal stabilizer. There is one logo light on the lower left side and one logo light on the lower right side.

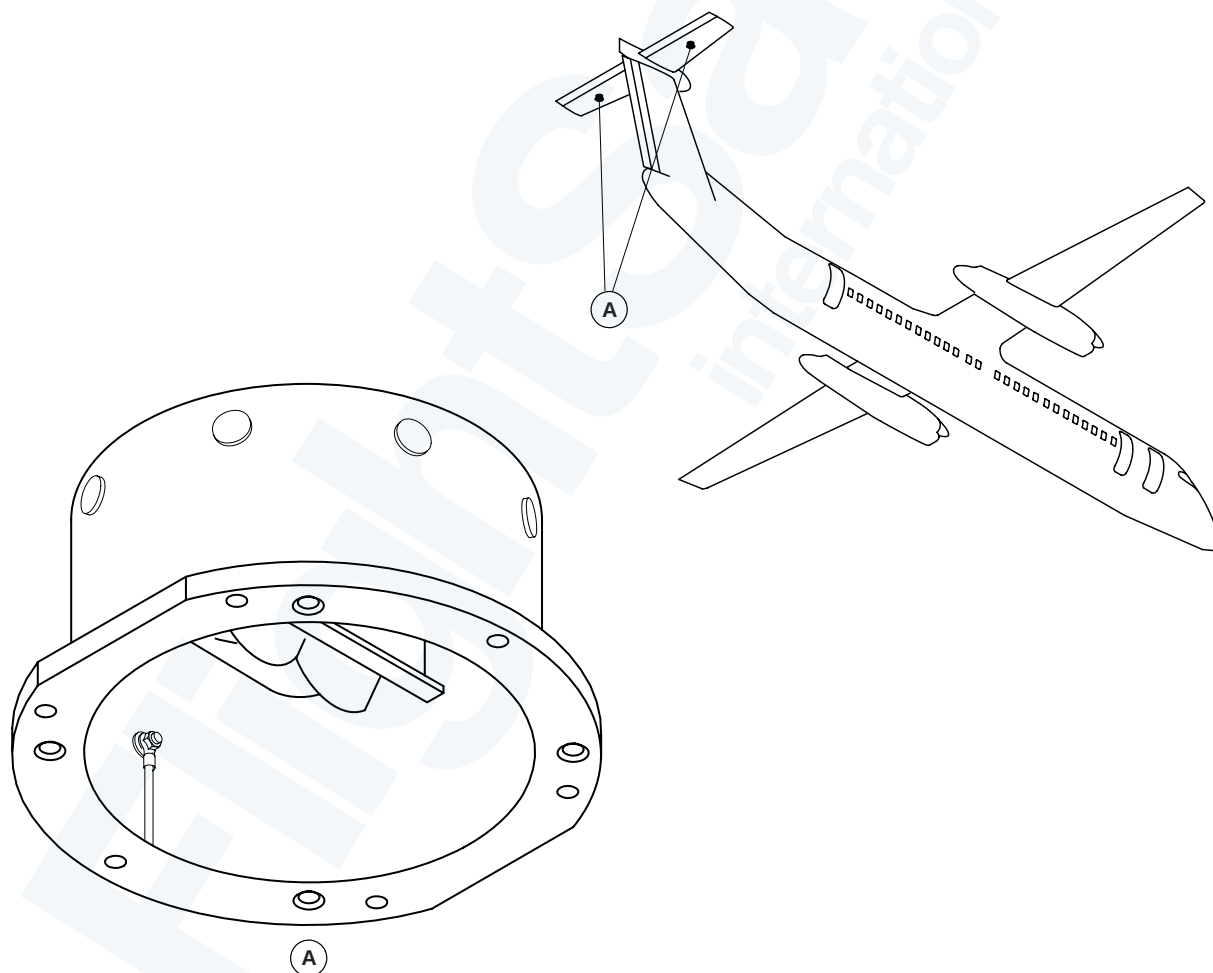


Figure 33-69. Logo Lights

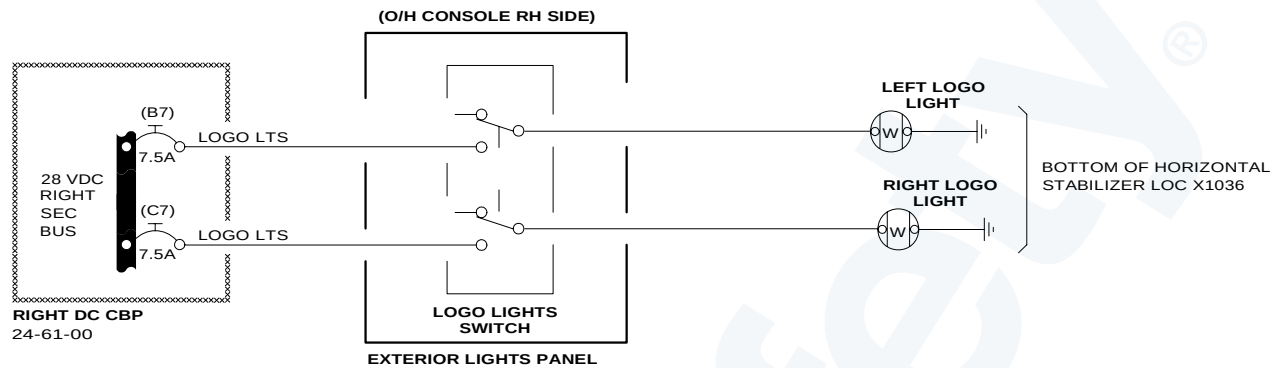
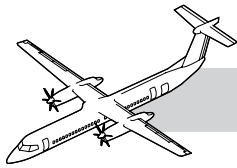


Figure 33-70. Logo Light Schematic

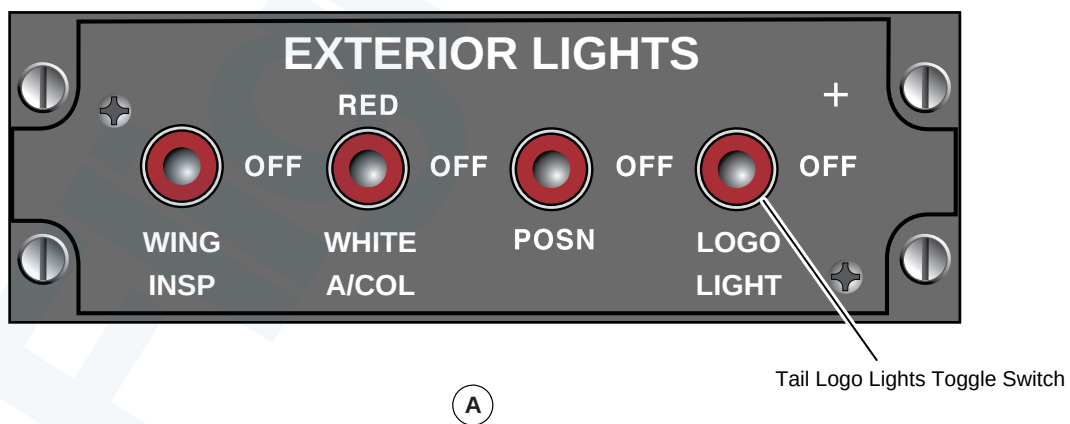
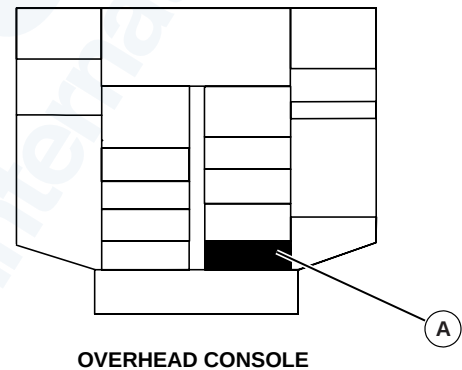
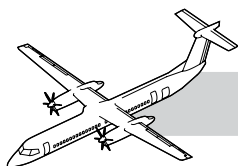


Figure 33-71. Logo Lights Switch - Locator



Refer to Figure 33-70. Logo Light Schematic.

NOTES

The right secondary bus supplies electrical power to the lights through a toggle switch.

Refer to Figure 33-71. Logo Lights Switch - Locator.

Logo lights are controlled using a two-position switch located on the overhead EXTERIOR LIGHTS control panel in the flight compartment.

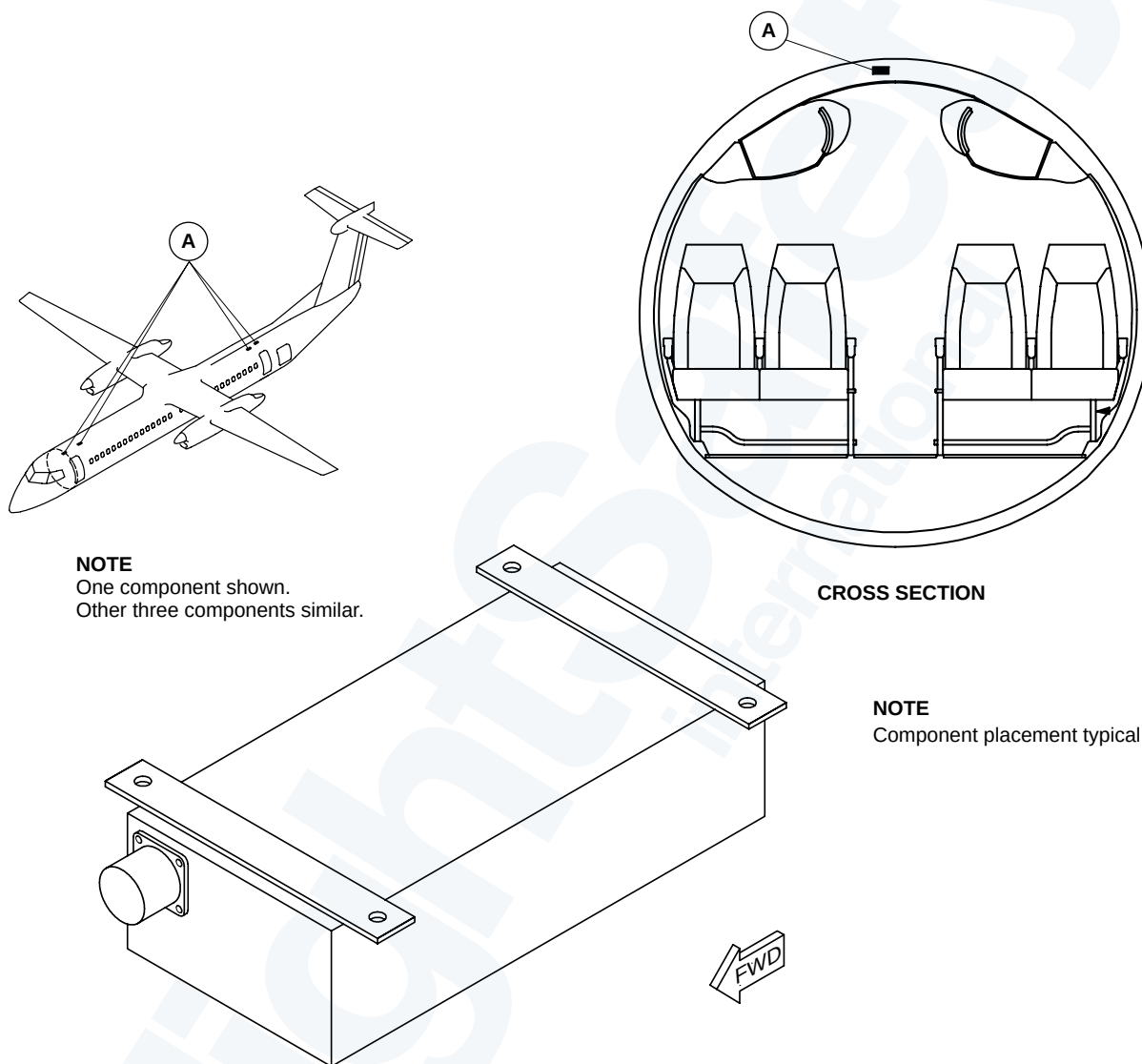
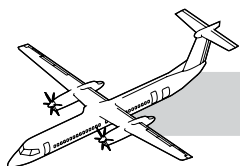
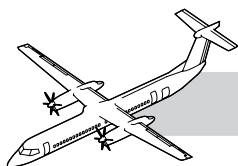


Figure 33-72. Emergency Lights Power Supply Unit



33-51-00 EMERGENCY LIGHT SYSTEM (NEXT GEN)

INTRODUCTION

The emergency light system supplies interior, and exterior lighting for emergency evacuation of the aircraft or if the DC power system fails.

GENERAL

The emergency lights system has the following components:

- Power Supply-Emergency Light Unit
- Emergency-Cabin Lights
- Emergency-Egress Lights
- Emergency-Floor Path Lights.

Power Supply, Emergency Light Unit

Refer to:

- Figure 33-72. Emergency Lights Power Supply Unit.
- Figure 33-73. Emergency Lights Power Supply Unit Electrical Schematic.

The emergency lights are powered by four self contained 6.4 V Ni-Cad battery packs, located above the passenger compartment ceiling, near each emergency exit.

The Four power supplies also have a solid state control circuit. The logic circuitry automatically de-energizes the Emergency Lighting System when the battery voltage drops below the battery cut-off level (5.4 VDC). Isolation diodes, internal to the EPS , prevent the batteries from discharging into the aircraft buses.

MAINTENANCE PRACTICES

Emergency-Light Power-Supply Units Removal

- There are four power supplies installed in the passenger compartment ceiling at Sta. X010.25, X083.00, X654.500 and X692.22 approximately
- Remove the power supply by disconnecting the electrical connector then the removing the attaching hardware from the mounting rack.

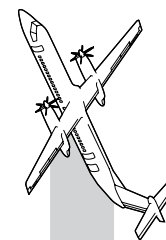
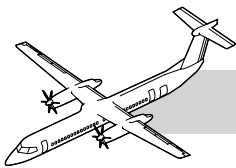


Figure 33-73. Emergency Lights Power Supply Unit Electrical Schematic



Emergency Cabin Lights

Refer to Figure 33-74. Emergency Cabin Lights.

There are 13 light assemblies installed on the ceiling panels in the passenger compartment. Eleven accent light assemblies provide spotlight illumination to the overhead storage bins. The accent light assemblies include 2 spotlights, 1 emergency flood light and a zone indicator light. (The zone indicator light is inoperative on the Q400).

Emergency EXIT signs are located near each of the four emergency exits.

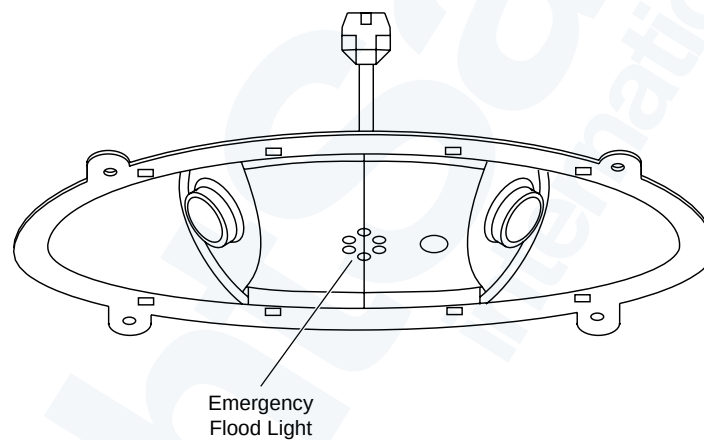


Figure 33-74. Emergency Cabin Lights

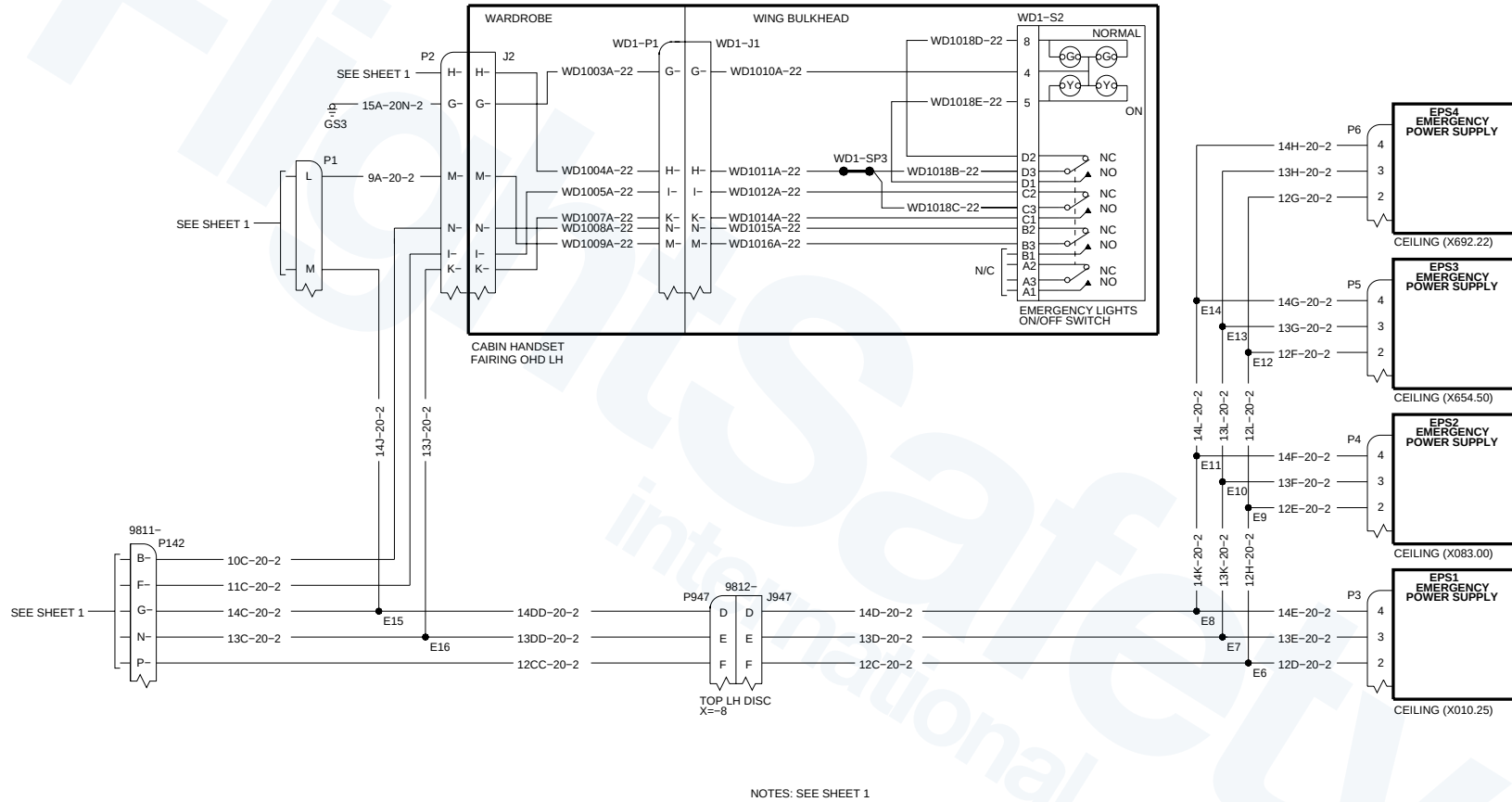
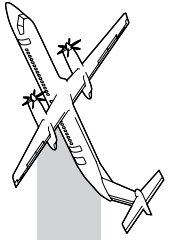
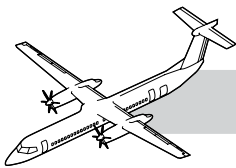


Figure 33-75. Emergency Egress Lights Electrical Schematic





Emergency Egress Lights

Refer to:

- Figure 33-75. Emergency Egress Lights Electrical Schematic.
- Figure 33-76. Emergency Egress Lights Locator. Locator.

Fixed exterior emergency egress lights are located at the left and right Aft exits and the front right emergency exit and the forward passenger door.

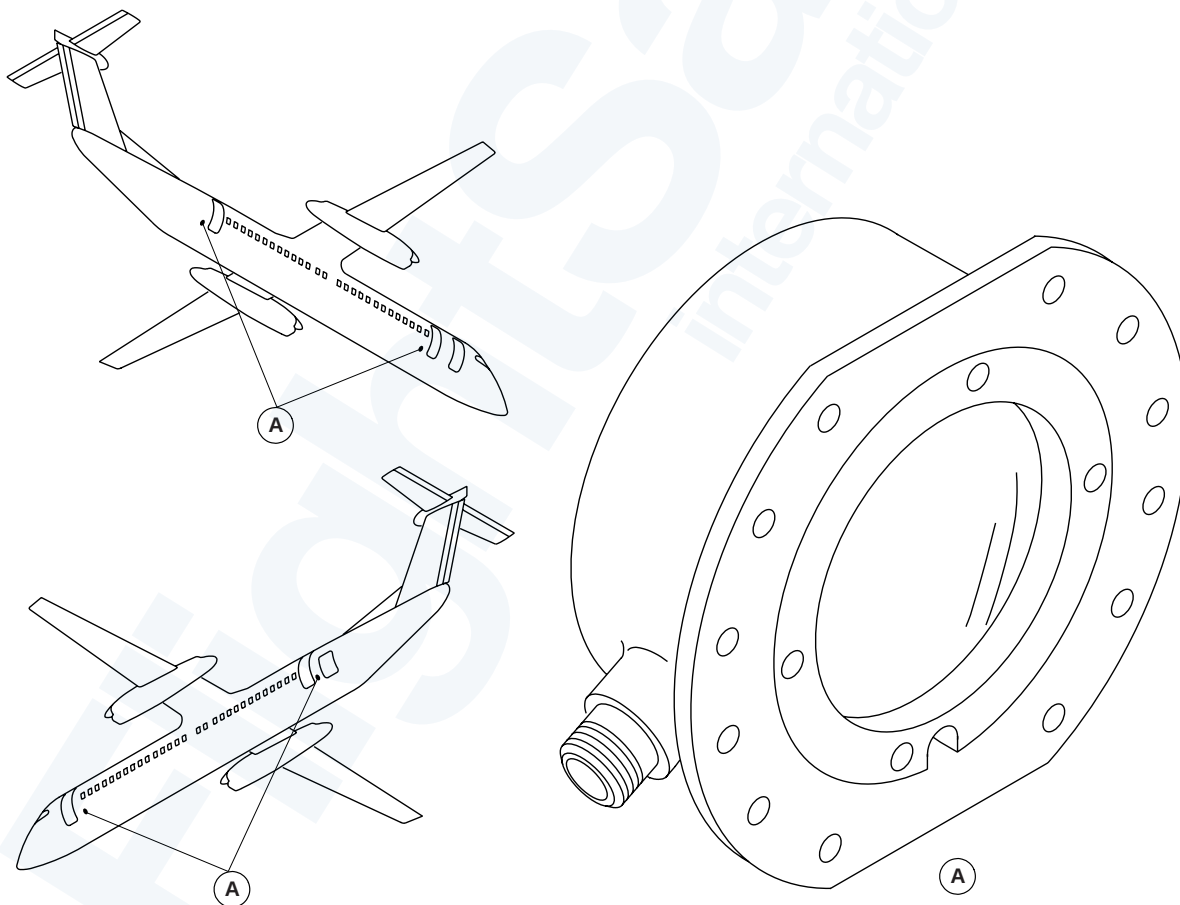


Figure 33-76. Emergency Egress Lights Locator

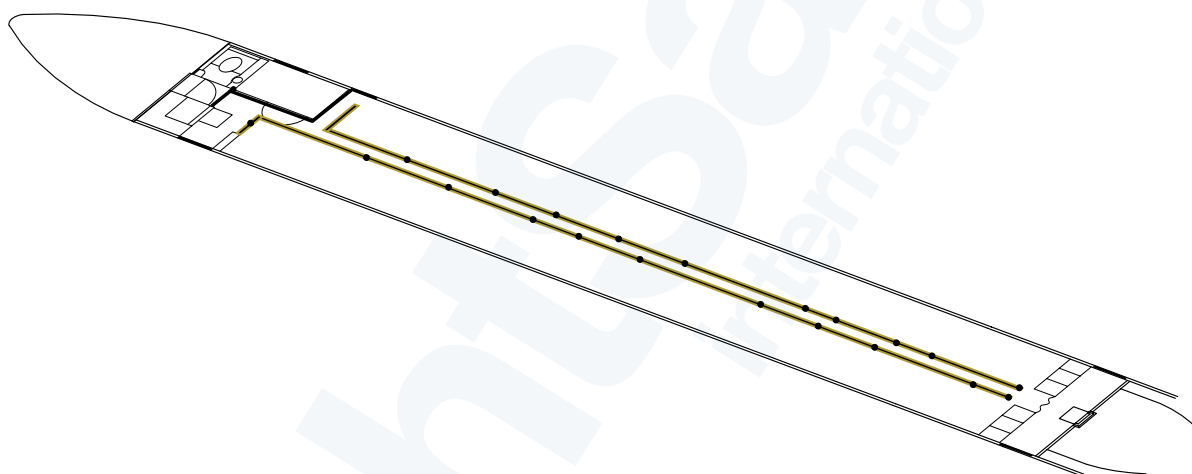
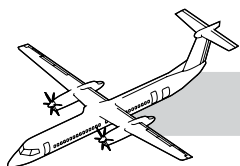
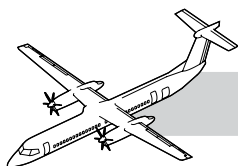


Figure 33-77. Emergency Floor Path Light Assembly



Emergency Floor-Path-Light Assemblies

NOTES

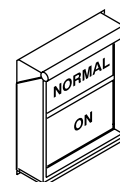
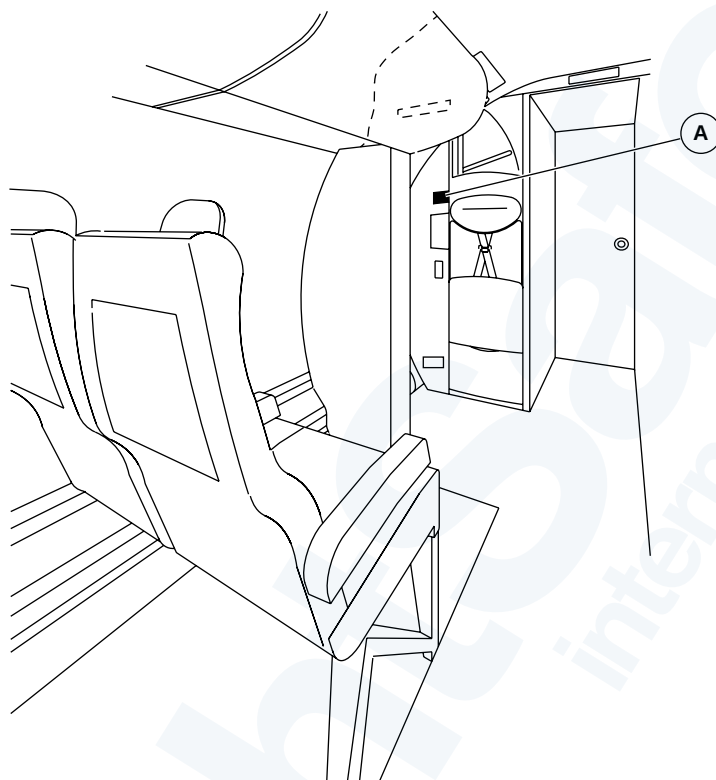
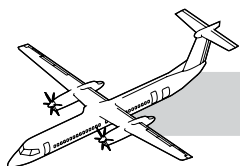
Refer to Figure 33-77. Emergency Floor Path Light Assembly.

Emergency floor-path-lighting is installed on the passenger compartment floor with silicone adhesive and double sided tape, to direct persons to the emergency exits. Each light assembly consists of a light cover, a photo-luminescent tape, a light base and end caps, where applicable.

The light strips are continuously charged by operation of the aircraft lighting system and available light. The system is considered to be sufficiently charged after 15 minutes of exposure to aircraft lighting. This is to meet the escape path lighting requirements for the longest permitted flight.

NOTE

Each light shows an expiry date on its mating side.



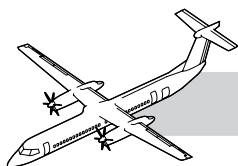
EMERGENCY LIGHTING

A

NOTE

View Looking Forward
From Fwd Passenger
Compartment.

Figure 33-78. Forward Passenger Information Sign Emergency Lights



CONTROLS AND INDICATIONS

Refer to:

- Figure 33-78. Forward Passenger Information Sign Emergency Lights.
- Figure 33-79. Emergency Lights Control Panel.

The emergency lights and EXIT signs are controlled by one of two switches:

- EMER LIGHTS-OFF/ARM/ON switch in the flight compartment
- EMERGENCY LIGHTING-NORMAL/ON switch on the main passenger entrance bulkhead.

When the EMER LIGHTS switch, in the flight compartment, is set to the ON position, the emergency lights come on. When the EMER LIGHTS switch is set to the OFF position, the emergency lights go off. When the EMER LIGHTS switch is set to the ARM position, the emergency lights automatically come on if there is a 28 VDC left main bus power failure.

The emergency power supply units receive charging power from the 28 VDC left essential bus with:

- The EMER LIGHTS switch in either the ARM or OFF position
- The EMERGENCY LIGHTING switch is set to NORMAL

When the EMERGENCY LIGHTING switch, on the main passenger entrance bulkhead, is set to ON, the emergency lights come on. When the EMERGENCY LIGHTING switch is set to NORMAL, the emergency lights go off, but will come on if there is a 28 VDC left main bus power failure. For this to occur, the EMER LIGHTS switch, in the flight compartment, must be in the ARM position.

An EMER LTS DISARMED caution light on the caution and warning panel comes on when the EMER LIGHTS-OFF/ARM/ON switch is not in the armed position.

If the emergency lights come on by the automatic function, they can be turned off by the EMER LIGHTS-OFF/ARM/ON switch first set to the ON and then the OFF position.

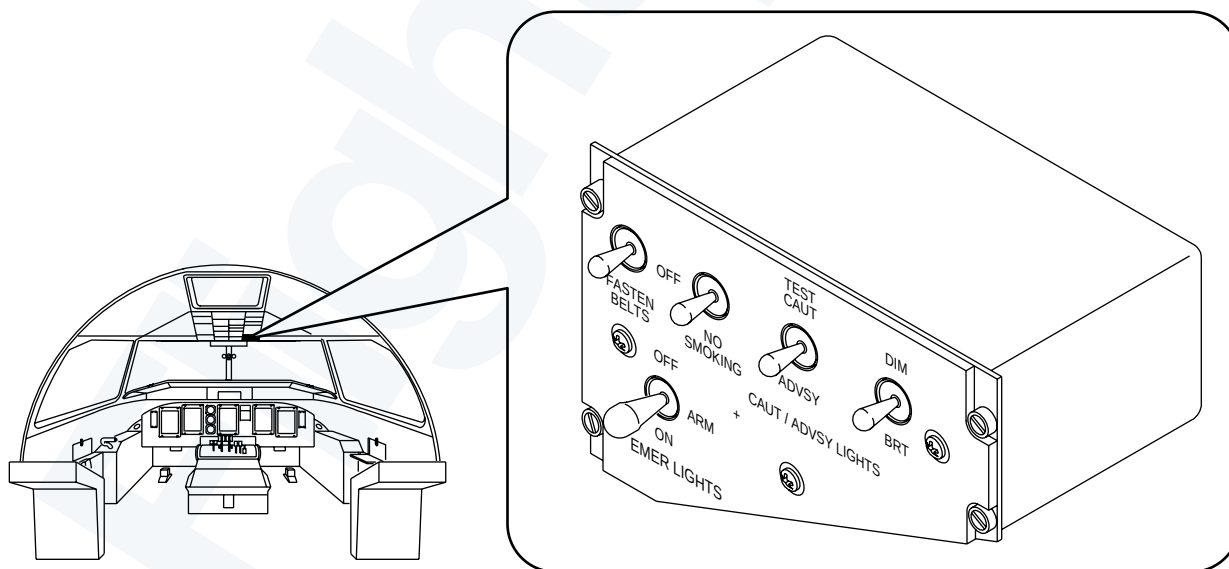
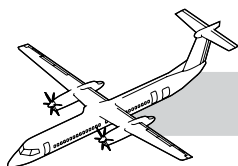


Figure 33-79. Emergency Lights Control Panel



33-00-00 SPECIAL TOOL & TEST EQUIPMENT

- 18-234 Extraction Tool (Vivisun)
- GSB3340001 Landing Light Alignment tool (Outboard)
- GSB3340002 Landing Light Alignment Tool (Inboard)
- GSB5180001 or GSB5180005 Bonding Tester
- None specified Light aiming tool (Locally manufactured, cardboard approximately 33 in. (83.82 cm) by 36 in. (91.44 cm) with cut-out for the shape of the leading edge)
- GSB3351001 T-light comparator
- Commercially available Micro-lambert meter or equivalent
- 58T-101 Capsule removal tool (Eaton)
- 58T-106 Flat blade screwdriver (Eaton) (or equivalent)
- 58T-107 Connector-block removal tool (Eaton)
- 14-44 Plastic scraper

33-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 33-12-11-000-801: Removal of the Dome Lights Lamp.
- AMM 33-12-11-400-801: Installation of the Dome Lights Lamp.
- AMM 33-21-06-960-801: Replacement of the Threshold or Ceiling-Panel Boarding-Light Lamps.
- AMM 33-22-00-960-801: Replacement of the Cabin Ceiling Light Lamp.
- AMM 33-23-00-960-801: Replacement of the Cabin Overhead Bin Fluorescent Light.
- AMM 33-23-00-960-802: Replacement of the Cabin Sidewall Lights (Next Generation).
- AMM 33-23-01-960-801: Replacement of the Cabin Overhead Lights (Next Generation).
- AMM 33-25-00-960-801: Replacement of the Lavatory Lights (Next Generation).
- AMM 33-24-00-000-801: Removal of the Reading Lights.
- AMM 33-24-00-400-801: Installation of the Reading Lights.
- AMM 33-24-00-710-801: Operational Test of the PSU and Information Signs.
- AMM 33-42-00-710-801: Operational Test of the Navigation Lights.
- AMM 33-51-01-000-801: Removal of the Emergency-Light Power-Supply Units.
- AMM 33-51-01-400-801: Installation of the Emergency Light Power Supply Units.